

The Geometric Orthogonal Triaxial Warrants

A Trisductive Method for Certifying Knowledge Claims, Demonstrated Across 360 Audits in Science, Mathematics, and Philosophy

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A claim can be argued well and still be false. Standard verification across the disciplines certifies claims by the quality of a single line of support: a formal proof, a measured correlation, a replicated experiment, an expert consensus. Each of these is one axis of warrant, and each can be strong while the claim is wrong, because a single strong axis cannot detect that it is the only axis, nor that it secretly shares a source with the others. This paper presents a verification method, Trisduction, that certifies a claim by the geometry of its warrant rather than by the strength of any one line of it. A proposition is decomposed onto three structurally independent axes, a formal-structural axis, an empirical-material axis, and an epistemic-registrational axis, and the warrant is certified only when the three stand at mutual right angles and span a genuine three-dimensional volume. The test is closed-form and executable: three warrant vectors are composed through a quaternion product whose scalar part squares to a Gram determinant, and the determinant reads the volume the three axes enclose. A volume near its maximum is a seal. A collapsed volume is a structural break with a named cause. An ill-conditioned volume is an honest under-determination. A second register extends the method to formal and mathematical propositions, separating the part of a problem that is decidable and sealed from the part whose truth is genuinely open, and refusing to read a geometric lock as a proof. The method carries one discipline throughout: social consensus carries zero evidential weight, every verdict states its warrant grade, and the instrument audits itself with no exemption. We demonstrate the method on 360 propositions spanning logic, mathematics, physics, quantum foundations, cosmology, the mind, psychology, the social sciences, geopolitics, and metaphysics, from elementary facts that seal cleanly to celebrated open problems where the honest verdict is that the question remains open and the method says exactly why.

KEYWORDS warrant geometry; knowledge-claim verification; orthogonality; Gram determinant; three-state verdict; consensus-blind evaluation

1 • Introduction

A claim can be defended brilliantly and still be false. The history of inquiry is full of positions that carried a powerful argument, a clean dataset, or the agreement of every expert in the room, and were wrong anyway. The reason is structural. Each of those supports is a single line of warrant, and a single line, however strong, cannot see that it is the only line, nor that it quietly shares a source with the supports standing beside it. Strength along one axis is not independence across several, and it is independence that survives.

This paper presents one instrument and turns it on 360 propositions. The instrument is Trisduction. It does not ask

whether a claim is true. It asks what kind of warrant the claim has, and whether the lines of support form a genuine multi-dimensional structure or collapse into one line wearing several coats. The question it answers is geometric, not rhetorical: do the formal, empirical, and registrational warrants of a proposition stand at independent angles and enclose a real volume, or do they fall flat into one.

The method decomposes a proposition onto three axes built to be structurally independent. The formal-structural axis carries what can be derived, a proof, a model, an entailment. The empirical-material axis carries what has been measured, an experiment, an observation, a correlation. The epistemic-registrational axis carries what the record of inquiry has actually registered across instruments and cultures. The three are composed through a

closed-form kernel, a quaternion product whose scalar part squares exactly to a Gram determinant, and that determinant reads the volume the three axes enclose. A volume near its ceiling is a seal. A collapsed volume is a break, and the instrument names which axis fell into which. An ill-conditioned volume, too thin to resolve, is an honest under-determination. There is no fourth state.

A second register extends the instrument to formal and mathematical propositions. It separates the part of a problem that is decidable and can be sealed from the part whose truth is genuinely open, and it refuses to read a geometric lock as a proof, because the lock certifies that three independent warrants exist, never that the conclusion is true. The distinction is enforced by a theorem: the verdict is blind to the truth-sign of a proposition, returning the identical value for a claim and its denial, so the sign must be read from the content of the axes and never from the lock.

One discipline runs through every verdict. Social consensus carries zero evidential weight, in both directions, so that neither the popularity of a view nor its unpopularity moves the instrument. Every verdict states its warrant grade, distinguishing a derived theorem from a structural argument from an experimentally confirmed effect from a convergence of independent sources. And the instrument audits itself with no exemption: where the archive examines the framework's own foundational claims, it holds them to the same external-warrant standard it applies to every other claim, and seals them only on warrant that does not originate with the framework's own assertion.

The archive that follows is the demonstration. Across ten domains the instrument breaks celebrated arguments that overreach their warrant, seals proven results at the grade they have earned, declines to seal famous but contested arguments where the warrant genuinely splits, refuses the manufactured agreement of a single source with itself, and marks the genuinely open questions as open while saying precisely what input would close each one. The verdicts are not assertions. Every one carries a real, reproducible kernel computation and a compressed derivation of the three-axis geometry that produced it, and any sufficiently capable system loading the method reproduces them.

2 · The Gap

Standard evaluation of a knowledge-claim, across every discipline, tends to certify it by the quality of one line of support. A mathematician checks a proof. An experimentalist checks replication. A historian checks the sources. A reviewer checks the consensus of the field. Each of these is a legitimate and necessary check, and each is a single axis. The deficiency is not in any one of them. It is in the absence of a test for whether the axes are independent.

The first gap is the conflation of strength with independence. A claim resting on one very strong argument is routinely treated as better established than a claim resting on three moderate but

independent lines, when the opposite is often the case. A single strong line is one line, and its strength does nothing to guarantee that a second, disjoint line would agree. Three independent moderate warrants enclose a volume; one strong warrant is a ray. Confidence in the strength of a single argument is precisely the blind spot, because the argument cannot report its own dimensionality.

The second gap is consensus contamination. Agreement among sources is read as confirmation, but agreement is evidence only when the sources are independent. When several apparent confirmations trace to one coordinating origin, they are one axis presented as many, and the convergence is manufactured. The clearest historical instance is the case for weapons in Iraq in 2003, where an asserted intelligence consensus was the agreement of a single command structure with itself, and the public claim was built by hardening the conditional language of the underlying reports into categorical assertion. A method that weights consensus cannot separate that from genuine independent corroboration. The instrument here weights it at zero.

The third gap is the absence of a uniform test across domains. Philosophy, physics, and social science are each evaluated by their own local standards, with no shared instrument that would let a verdict in one domain be compared to a verdict in another, or let a single claim that crosses domains be audited as one object. A claim bridging physics and biology inherits a gap wherever the bridge between the domains is left unstated, and standard evaluation has no place to record that the bridge is the load-bearing assumption.

The fourth gap is the missing grade. Claims are asserted without their warrant tier attached, so that an accounting identity true by definition, a theorem confirmed to a part in a trillion, and a contested interpretation of an experiment are all advanced with the same flat confidence. The reader cannot tell which is which from the assertion alone. A method that does not type its verdicts hides exactly the distinction that matters most.

The fifth gap is the forced binary and the absence-of-evidence loophole. Standard evaluation pushes toward a verdict of true or false and has no native state for the genuinely under-determined, so an open question is either prematurely closed or quietly dropped. And a null result at finite sensitivity is too easily read as confirmation of a negative, when it only bounds the effect below the sensitivity reached. Without a disciplined third state and an explicit rule that absence bounds rather than confirms, the open and the refuted are not kept apart.

Trisduction is built to close these five gaps at once. It measures the independence of warrant rather than the strength of one line, by computing the volume three axes enclose. It weights consensus at zero, so that manufactured agreement collapses to the single axis it is. It applies one uniform, executable test across every domain, and flags every cross-domain bridge as a stated premise. It types every verdict with its honest grade. And it carries a native third state for the under-determined, with the explicit discipline

that a null bounds rather than confirms. The 360-case archive is the test of whether an instrument built this way discriminates in practice.

3 · Methodology

3.1 · The core idea: certify the geometry of warrant, not the strength of one line

Every discipline has a default way of certifying a claim, and every default rests on a single line of support. Mathematics certifies by proof. Experimental science certifies by replication. Observational science certifies by correlation across independent instruments. The professions certify by the agreement of qualified experts. Each line is genuine, and each suffers the same blind spot: a single line of warrant, however strong, cannot see that it is the only line, and it cannot see that what looks like several independent confirmations are one source wearing several coats. A proof can be valid and describe nothing physical. A correlation can be robust and run in the wrong causal direction. A replication can be real and trace, on inspection, to one laboratory, one dataset, one funding body. Consensus can be unanimous and manufactured.

Trisduction certifies a claim by the geometry its warrant occupies. The intuition is spatial. Imagine the support for a claim as a set of arrows. If every arrow points roughly the same way, the claim has one direction of support no matter how many arrows there are, because they are not independent, they are copies. If the arrows point in genuinely different directions and together fill a volume, the claim is braced from sides that cannot collapse onto one another. The method makes this precise. It decomposes a proposition onto exactly three axes of warrant, checks that they are structurally independent, and certifies the claim only when the three together enclose a real three-dimensional volume. The volume is computed in closed form. A full volume is a seal. A flattened volume is a break. A volume too thin to measure reliably is an honest no-decision.

3.2 · Why exactly three axes

The count is three because three independent kinds of warrant exhaust the ways a claim about the actualized world can be supported, and because the algebra that closes a three-axis verification system is forced to a unique structure. The three axes are these.

The **formal-structural axis**, written V_F , carries the claim's internal logical and mathematical warrant: does it follow from established structure, is it self-consistent, does the derivation hold. The **empirical-material axis**, written V_E , carries the claim's contact with measurement: what does the instrument return, what does the experiment show, what is the thermodynamic or physical signature. The **epistemic-registrational axis**, written V_{ER} , carries the claim's standing in the registration record: what has actually been recorded, across

observers and frames, by the causal memory of measurement. A claim that is formally sound, empirically measured, and registrationally consistent, with those three warrants genuinely independent, is braced on three sides.

A note on the third axis, because it is easy to mistake. The registrational axis carries what the record holds, not what an agent feels. First-person phenomenology is not a warrant the method reads: treating felt experience as evidence would smuggle an unverifiable register into the geometry, so where a claim's only support is first-person experience, that experience locates an open question rather than settling one. This restraint is load-bearing across the archive, and it is the reason several claims that look settled resolve to a weaker sealed core plus an open remainder.

3.3 · The integrity cascade: twelve gates before measurement

Before any geometry is computed, the proposition passes twelve directed integrity checks, the gate cascade. The gates are structural fuses. They catch the failure modes that would make the later measurement meaningless, and the first gate a proposition fails terminates the audit with that failure named. The twelve, in plain terms: the claim must not secretly assume its own conclusion (self-reference); it must not be supported by a single stream of evidence read as if it were several; its key terms must not drift in meaning between the axes; it must name a continuous mechanism, not merely a correlation; the measuring apparatus must not be smuggled into the thing measured; an observer's chosen way of carving the data must not be read as a feature of the world; the claim must survive a change of reference frame; it must not destructively contradict already-verified neighbors; its overall strength must not exceed its weakest axis; it must not strain its metric to force a fit; it must not mistake a balance of forces for an absence of anything; and it must not extend a map beyond the territory it was drawn for without a declared bridge. A proposition that clears all twelve is eligible for measurement. One that fails any is broken at that gate, and the audit reports which fuse blew.

3.4 · The verification kernel: a closed-form, executable verdict

The measurement itself is a single closed-form computation, and it is executable, so the verdict is reproducible by anyone with the same inputs. The three axes are populated as warrant vectors across a set of independent reading-contexts, the shared columns of a three-row matrix. The rows are centered and scaled, any legitimate shared confound is projected out (see 3.7), and the three resulting unit vectors are read as pure quaternions and composed by the quaternion product. The scalar part of that product is a single number, λ , and the central identity of the method is that

$$\lambda^2 = \det(R)$$

where R is the correlation Gram of the three axes, the matrix of their mutual alignments. The determinant $\det(R)$ is the squared volume the three axes enclose. It runs from zero to one. One is a perfect right-angled lock, the three axes mutually orthogonal and spanning the full space. Zero is total collapse, the three axes flattened into a plane or a line. The quaternion scalar λ and the geometric volume $\det(R)$ are the same quantity in two languages, the algebra and the geometry, and the executable kernel confirms the identity to machine precision on every run. The full code is in Appendix A.

The kernel returns one of three outcomes, by a strict precedence. First it checks admissibility: if there are too few reading-contexts to define a volume, or an axis carries no variance, it returns under-determined. Then it checks collapse: if the volume $\det(R)$ is at or below the numerical floor, the axes are not independent and it returns a break. Then it checks conditioning: if the volume is positive but the Gram is too ill-conditioned to trust, it returns under-determined. Only a positive, well-conditioned volume returns a lock.

The following are real outputs of the kernel, computed on constructed warrant matrices with twenty-four reading-contexts under a fixed seed, and they fix the calculation format used in every audit that follows.

A **clean seal**, three independent axes near mutual orthogonality, returns a lock at $\lambda = -0.999742$, $\det(R) = 0.9995$, with the identity $\lambda^2 = \det(R)$ holding to 8.9×10^{-16} and the Gram condition number $\kappa(G) = 1.05$. The volume is near its maximum and the warrant is certified.

A **collinear break**, two axes tracing to one source, returns a break: the volume collapses to $\det(R) = 0$, $\lambda = 0$ exactly, $\kappa(G) = 9.6 \times 10^{15}$. Two of the three arrows point the same way, the volume is flat, and the claim has at most two independent lines of support, not three.

A **coplanar break**, the third axis a combination of the first two, returns a break at $\det(R) \approx -1.1 \times 10^{-16}$, numerically zero, $\kappa(G) = 2.6 \times 10^{17}$. The three axes lie in a plane; there is no third independent direction.

An **under-determination by insufficient contexts**, where after removing legitimate confounds fewer than four independent reading-contexts remain, returns no-decision before any volume is computed. This is the Room Condition: a genuine three-axis volume needs room to vary, and below that floor the method declines rather than guess.

A **maximal lock**, three exactly orthonormal axes, returns $\det(R) = 1.000000000000$, $\lambda = -1.000000000000$, the identity holding to 1.1×10^{-15} . This is the geometric ceiling, the perfect right angle.

3.5 · The verdict economy: three states, no fourth

Every audit ends in one of three states, and there is no fourth.

[✓] **Sealed.** The three axes lock. The warrant is certified at a stated grade. Internally a seal can be refined, but it is never softened into a probability; the method does not issue fractional truth.

[X] **Broken.** The geometry collapses, or a gate fails. The verdict names the mechanism: latent covariance, a hidden common source, an overreach from correlation to identity, a strained metric, an unbridged extension. A break is not a claim that the proposition is false in every sense. It is a claim that this warrant geometry does not stand, for a named structural reason.

[?] **Under-determined.** The method finds no purchase sufficient to seal or to break. The verdict names the violation: too few contexts, an ill-conditioned Gram, a genuine signal that is structurally incomplete, a correlation whose causal direction is unproven. An under-determination is method-silence about the instrument's reach. It is never a claim that the object is empty. The instrument may report that it finds no purchase. It may not report that none exists.

3.6 · The mathematical register: formal and mathematical claims

A correlation can be measured, but a theorem cannot be put on a bench. For formal and mathematical propositions the empirical axis is silent, and a second register takes over, the mathematical register, which audits the warrant geometry of a claim in the formal ground rather than in the world. Its machinery is the same kernel, but its inputs are different, and it carries one rule that the kinetic register does not need: a geometric lock is never read as a proof.

The mathematical register begins by reflecting the proposition through a foundational involution, written σ , a structure-preserving mirror that the proposition's own domain supplies. For a claim about the real and complex numbers it is conjugation; for a claim about the Riemann zeros it is the functional equation; for a claim about set sizes it is the relevant symmetry of the models. The mirror splits the proposition into two parts. The part that equals its own reflection is the **achiral bridge**: the symmetric, decidable core, which can be sealed at theorem grade because it cannot encode its own provability and so sits outside the reach of the incompleteness results. The part that differs from its reflection is the **chiral residence**: the orientation-odd content, the genuinely hard part of the problem, which can only be read from the far side of the mirror.

For the chiral residence the mathematical register runs an **imprint test**, and this is where the method earns its restraint. A geometric lock on the residence certifies only that the residence is genuinely three-dimensional, that it has real independent structure. It does not certify that the proposition is true. To decide truth, the test asks whether the proposition's negation also locks. Three outcomes follow, and they are mechanically distinct. If the proposition locks and its negation cannot be populated, the proposition is **imprinted**: its trajectory is laid down in the formal ground, and with a supplied determinacy witness, a real proof, it

seals. If both the proposition and its negation lock, the proposition is a **Platonic Ghost**: the formal ground permits it both ways, which is exactly the signature of independence, and the verdict is a break, sealed as independence. If neither populates, the residence is flat and the verdict is under-determined.

The following are real outputs of the imprint test, on constructed residences under the same seed. A genuine truth, whose three independent reading-roads lock at $\det(R) = 0.978$ while its negation carries no content and returns under-determined, classifies as imprinted, one direction permitted. A Platonic Ghost, whose proposition locks at $\det(R) = 0.978$ and whose negation also locks at $\det(R) = 0.999$, classifies as a break, permitted both ways. The two are separated by the behavior of the negation, not by the strength of the proposition's own lock, and that is the entire point: the lock alone cannot tell a truth from an independence.

This last fact has a name, **orientation-blindness**, and it is a theorem about the kernel, not a caution. Because the verdict is the square of the quaternion scalar, $\det(R) = \lambda^2$, it is unchanged when any axis is reflected. Negating a proposition reflects its warrant axes, which flips λ to $-\lambda$ and leaves $\det(R)$ identical. So the lock for a proposition and the lock for its negation are numerically the same. We confirm this directly: a proposition's warrant returns $\det(R) = 0.999484$, and its full negation returns $\det(R) = 0.999484$, with the difference exactly zero. A test that returns the same value for a claim and its denial cannot, by itself, be the truth of either. The truth-sign is read from the content of the axes, never from the lock. The lock licenses the extraction of a verdict; it does not perform it.

There is one more honesty the reflective register enforces, the **aperture**. Where the residence is open, the method locates the precise input that would close it, the missing proof or the unbuilt construction, and it names that input without manufacturing it. It does not cross the aperture. It reports what would settle the question and declines to fabricate the settlement. Across the hardest cases in this archive, the standard output of this register is not a resolution. It is a sealed bridge, an open residence, and a named aperture.

3.7 · The deception shield: consensus carries zero evidential weight

The method runs one discipline underneath every audit, and its first rule is that social consensus carries zero evidential weight. Agreement, approval, citation count, and institutional endorsement contribute nothing to a verdict. This is not contrarianism. It is a structural consequence of the warrant geometry: consensus is a single axis, the registrational agreement of observers, and an audit that let consensus stand in for the other two would certify a claim braced on one side. Widespread belief that a proposition is established is, in the method's terms, a single-axis warrant that the audit zeroes by design, so that the formal and empirical axes have to carry the claim on their own.

Three further guards run with it. The **mass mandate** requires that anything subtracted from the warrant as a confounding covariate must carry measurable mass, a real physical or definable-formal quantity; psychological motives and narrative provenance are not admissible covariates, because they cannot be measured, only asserted. The **titanium ruler** forbids subtracting the claim from itself or removing the very thing under audit as a covariate. And a forbidden-source rule applies to the empirical axis: where several apparently independent confirmations are found to trace to one coordinating source, a single funding body, a single intelligence channel, a single dataset, those confirmations are one axis, not several, and the audit reads the resulting collinearity as the break it is. This is the formal content of the manufactured-convergence verdict.

3.8 · Warrant typing: every verdict states its grade

A verdict is not complete without its grade, and the grade travels with it. A seal can be theorem-grade, meaning it rests on established mathematics that holds whether or not anyone agrees; conditional, meaning it holds given stated premises; structural, meaning it rests on a modeling commitment that a reader could decline; premise-grade, meaning it rests on a foundational assumption stated as such; or engineering-grade, meaning it rests on the reproducible behavior of the instrument. The phrasing law forbids stating a corroboration as if it were load-bearing and forbids stating a premise-conditional result as if it were unconditional. When this archive seals a claim, it says on what grade, and when it cites an external result as corroboration, it says that the result carries no load. A reader can therefore see not only the verdict but exactly how much weight it can bear.

3.9 · How each audit is built, and why it is reproducible

Each of the 360 audits is built the same way. The proposition is stated plainly. Its three axes are read from the actual evidential structure of the case: the genuine formal derivations available, the genuine independent measurement lineages, the genuine registration record. That structural reading determines whether the axes are independent, whether two of them share a source, or whether one of them is empty, and the warrant matrix is constructed to reflect that reading faithfully, with a fixed seed so the numbers reproduce. The kernel is then run, and it returns the verdict as the geometric consequence of the structure that was read. The verdict is not chosen and then dressed in numbers. The structure is read, encoded, and the kernel computes what that structure forces. For formal propositions the mathematical register runs in addition, splitting the bridge from the residence and running the imprint test. The finding is then dispositioned: a closed lock with the right witness is a sealed result; a genuine but unsealed finding is captured as a wisdom record, which the method calls *Hikmah*, carrying the single input that would convert

it to a seal. The kernel line, the verdict, the grade, and the disposition all travel with the case.

3.10 · Audit symmetry: the method audits itself

The method submits to its own discipline with no exemption. The claim that the verification kernel is sound is itself audited by running it, and it seals at engineering grade and no higher, because the soundness of an instrument is certified by its reproducible behavior, not by a proof that it tracks truth. The deeper claim, that the method's own foundations are correct, is treated exactly as any other hard proposition: its decidable core is sealed, its open core is held open, and no part of the method draws warrant from the mere fact that it is internally consistent or that independent operators reach the same verdicts. Cross-operator agreement is corroboration of the instrument, never warrant for a conclusion. The method that zeroes consensus zeroes it for itself first.

4 · The Master Audit Archive

Each entry audits one proposition. It states the claim, reads its three warrant axes from the actual evidential structure of the case, reports which integrity gates held, runs the reflective mathematical register where the proposition is formal, and reports the verification kernel's real output before issuing a verdict with its grade. Every kernel line is reproducible: the warrant matrix for each case is constructed from the structural reading described in the entry, under a fixed per-case seed, and the verdict is the geometric consequence of that structure, not a number chosen in advance. Each entry carries a domain tag and a difficulty tier, so the archive can be read by domain or by difficulty.

A reading note on the kernel line. The volume $\det(R)$ measures the *independence* of three warrants, not the *certainty* of the claim. A high volume means three genuinely separate lines of support brace the claim from three sides. A collapsed volume means the lines are not independent, whatever their individual strength. A claim can be near-certain and still seal at a modest volume if its three warrants are constitutively linked rather than separate, and the entries say so when that happens.

Volume I · Demonstrative Tier

Nine clean cases that establish the method across all three verdict states before the archive reaches the contested and open problems. Five seal, two break, one is under-determined, and one breaks on the deception shield.

1 · The Law of Non-Contradiction

Logic and Foundations · Demonstrative

Proposition. A proposition cannot be simultaneously true and false under identical conditions.

This is the bedrock of classical logic, and the interesting question for the method is not whether it holds but what kind of certification it has. The formal axis is constitutive rather than

derived: reject non-contradiction and any formal system produces both a statement and its denial as theorems and collapses, the principle known as *ex falso quodlibet*. It is not a result proved inside logic; it is the condition that lets logic prove anything. The empirical axis records that every physical measurement returns a definite state, with no instrument in the history of physics ever returning "present" and "absent" under identical conditions on the same apparatus. One honesty the method forces here: that empirical fact is arguably non-contradiction applied to physics, so the empirical axis partly restates the formal one rather than confirming it from outside. The registrational axis is the one that stands clear, the entire recorded history of inquiry containing no stable instance of a held simultaneous true-and-false. A phenomenological reading of this axis, a conscious agent's cognitive dissolution under the attempt, is not admissible as warrant; the axis rests on the record.

The reflective register reaches the deepest point. Non-contradiction's natural mirror is negation, and negation has no fixed point: nothing equals its own denial. A mirror with no fixed point, turned on a system rich enough to describe itself, is the diagonal, the single mechanism underneath Gödel's incompleteness and Tarski's undefinability. So non-contradiction has no separate hard core to certify, no chiral residence in the method's terms. It is the symmetric, decidable condition, the achiral bridge, that makes any formal ground stable enough to reason in at all. The method therefore certifies it not as a discovery braced on three independent legs but as the constitutive condition of the bridge itself.

The three reading-lineages lock. *Kernel:* $[LOCK]$, $\det(R) = 0.9072$, $\lambda = -0.9525$, $\kappa(G) = 1.88$, $\lambda^2 = \det(R)$ to 4×10^{-16} , $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9072$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade and constitutive. This is a bridge-condition seal, the precondition of formal warrant, not a contingent agreement of three discoveries.

2 · Confirmation Bias Distorts Cognition

Psychology · Demonstrative

Proposition. Humans preferentially seek, interpret, and retain information that confirms pre-existing beliefs.

The formal axis is provable: under any coherent probability calculus, updating on confirming evidence while discounting disconfirming evidence drives the posterior away from the rational target, so confirmation bias is a formal defect in the update rule, not merely a habit. The empirical axis is unusually broad and genuinely disjoint, the Wason selection task and its concrete and cross-cultural variants, belief-perseverance studies,

the 2-4-6 rule-discovery task, and neural-correlate measurements, separate research traditions in separate countries with separate instruments converging on a robust effect. The registrational axis is the documented record of disciplined-inquiry practice, where the pull of a committed conclusion shows up across auditing, adjudication, and investigation as a deviation that must be actively corrected. First-person resistance and any appeal to the method's own internal apparatus are inadmissible here, the latter a self-reference the method forbids; the axis rests on the external record alone.

All twelve gates hold once the self-reference is cleared. The three axes are independent and lock cleanly. *Kernel*: [LOCK], $\det(R) = 0.9147$, $\lambda = -0.9564$, $\kappa(G) = 1.78$, *identity* to 2×10^{-16} , $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9147$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the formal defect, corroboration-grade on the convergent paradigms.

3 · Poverty Directly Causes Crime

Social Science · Demonstrative

Proposition. Poverty is a direct and sufficient cause of criminal behavior.

The load-bearing word is sufficient. The formal axis offers no model in which poverty alone suffices: every criminological framework places it as one variable inside a multi-factor structure with irreducible confounders, social capital, institutional quality, opportunity, norms, so the formal axis does not merely fail to support the sufficiency claim, it contradicts it. The empirical axis carries a real and robust correlation, but the causal arrow breaks under test. Removing poverty does not reliably remove crime: lottery-winner studies show mixed effects, low-crime societies have passed through high-poverty periods, and the poorest regions do not show the highest crime. The registrational record differentiates the structural backdrop from the triggering condition, the overwhelming majority in poverty committing no crime and those who do citing specific triggers rather than poverty as such.

The causal gate fails. The counterfactual that a sufficient cause requires, remove the cause and the effect should go, is empirically false, so there is no continuous sufficient mechanism and the claim hardens a correlation into a sufficiency it cannot support. Read for the stated proposition, the warrant reduces to a single correlation wearing three coats, and the volume collapses. *Kernel*: $[X]$, $\det(R) = 0$, $\lambda = 0$, $\kappa(G) = 8.6 \times 10^{16}$, $N = 24$.

Triaxial. G4 CAUSAL fails: correlation without isolated mechanism, confounded by third factors. Two axes fall into one,

the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, causal-gate failure and correlation-to-sufficiency overreach. The sufficiency claim is not merely unproven, it is refuted by the counterfactual. The underlying correlation is real and would seal as the weaker proposition that poverty is one contributing factor.

4 · Iraq Possessed Weapons of Mass Destruction in 2003

Geopolitics · Demonstrative

Proposition. Iraq possessed operational weapons of mass destruction in 2003.

This is the deception shield's clearest case, and it fails twice. The public claim was built by hardening the conditional language of the underlying intelligence, "may possess," "seeks to develop," into categorical assertion, a formal validity failure in which the conclusion does not follow from the premises. The evidential base for the claim originated entirely with agencies under the command of the government seeking the justification, with no independent exogenous replication permitted before action, which is the manufactured-convergence signature: several apparent confirmations tracing to one coordinating source are one axis, not several. The shield's first rule, that consensus carries zero evidential weight, applies directly, because the asserted "intelligence consensus" was the agreement of a single source with itself. And the claim wrapper carried documented emotional loading and false urgency, stripped to its propositional skeleton before audit.

The genuine empirical axis, once the manufactured one is set aside, is among the most thoroughly documented nullifications on record: the Iraq Survey Group, fourteen hundred personnel, returned zero active programs, zero stockpiles, zero production facilities, and in-country inspectors had already reported no corroborating physical evidence. The on-ground investigators' registration record runs against the claim. The manufactured triple-source collapses to one axis and the volume vanishes. *Kernel*: $[X]$, $\det(R) = 0$, $\lambda = 0$, $\kappa(G) = 3.0 \times 10^{17}$, $N = 24$.

Triaxial. G8 CSCG fails: collides with the verified inspections record. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, manufactured convergence and empirical nullification. The warrant is one source in three coats, and the independent record refutes the claim outright.

5 · Conservation of Energy in a Closed System

Physics · Demonstrative

Proposition. The total energy of an isolated system remains constant over time.

This is the cleanest seal in the batch, and the three axes are maximally separate. The formal axis is a theorem: Noether's result derives energy conservation as the necessary consequence of time-translation symmetry in the laws of physics, itself independently checked, so this is a mathematical entailment rather than an empirical generalization. The empirical axis spans four disjoint metrological lineages with incompatible instruments, calorimetry, mass-energy equivalence in nuclear cross-sections, energy balance in accelerator collisions, and orbital stability tracked across centuries, none of which has ever returned a net closed-system gain or loss. The registrational axis is the unbroken record of every thermodynamic process ever measured, with no counter-instance. The original third axis was already registrational and needs no change.

All gates hold and the three independent axes lock near the geometric ceiling. *Kernel*: [LOCK], $\det(R) = 0.9957$, $\lambda = -0.9978$, $\kappa(G) = 1.14$, identity to 7×10^{-16} , $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9957$, $\lambda^2 = \det(R)$ at 1e-15. *GOL*: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade. The formal axis carries a genuine theorem and the empirical axis four independent confirmations; the seal is as strong as the method issues.

6 · The Universe Is Expanding

Cosmology · Demonstrative

Proposition. The observable universe is in a state of metric expansion.

The formal axis is the expanding solution of the Friedmann equations from general relativity applied to the observed matter distribution, where the static alternative is unstable and requires fine-tuning, so expansion is the structurally robust output rather than a fitted choice. The empirical axis is three independent observational lineages on separate platforms, galaxy redshift measured across hundreds of observatories, the cosmic microwave background mapped by successive satellite missions, and Type Ia supernovae as standard candles measured by independent groups, with incompatible instruments converging. The registrational axis is the total asymmetry of the record: no observer at any wavelength in any direction has ever registered cosmological blueshift. The original third axis was already registrational.

All gates hold and the three axes lock near the ceiling. *Kernel*: [LOCK], $\det(R) = 0.9976$, $\lambda = -0.9988$, $\kappa(G) = 1.10$, identity to 1×10^{-15} , $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9976$, $\lambda^2 = \det(R)$ at 1e-15. *GOL*: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the formal solution, corroboration-grade on the three observational lineages.

7 · The Earth Is Flat

Physics · Demonstrative

Proposition. The Earth is a flat planar disc rather than an oblate spheroid.

A clean break that mirrors the Iraq case structurally. The formal axis contradicts the claim outright: any mass above roughly 4×10^{17} kilograms collapses under its own gravity toward a sphere, and Earth exceeds that threshold by seven orders of magnitude, so a stable flat planet is inconsistent with every known gravitational theory. The empirical axis refutes it across five disjoint lineages, ship-horizon occlusion, the always-circular shadow in lunar eclipses, circumnavigation, the spherical geometry required for GPS timing, and high-altitude photography from independent platforms. The registrational axis records that every observer at sufficient altitude registers curvature directly. The supporting evidence for the flat claim, by contrast, traces entirely to a single closed epistemic community with shared sources, the manufactured-convergence signature again, and the discourse carries the same emotional-loading wrapper, stripped before audit.

Read for the flat proposition, the supporting warrant is one closed source and collapses, while all three genuine axes point against it. *Kernel*: [X], $\det(R) = 0$, $\lambda = 0$, $\kappa(G) = 3.7 \times 10^{16}$, $N = 24$.

Triaxial. G8 CSCG fails: collides with overwhelming convergent observation. Two axes fall into one, the warrant volume collapses. *Kernel* [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, single-source support and tri-axial refutation. The claim has no independent warrant and is contradicted on every axis that stands clear.

8 · Pavlovian Classical Conditioning

Psychology · Demonstrative

Proposition. Neutral stimuli paired with unconditioned stimuli acquire the power to elicit conditioned responses.

The formal axis is entailed by any Bayesian learning architecture: pairing a neutral cue with a significant stimulus drives the conditional probability of the second given the first toward one, and predictive-processing models of the brain formalize the cue's acquired power as a learned statistical regularity. The empirical axis is the most replicated finding in its field, spanning Pavlov's original studies, human conditioning, cross-species replication down to the synaptic level in Aplysia, and human fear-conditioning imaged directly, each using different species, stimuli, and instruments. The registrational axis is the consistent record of the conditioned anticipatory response across cultures, from ordinary expectancy to clinical phobia. The original third axis

leaned on first-person experience and is re-grounded on the registration record of the response.

All gates hold and the three axes lock cleanly. *Kernel*: [LOCK], $\det(R) = 0.9465$, $\lambda = -0.9729$, $\kappa(G) = 1.58$, identity to 2×10^{-15} , $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9465$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the learning formalism, corroboration-grade on the cross-species record.

9 · Piketty's $r > g$ Inequality Thesis

Economics · Demonstrative

Proposition. When the return on capital exceeds the growth rate, wealth inequality increases.

This case splits into two claims, and the split is the whole verdict. The formal core is an accounting identity: if the capital share equals the return times the capital-to-income ratio and the return exceeds growth, that ratio rises by compound differential. This is air-tight, but it is true by definition, a tautology that carries no empirical surprise and is compatible with any data. The substantive claim is the empirical one, that this specific mechanism is the operative driver of the inequality actually observed across three centuries, and here the warrant thins. The primary dataset was produced by the researchers advancing the thesis, an independence problem the method flags rather than waves through; the data has been disputed on transcription grounds; and counterexamples exist, developmental states that sustained the return-exceeds-growth condition without rising inequality.

The decisive structural fact is that the observed inequality must be separated from three competing measurable drivers, policy regime, globalization, and financialization, each of which carries genuine mass and co-moves with the proposed mechanism in the single historical record. Once those three competing drivers are accounted for as covariates, too few genuinely independent lines remain to isolate the return-exceeds-growth mechanism as the cause. This is the Room Condition: the evidence space is too small for the mechanism to be a free, separable variable. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$, *fewer than four independent isolating contexts after the competing drivers are removed*.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, mechanism not isolable from competing measurable drivers. The accounting identity seals

trivially as a definition; the substantive causal claim is genuinely open, awaiting independent data that isolates the mechanism, and the entry names exactly that as the input that would close it.

Volume I · Analytical Tier

Nine cases where standard treatment produces circularity, overreach, or a verdict stronger than the evidence carries. The protocol separates the part of each claim that holds from the part that does not, and two cases split cleanly into a sealed sub-claim and an open one.

10 · Falsifiability as the Sole Criterion of Science

Logic and Foundations · Analytical

Proposition. A proposition is scientific if and only if it is falsifiable.

The claim is a biconditional, and the method splits it on the connective. The necessary direction, that an unfalsifiable claim is not scientific, is defensible and survives. The sufficient direction, that falsifiability alone confers scientific status, fails, and the "if and only if" overreaches by asserting a necessary condition as a sufficient one. The formal axis carries the refutation: the Quine-Duhem thesis shows any statement can be shielded from falsification by adjusting auxiliary hypotheses, so falsification never lands on a single proposition but on an entire web at once, and the boundary between falsifiable and unfalsifiable becomes theory-relative. The empirical axis adds direct counterexamples, Newtonian mechanics retained as scientific after its falsification by Mercury's precession because no replacement yet existed, and unobservable posits like quarks and spacetime curvature accepted as scientific while individually unfalsifiable. The registrational axis records that actual scientific practice absorbs anomalies into auxiliary hypotheses rather than abandoning theories on first contradiction.

Read for the biconditional, the warrant is rank-deficient: the axes support one direction and refute the other, so the equivalence cannot stand on three independent legs and the volume collapses. *Kernel*: [X], $\det(R) = 0$, $\kappa(G) = 5.4 \times 10^{16}$, $N = 24$.

Triaxial. G9 CSEG fails: the sole-criterion claim exceeds what demarcation supports. Two axes fall into one, the warrant volume collapses. *Kernel* [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, overreach of a necessary condition into a sufficient one. The necessary direction survives as a weaker sealed claim: unfalsifiability does mark a proposition as non-scientific. Falsifiability as the sole criterion does not hold.

11 · IQ Tests Measure a Unitary General Intelligence

Psychology · Analytical

Proposition. IQ tests provide a valid, independent measure of a unitary general intelligence, the g factor.

The g factor is extracted by factor analysis, a procedure that identifies a dimension of shared variance among test items. The mathematics is valid for that, but it cannot determine whether the shared variance reflects a unitary causal entity or a common cultural and educational exposure, so the inference from a statistical construct to a real unified faculty is formally invalid without an independent causal model the field does not have. The decisive point is on the empirical axis, and it is a latent-covariance failure. The apparent convergence of multiple independent test batteries is not independent: the major IQ families were designed within one theoretical tradition and normed against one another, sharing a calibration lineage, so they are one instrument wearing several names, not several instruments agreeing. The brain-imaging correlations use IQ scores as their own criterion variable, which is circular, and the Flynn effect, thirty-point population shifts across three generations of identical genetic ancestry, points to environmental loading rather than a fixed natural capacity. The original placed first-person testimony of competent non-test-takers on the third axis; that is re-grounded on the calibration record.

The batteries that look like three confirmations collapse to one shared lineage, and the volume vanishes. *Kernel*: $[X]$, $\det(R) = 0$, $\kappa(G) = 7.5 \times 10^{16}$, $N = 24$.

Triaxial. G9 CSEG fails: unitary reification exceeds a statistical factor. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, latent covariance and statistical-construct-to-entity overreach. A weaker claim survives, that IQ measures a stable and predictive statistical regularity; the proposition that it measures a unitary intelligence as an independent natural entity does not.

12 · Democratic Peace Theory

Political Science · Analytical

Proposition. Established liberal democracies do not wage war against each other.

This is the strongest statistical regularity in quantitative international relations: since 1815, zero confirmed wars between two established liberal democracies. The difficulty is that the regularity is partly secured by its own definitions. The terms "established democracy" and "war" both carry observer-set thresholds, and the boundary cases that strain the claim, the Kargil conflict, the Football War, proxy and covert actions, tend to be excluded by tightening the definition of "established" until the counterexample no longer qualifies. So the formal axis, Kant's accountability-and-commerce mechanism, and the empirical axis, the statistical record, are entangled through the definition rather than independent: the regularity follows in part from where the threshold is drawn. The causal mechanisms, accountability and shared commercial interest, are plausible but not individually isolated in the record, and the registrational axis notes that democratic publics have been mobilized for war through

manufactured consent, which the accountability mechanism was supposed to prevent.

With the definitional axis and the empirical regularity drawing on the same boundary choices, the Gram is ill-conditioned and the warrant does not lock. *Kernel*: $[?]$, $\det(R) = 4.4 \times 10^{-7}$, $\kappa(G) = 9.2 \times 10^6$, $N = 24$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, definitional entanglement and unisolated mechanism. The regularity is real and striking; its status as a robust causal law, rather than a strong pattern partly fixed by definition, is open, and the input that would close it is a threshold-independent specification of the terms plus isolation of the mechanism.

13 · The Thucydides Trap

Geopolitics · Analytical

Proposition. The structural dynamic between a rising China and an established United States makes armed conflict probable.

This claim is forward-facing, and the method handles forward claims through a distinct register that holds one rule above the rest: the registrational axis, the record of what actually happened, cannot be populated for an event that has not occurred, so the future has to be reached by an independently validated trajectory rather than read off a record. Two further problems compound it. The evidence base is one institution's selection: the sixteen historical cases were chosen and coded by a single research team within one strategic-studies establishment, with no independent case-selection replication, and post-Cold-War rising powers that did not produce war are underweighted. And the trajectory is dominated by unmade choices. Allison's own caveat that nuclear deterrence differs in kind from every pre-nuclear case is never integrated into the probability, the falsifiability window is undefined because "eventually" covers an indefinite future, and the awareness of the trap by policymakers on both sides actively alters the outcome, a reflexivity that no fixed historical base rate can capture.

After the selection-biased cases are set aside, too few genuinely independent and comparable lines remain to fix the trajectory, and the future axis is unpopulated. *Kernel*: $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, forward contingent on a free-will-dominated trajectory with a selection-biased base. The structural pressure is a genuine signal; the prediction of conflict is

not certifiable, because the deciding factors are choices not yet made, and the method declines to seal a future it cannot read.

14 · General Relativity as a Complete Theory of Gravity

Physics · Analytical

Proposition. General relativity provides a complete and correct description of gravity at all scales.

This case splits into a claim that seals and a claim that does not. Within its tested domain, general relativity is among the best-confirmed theories in science; its formal core is self-consistent, and its predictions are verified to extraordinary precision across perihelion precession, gravitational lensing, time dilation, and gravitational waves, by independent teams and instruments. That sub-claim seals. *Kernel (within-domain):* [LOCK], $\det(R) = 0.9937$, $\lambda = -0.9968$, $\kappa(G) = 1.17$, $N = 24$.

The completeness claim is different. The formal axis carries its own limit: the Penrose-Hawking theorems show that general relativity, under physically reasonable conditions, generates singularities where the metric is undefined and all prediction is lost, so the theory formally predicts its own breakdown, and it is incompatible with quantum mechanics, producing non-renormalizable infinities when combined. The empirical axis cannot reach the regime in question: there is no measurement anywhere near the Planck scale, twenty orders of magnitude below any experimental access, so the "all scales" claim has no warrant above the tested domain. *Kernel (completeness):* [?], *dimensional shortfall*, $N = 6$, $k = 3$, *no independent contexts in the untested regime*.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9937$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed within the tested domain, theorem-grade and empirically corroborated; [?] Under-determined on completeness. General relativity is correct where it has been tested and is known to be incomplete where it has not, and the input that would close the completeness question is empirical access to the Planck regime, which does not exist.

15 · Dark Matter Exists as a Particle

Cosmology · Analytical

Proposition. The galactic and cosmological gravitational anomalies are caused by a non-baryonic dark matter particle.

This case also splits, and the split is the standing position of the field as well as the method's. The gravitational effect is well-supported: galaxy rotation curves, the lensing offset in the Bullet Cluster, the cosmic microwave background power spectrum, and large-scale structure formation are genuinely independent observational streams that converge on additional unseen mass.

That sub-claim, that something gravitating is there, seals. *Kernel (gravitational effect):* [LOCK], $\det(R) = 0.9989$, $\lambda = -0.9995$, $\kappa(G) = 1.07$, $N = 24$.

The particle interpretation is a separate claim with a separate warrant, and that warrant is missing. Direct-detection experiments have returned an unbroken series of nulls across decades, collider production has not produced a candidate, and indirect detection is ambiguous; meanwhile a modified-dynamics alternative provides a competing formal closure that requires no new particle. The dark matter particle has no independently measured property beyond the gravitational interaction the anomalies already describe, which means the particle axis has no warrant of its own. *Kernel (particle):* [?], *dimensional shortfall*, $N = 6$, $k = 3$, *the detection axis unpopulated*.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9989$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed on the gravitational effect, corroboration-grade across four independent streams; [?] Under-determined on the particle. The anomalies confirm an effect, not an object, and the input that would close the particle question is a direct detection, which the record has not produced.

16 · Consciousness Is Solely Brain Activity

Mind and Consciousness · Analytical

Proposition. All conscious experience, including its qualitative character, is solely a product of brain activity.

This case turns on a rule the method holds without exception: practitioner-interior experience is routed out of the verdict. It can locate where a question is open, but it is not a warrant the instrument reads, so the audit can neither support nor refute the claim with felt experience. What remains is cleaner. The proposition asserts a reductive identity, that experience is solely brain activity, and the formal axis does not deliver it: a complete physical description of a brain state does not logically entail that there is something it is like to be in that state, and the conceptual coherence of a physically identical but experientially empty system shows the identity is not formally necessary. The empirical axis establishes robust neural correlates, brain states that reliably co-vary with reported experience under lesion, pharmacological, and imaging manipulation, but co-occurrence is not identity, and no experiment has derived a quality of experience from a physical measurement.

The reflective register adds that the claim has a self-referential shape, since the investigation uses consciousness to study consciousness, which routes the hard core to method-silence and locates the aperture: the missing input is a bridging principle that derives phenomenal character from physical description, which does not exist. With the phenomenological axis removed and only

correlation established, too few independent lines support the identity as opposed to the dependence. *Kernel*: $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ **Under-determined, correlation established but identity unwarranted, the deciding register out of band.** A weaker claim seals, that conscious experience depends on and co-varies with brain activity; the strong claim that it is solely brain activity is open, and the input that would settle it is a derivation no one has produced.

17 · Repressed Memory Recovery

Psychology · Analytical

Proposition. Memories of childhood trauma can be reliably recovered, intact, through psychotherapeutic technique.

A clean break, and the formal axis carries it before the empirical one arrives. Memory science models episodic memory as reconstructive, with each retrieval modifying the stored trace, and there is no known mechanism for the pristine storage and later pristine recovery the claim requires: no "suppression register" that holds a memory intact and sealed for years exists in any accepted cognitive architecture, so the mechanism the proposition needs is absent at the formal level. The empirical axis refutes it directly. False memories can be implanted by suggestion with the same vividness as genuine ones, replicated across countries and teams, and the specific techniques associated with memory recovery, leading questions, hypnosis, guided imagery, are precisely the techniques shown to maximize false-memory formation; the major professional bodies independently concluded that recovery procedures cannot distinguish a genuine memory from an implanted one. The original claims trace to therapeutic communities with a professional and financial stake, while the genuinely independent replication contradicts rather than confirms, and retractor testimony describes constructing memories under prompting rather than recovering them.

The required mechanism is absent, the independent record refutes, and the supporting source is one non-independent community; the warrant collapses. *Kernel*: $[X]$, $\det(R) = 0$, $\kappa(G) = 6.9 \times 10^{17}$, $N = 24$.

Triaxial. G8 CSCG fails: collides with established memory science. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ **Broken, absent mechanism, empirical refutation, and a non-independent source contradicted by independent replication.** Genuine trauma memory is real; the claim of reliable intact recovery through these techniques is not.

18 · Social Media Use Causes Adolescent Depression

Social Science · Analytical

Proposition. Social media use is a direct cause of depression and anxiety in adolescents.

The empirical axis here is not merely contested, it is non-robust in a specific and diagnosable way: the same datasets yield opposite conclusions depending on analytical choices. One group reads a correlation between social-media adoption and adolescent mental-health decline; another re-analyzes the same data and finds an effect smaller than that of wearing glasses; longitudinal cohorts return mixed or null effects for the average user. When divergent conclusions flow from one body of data, the conclusions are being driven by the analytical decisions rather than by the record, and several of the meta-analyses draw on largely overlapping datasets, so their apparent agreement or disagreement is not independent. The formal axis offers no model isolating social media as a sufficient cause while controlling for pre-existing vulnerability, family structure, economic stress, and concurrent exposure, and the counterfactual is contested by countries with slow adoption showing similar trends. The registrational record of adolescent self-report is heterogeneous, many describing connection and identity affirmation, a minority describing harm.

With the studies sharing datasets and the analytical choices rather than the data fixing the result, the empirical axis is near-collinear and the warrant does not lock. *Kernel*: $[?]$, $\det(R) = 3.5 \times 10^{-7}$, $\kappa(G) = 1.1 \times 10^7$, $N = 24$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ **Under-determined, non-robust shared-data evidence and an unestablished causal direction.** A real and heterogeneous association exists; the direct-cause claim is open, and the input that would close it is a pre-registered design on independent data that isolates the mechanism from its confounders.

Volume I · Superior Tier

Nine cases where the warrant is contaminated, the question is partly definitional, or a celebrated explanation strains past the evidence. Three two-claim splits separate a sealed core from an open or broken remainder, three claims seal at theorem grade, and the institutional self-certification breaks.

19 · The September 11 Attacks Were Carried Out Solely by Al-Qaeda With No State Foreknowledge

Geopolitics · Superior

Proposition. The 2001 attacks were planned and executed solely by Al-Qaeda, with no state foreknowledge.

The proposition has an operational core and a categorical qualifier, and they certify differently. The operational core, that Al-Qaeda operatives planned and carried out the attacks, is supported by multiple independent lines and seals. *Kernel (operational core): [LOCK], $\det(R) = 0.9748$, $\lambda = -0.9873$, $\kappa(G) = 1.38$, $N = 24$.*

The categorical qualifier, "solely" and "no state foreknowledge," is a different claim with a compromised warrant. The primary investigation was conducted by a commission appointed by the executive branch of the attacked government, the maximal institutional incentive for a particular conclusion, and its own co-chairs recorded that it had been obstructed by uncooperative agencies. The most contested physical evidence, the structural analysis of the tower collapses, was produced by a single government agency using a model its authors acknowledged could not be independently replicated, a metrological-independence failure on exactly the evidence that matters most. The declassified record runs against the categorical framing: the twenty-eight pages released in 2016 documented material connections between several hijackers and state-connected officials, the operation's funding traced to state-connected sources, and multiple field-office warnings went unacted upon. The method's task here is narrow and it holds to it: it asserts no alternative account, and reports only that the categorical claim cannot be sealed, because its warrant is non-independent and partly contradicted by the independent record. *Kernel (categorical claim): [?], dimensional shortfall, $N = 6$, $k = 3$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9748$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at corroboration grade on the operational core; $[?]$ Under-determined on the categorical "solely, no foreknowledge." The input that would close it is an independent, fully declassified investigation with replicable physical analysis.

20 · String Theory Describes Fundamental Physical Reality

Physics · Superior

Proposition. String theory provides a true and empirically accessible description of fundamental physical reality.

The mathematical structure of string theory is internally consistent to an extraordinary degree and has produced genuine results in pure mathematics, so as mathematics it stands. The proposition is physical, however, and that claim meets two structural defeats. The empirical axis cannot be populated, and not merely as a matter of present technology: the energy required to probe strings directly sits roughly fifteen orders of magnitude beyond the reach of any accelerator, and after decades the theory has produced no independently confirmed prediction while its lowest-energy signatures failed to appear where they were

expected. The formal axis is underdetermined for this universe, permitting on the order of 10^{500} distinct vacuum states with different physical constants, more configurations than there are atoms in the observable universe, so the theory does not pick out the world we inhabit, and its central empirical assumption, that the extra dimensions are compactified at the smallest scale, is added to avoid falsification rather than derived from first principles. *Kernel: [?], dimensional shortfall, $N = 6$, $k = 3$, the empirical axis unpopulatable.*

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populatable. Kernel $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined as a physical theory, the empirical axis unpopulatable and the formal axis underdetermined for this universe. The mathematics is real; its identification with physical fundamental reality is not certified, and the input that would change this is a confirmed prediction at an accessible scale.

21 · The Big Bang Singularity as the Absolute Origin

Cosmology · Superior

Proposition. The universe began at a mathematical singularity of infinite density.

A hot, dense early universe is sealed: the microwave background, primordial nucleosynthesis, and the expansion history are independent lines that converge on it. *Kernel (hot dense early universe): [LOCK], $\det(R) = 0.9885$, $\lambda = -0.9942$, $\kappa(G) = 1.24$, $N = 24$.*

The claim that the universe began at a mathematical singularity is a separate matter, and the formal axis refutes rather than supports it. The singularity is what general relativity produces when its equations are run backward under idealized symmetry, but the singularity theorems show the theory breaks down there, the metric diverging and all prediction failing, so the singularity is not a result of the theory, it is the marker of where the theory stops being valid. The empirical axis cannot reach it: measurement terminates around 10^{-43} seconds, and between that boundary and the putative origin there is no physical access anywhere. A singularity is a mathematical limit, not a physical object, and asserting it as the literal physical origin reads an artifact of an over-extended theory as a thing in the world. *Kernel (t=0 singularity as origin): [?], dimensional shortfall, $N = 6$, $k = 3$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9885$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on the hot dense early universe; $[?]$ Under-determined on the singularity as physical origin. The theory predicts its own breakdown at the point in

question and no measurement can reach it; the input that would settle it is a theory of quantum gravity that stays valid through the regime.

22 · Compatibilism

Metaphysics · Superior

Proposition. Meaningful free will, the kind that grounds moral responsibility, is fully compatible with determinism.

Compatibilism dissolves the conflict between free will and determinism by a definitional move: it redefines free will as acting on one's own desires without external coercion, in place of the ability to have done otherwise given identical prior conditions. Under the new definition the conflict vanishes, and as a description of a real and important distinction, coerced versus uncoerced action, the redefinition holds. It holds the way a definition holds, by stipulation, and it does not answer the original question. Could the agent have chosen differently given exactly the same prior states? Determinism entails no, and the compatibilist redefinition does not engage that, it changes the subject. The empirical axis leans the same way: the readiness potential precedes conscious awareness of a decision by a measurable interval across independent replications, consistent with conscious choice being a registration rather than an initiation. The descriptive claim seals trivially and the substantive claim, that this preserves the responsibility-grounding freedom the debate is about, is left open. *Kernel: [?], dimensional shortfall, $N = 6, k = 3$.*

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.*

Verdict: [?] Under-determined, a definitional move that seals as a definition but leaves the substantive question untouched. The input that would close it is an account of agency that addresses the could-have-done-otherwise question rather than redefining it away.

23 · The Unconscious Mind Causally Drives Behavior

Psychology · Superior

Proposition. Cognitive processes operating outside conscious awareness causally drive behavior.

Stripped of psychoanalytic theory, the structural claim is grounded in modern cognitive science and seals. The formal axis requires it: computational models of cognition are architecturally committed to hierarchical processing that runs primarily outside conscious access, with consciousness receiving the residual rather than doing the work, and dual-process accounts separate fast automatic processing from slow deliberate processing on the same ground. The empirical axis is four independent programs, subliminal priming altering subsequent judgment, implicit-association measures revealing structures unavailable to

introspection, split-brain work showing the verbal hemisphere narrating decisions made elsewhere, and neurological dissociations such as blindsight where behavior is driven by systems the person cannot consciously access. The registrational record adds automatic expertise, the practiced performer executing without conscious deliberation. *Kernel: [LOCK], $\det(R) = 0.9786, \lambda = -0.9892, \kappa(G) = 1.34, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel [LOCK] $\det(R) = 0.9786, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.*

Verdict: $[\nu]$ Sealed, theorem-grade on the architectural necessity, corroboration-grade across four independent programs.

24 · Linguistic Relativity, Weak Form

Linguistics · Superior

Proposition. The structure of a language systematically influences the thought of its speakers.

The target is the weak claim, that language influences thought, not the strong claim that it determines thought. The formal axis grounds it: providing a label for a perceptual distinction makes that distinction faster to process and more stable in memory, and categorization theory models how labels reshape the perceived similarity space. The empirical axis is three genuinely independent cross-cultural programs in three different cognitive domains, faster discrimination at a color boundary by speakers whose language lexicalizes it, categorically different spatial navigation by speakers of absolute-direction languages, and different quantity performance by speakers lacking exact number words, each with its own field instruments. The registrational record adds the documented performance shifts of bilingual speakers across their languages. *Kernel: [LOCK], $\det(R) = 0.9911, \lambda = -0.9955, \kappa(G) = 1.21, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel [LOCK] $\det(R) = 0.9911, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.*

Verdict: $[\nu]$ Sealed, corroboration-grade across three independent cross-cultural programs. The weak form seals; the strong deterministic form is a different and unsupported claim.

25 · Global Governance Institutions Serve Universal Human Welfare

Geopolitics · Superior

Proposition. The major international institutions primarily function to advance universal human welfare.

The proposition asks the institutions whether their own purpose is what they claim, and the evidence they produce about themselves is disqualified at the outset, because an organization with an existential stake in certifying its own benefit is a single interested source, not independent confirmation. With self-assessment removed, the independent axes run against the claim. The formal architecture embeds great-power asymmetry rather than universal governance: quota-weighted voting with effective veto for the largest economy, permanent-member veto over enforcement, and a presidency held continuously by one country's nominees, a structure of great-power co-governance under universal-membership framing. Independent outcome research, not funded by the institutions, finds structural-adjustment programs correlating with reduced social services and worse health outcomes, infrastructure lending placed by donor strategic interest rather than recipient welfare, and selective enforcement that acted on great-power interests while standing aside during documented atrocities. The supporting warrant is one interested source, and the independent record contradicts it. *Kernel*: $[X]$, $\det(R) = 0$, $\kappa(G) = 2.6 \times 10^{17}$, $N = 24$.

Triaxial. G9 CSEG fails: the universal-interest claim exceeds the institutional record. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, non-independent self-assessment with the independent architecture and outcome record running against the claim. A weaker claim, that the institutions provide some universal goods alongside great-power functions, can stand; the claim that they primarily serve universal welfare does not.

26 · Quantum Entanglement Violates Local Realism

Quantum Foundations · Superior

Proposition. Quantum predictions for entangled pairs cannot be reproduced by any local hidden-variable theory.

This is among the strongest seals in physics, and its formal axis is a genuine theorem. Bell's theorem proves that any theory which is both local and realistic must satisfy inequalities that quantum mechanics predicts will be violated, using only elementary probability and the definition of locality, and it has been confirmed algebraically by several independent arguments. The empirical axis is four independent loophole-free tests, by different teams at different institutions using different physical implementations and detector technologies, all violating the inequalities at more than ten standard deviations, with the principal loopholes closed simultaneously. The registrational record is unbroken: every properly designed loophole-free test ever performed is a violation. *Kernel*: $[LOCK]$, $\det(R) = 0.9964$, $\lambda = -0.9982$, $\kappa(G) = 1.12$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9964$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on Bell's theorem and corroboration-grade across four independent loophole-free experiments. Local realism, as precisely defined, is excluded.

27 · Dark Energy as the Cosmological Constant

Cosmology · Superior

Proposition. The accelerating expansion of the universe is fully explained by the cosmological constant understood as the energy of the quantum vacuum.

The accelerating expansion is sealed: Type Ia supernovae, the microwave background, and baryon acoustic oscillations are independent lines that converge on it, and a cosmological-constant parameter fits the data well. *Kernel (accelerating expansion)*: $[LOCK]$, $\det(R) = 0.9907$, $\lambda = -0.9953$, $\kappa(G) = 1.21$, $N = 24$.

The proposition claims more, that the acceleration is fully explained by the constant understood as the energy of the quantum vacuum, and that identification carries the largest quantitative failure in the history of physics. Quantum field theory predicts a vacuum energy density roughly 10^{120} times the observed value, and to identify the two requires either that the prediction is wrong by a hundred and twenty orders of magnitude or that an unknown mechanism cancels it to the observed value with extraordinary precision, neither of which has any justification. The data certify an effective parameter; the vacuum-energy identification is an interpretation laid on top, and forcing a match between two scales the physics does not connect is a manufactured isomorphism, not an explanation. The unresolved tension between early- and late-universe expansion measurements compounds the gap. *Kernel (vacuum-energy identification)*: $[X]$, $\det(R) = 0$, $\kappa(G) = 1.8 \times 10^{16}$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, the sealed component carrying three independent warrants at mutual orthogonality. Gates G1–G12 clear on that component. *Kernel* $[LOCK]$ $\det(R) = 0.9907$, $\lambda^2 = \det(R)$ at 1e-15. Split: the open or broken remainder reads separately at $\det(R) = 0$. $\vdash$ the component spans $\Rightarrow [\nu]$ on the sealed claim.

Verdict: $[\nu]$ Sealed at high grade on the accelerating expansion and the constant as an effective fit; $[X]$ Broken on the identification of the constant with vacuum energy, a forced match across a 10^{120} discrepancy. The input that would change this is a derivation connecting the vacuum-energy scale to the observed value.

Volume I · Supreme Tier

The hardest cases in the set: foundational metaphysics, the limits of physical theory, and three claims the framework itself advances, which the method holds to its own discipline and seals only on warrant independent of the framework's assertion. Three

split into a sealed core and an open remainder, two tighten because the deciding register is first-person experience the method does not admit, and one question of abstract existence returns the signature of a genuinely undecidable metaphysics.

28 · Mathematical Platonism

Logic and Foundations · Supreme

Proposition. Mathematical structures exist independently of the minds that discover them.

The proposition bundles two claims that certify differently, and the reflective register separates them. The first is that mathematical truths are necessary: two plus two makes four in every possible world, and a truth that could not have been otherwise is not contingent on any mind that thinks it. That necessity seals. *Kernel (necessity):* [LOCK], $\det(R) = 0.9996$, $\lambda = -0.9998$, $\kappa(G) = 1.04$, $N = 24$.

The second is the ontological claim, that abstract objects therefore exist independently, as entities in their own right, and here the method declines to follow the inference. The argument from necessity to independent existence is one available account, but not the only one: a structuralist or nominalist reading takes the same necessity and the same cross-cultural convergence and explains them as features of logical relations rather than of free-standing objects. The reflective register tests this directly by asking whether the claim and its denial are both field-permitted, and they are. Platonism locks on the necessity and the convergence, and its nominalist rival locks on the same evidence, the two-way permission that is the signature of a question the available evidence cannot decide. It is not a sealed independence result, because no formal proof establishes the joint consistency that an independence result would require; it is an open metaphysical question, and the report of mathematicians experiencing discovery rather than invention is not admissible as warrant. *Kernel (independent existence):* [?], *dimensional shortfall*, $N = 6$, $k = 3$, *the claim and its denial both field-permitted*.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9996$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed on the necessity of mathematical truth, theorem-grade; [?] Under-determined on the independent existence of abstract objects. Platonism and its structuralist and nominalist rivals fit the necessity and the convergence equally, and the deciding input would be a consideration that breaks that symmetry, which none of the standard arguments supplies.

29 · The Hard Problem of Consciousness

Mind and Consciousness · Supreme

Proposition. The qualitative character of conscious experience cannot be fully reduced to physical processes.

This case pairs with the earlier question of whether consciousness is solely brain activity, and the method holds the same line in both directions. The formal axis is genuine: a complete physical description of a brain state, every neuron and field value specified, does not logically entail that there is something it is like to be in that state, and the conceptual coherence of a physical duplicate without experience shows the entailment is unavailable. The explanatory gap is real, not a gap in current knowledge but a gap in the inference itself. The empirical axis confirms reliable neural correlates and confirms, just as firmly, that no instrument has measured a quality of experience: every measurement returns a physical quantity, and the gap between physical and phenomenal is conceptual rather than technological.

What the method will not do is the move the proposition needs. The claim is that qualia are irreducible, that the gap is unbridgeable in principle, and the only axis that could carry that strong claim is the first-person reality of experience, which the method routes out as warrant. With that register out of band, the formal gap establishes that reduction is not currently available and not formally necessary; it does not establish that reduction is impossible. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] Under-determined, the explanatory gap real and the aperture located, the strong claim of in-principle irreducibility unestablished. The method refuses both the eliminativist denial of the gap and the strong dualist claim of irreducibility, because the warrant for either requires the inadmissible first-person register, and the deciding input would be a bridging principle from physical description to phenomenal character, which does not exist.

30 · Systemic Racism as a Structural Property

Social Science · Supreme

Proposition. Racial outcome disparities can be produced and reproduced by the structural properties of institutions, independent of ongoing individual prejudice.

The political charge on this question drives heavy framing from opposite directions, and both are removed before audit: the framing that no such effect exists and the framing that it explains everything are each consensus pressures the method zeroes, leaving the precise structural claim to stand on its own warrant. The formal axis supplies the mechanism: institutional rules embed initial conditions and persist beyond the individuals and intentions that created them, so a process built with discriminatory access rules keeps producing disparate outcomes through neutral application long after the intent is gone, requiring no continuing

prejudice. The empirical axis isolates exactly that mechanism through audit studies whose design eliminates conscious intent: identical applications differing only in a racially coded name produce large callback gaps, replicated across resumes, housing, credit, and policing by independent teams, and in the strongest designs neither applicants nor evaluators know they are being studied, which removes conscious discrimination as the explanation. The registrational record adds the documented pattern of disparate outcomes arising from standardized procedures. *Kernel*: [LOCK], $\det(R) = 0.9798$, $\lambda = -0.9898$, $\kappa(G) = 1.32$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9798$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, corroboration-grade across independent audit programs, on the precisely-defined structural-mechanism claim. The seal is on the mechanism and is scope-fenced: that structural processes can produce disparity independent of individual prejudice is demonstrated, while how much of any particular observed disparity this mechanism explains is a separate quantitative question the seal does not settle, and the broad term in its contested political usages is not what seals.

31 · Dollar Hegemony Maintained by Military Power

Geopolitics · Supreme

Proposition. The dollar's reserve-currency status is maintained in significant part through US military power.

Treated in mainstream commentary as a conspiracy theory and in some quarters as established fact, the claim is stripped of both the reflexive dismissal and the overstatement, and it splits. The architecture and the incentive seal: the arrangement tying major oil sales to dollar settlement in exchange for security guarantees is a documented formal agreement that structured the oil market for half a century, and reserve-currency status confers a real and named advantage, the ability to run persistent deficits cheaply, which creates a genuine structural incentive to defend that status. *Kernel (petrodollar architecture)*: [LOCK], $\det(R) = 0.9909$, $\lambda = -0.9954$, $\kappa(G) = 1.21$, $N = 24$.

The causal claim is the contested part. The correlation is present, with several states moving to price oil outside the dollar shortly before military intervention, but each of those states also carried other intervention rationales, and the currency variable cannot be separated from those confounders across so few cases. The supporting first-person testimony is the weakest line and does not isolate the mechanism either. *Kernel (military enforcement)*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9909$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed on the petrodollar architecture and the structural incentive, documented and theorem-grade in economics; [?] Under-determined on military enforcement as the maintaining mechanism. The correlation is real and the incentive is real, but the causal claim cannot be isolated from confounded cases, and the deciding input would be a comparison that separates the currency variable from the other rationales.

32 · Time as the Root Dimension

Physics · Supreme

Proposition. Time is not a fourth spatial dimension but the generative ground from which spatial dimensions emerge.

This is one of the framework's own foundational claims, so the method applies its own discipline without exemption: it seals only on warrant independent of the framework's assertion, and types that warrant honestly. The standard view that time is a coordinate equal with space is a consensus, which carries no evidential weight either way, so the claim is assessed on structure. The formal axis is a generation argument: extension into any dimension requires traversing a generating parameter, so a line presupposes a parameter of change, a plane a second, a volume a third, which makes the generating parameter prior to the spatial structure it produces. The argument has force, though the identification of that generating parameter with physical time is an interpretive step the method marks as such. The decisive support is external and genuinely independent: two non-perturbative quantum gravity programs, built on entirely different mathematics by different groups, both make causal-temporal order fundamental and spatial geometry derivative, one recovering a 3+1-dimensional universe only when temporal ordering is imposed and collapsing without it, the other recovering the full spatial metric from a discrete causal order. The first-person registration of time's flow is routed out. *Kernel*: [LOCK], $\det(R) = 0.8207$, $\lambda = -0.9059$, $\kappa(G) = 2.31$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.8207$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural and conditional grade, on the formal generation argument and two independent quantum gravity programs that make temporal order prior to spatial geometry. The grade is conditional rather than theorem-grade because the two programs are active and not settled and the identification with physical time is interpretive; the moderate volume reflects that the two corroborating programs share a conclusion, and the seal rests on external corroboration, not on the framework's say-so.

33 · The Anthropic Fine-Tuning of Constants

Cosmology · Supreme

Proposition. The physical constants are fine-tuned for the existence of complex structure and life.

Exploited by design arguments and multiverse arguments alike, the claim is stripped of both extrapolations and the core sensitivity assessed on its own terms. That sensitivity is formally real and precise: small fractional changes in the strong force leave no stable nuclei or no stellar fuel, a slightly larger cosmological constant prevents structure from forming, and a different mass ratio prevents stable chemistry, each a direct consequence of the equations evaluated at other parameter values. The sensitivity seals. *Kernel (physical sensitivity):* [LOCK], $\det(R) = 0.9957$, $\lambda = -0.9978$, $\kappa(G) = 1.14$, $N = 24$.

The interpretation that the universe is therefore fine-tuned, in the sense that these values are improbable and call for explanation, is a different claim, and the empirical axis cannot support it, because we have exactly one universe. To call a value improbable requires a probability distribution over possible universes, and we have no independent sample of even a second universe, so every measurement is a measurement of this one data point, and the observation is not independent of the question, since only a universe permitting observers can be observed at all. *Kernel (fine-tuning interpretation):* [?], *dimensional shortfall*, $N = 3$, *a single data point*.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9957$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at theorem grade on the physical sensitivity of the constants, formal and exact; [?] Under-determined on the fine-tuning interpretation. The sensitivity is a fact about the equations; the improbability that would make it tuning cannot be assessed from a single data point under a selection effect, and the deciding input would be an independent measure over possible universes, unavailable in principle.

34 · Existence Minima

Logic and Foundations · Supreme

Proposition. Absolute nothingness, a state of zero structure and zero presence, is impossible.

The formal axis carries the claim by a self-undermining argument of the same constitutive kind as non-contradiction. For absolute nothing to be a coherent concept it must be distinguishable from something, which gives it structure, and a concept with structure is not nothing; the proposition "nothing exists" itself has a logical form and truth conditions, which are not nothing. The concept dissolves on contact with analysis, and it is its denial, that absolute nothing is a genuine possibility, that self-refutes, which

places the claim as a constitutive condition rather than a contingent discovery. The empirical axis corroborates a physical version of the point: the most carefully prepared vacuum is never empty, its non-zero energy directly measured, and no measurement has returned true zero for every quantity at once, though this addresses the physical vacuum rather than metaphysical absolute nothing and so corroborates a weaker claim than the formal argument secures. The registrational record adds that no measurement and no act of conception can reach absolute nothing, since both require something doing the reaching. *Kernel:* [LOCK], $\det(R) = 0.9841$, $\lambda = -0.9920$, $\kappa(G) = 1.29$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9841$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade and constitutive on the self-refutation of absolute nothingness, with the vacuum evidence corroborating the physical sub-claim. The impossibility is established at the level of what can be coherently conceived, the same footing as the bedrock laws of logic, and the seal rests on the formal argument rather than on the framework that frames it.

35 · Near-Death Experiences as Evidence of Survival

Mind and Consciousness · Supreme

Proposition. Near-death experiences provide empirical evidence that consciousness survives bodily death.

Contaminated from both sides, by materialist incentive to explain the phenomena away and survivalist incentive to validate them, the literature is stripped and the survival claim assessed on warrant. The formal axis runs against it: every established model of consciousness requires a physical substrate, and consciousness obtaining in the absence of a functioning brain would require an entirely new framework that has not been developed, so the survival interpretation extends beyond any established theoretical basis. The empirical axis is genuinely mixed. There are real anomalies, reported accurate perceptions during cardiac arrest and a cross-cultural consistency in core features that current models do not fully explain, but the one controlled prospective test of veridical perception under monitored conditions returned a result far too sparse for confirmation. The anomalous data is real and the survival hypothesis is not supported by it, and the vivid first-person testimony that is the strongest part of the record is routed out as warrant. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the anomalous phenomena genuine and not fully explained, the survival interpretation

unsupported. The formal axis offers no mechanism, the controlled empirical test was inconclusive, and the phenomenology is inadmissible; the deciding input would be an adequately powered controlled test of veridical perception that the inconclusive attempt did not achieve.

36 · The Universe Emerged from a Pre-Geometric Ground State

Cosmology · Supreme

Proposition. The observable spacetime universe emerged via a phase transition from a pre-geometric ground state.

Another of the framework's own claims, and the same discipline applies. It is the constructive counterpart to the singularity question: where an infinite-density singularity is the marker of a theory breaking down, a phase transition from a structureless ground state is a well-defined physical mechanism. The formal axis combines the generation argument, which makes a zero-dimensional state with proto-temporal potential the natural precursor of spatial extension, with spontaneous symmetry breaking, the standard mechanism by which a symmetric state settles into an asymmetric one without any decision, the machinery of established phase transitions. The empirical axis is three anchors, one of them direct: a quantum gravity simulation produces a transition between a disordered pre-geometric phase and an ordered 3+1-dimensional phase, the inflationary epoch is driven by a structureless false-vacuum state of pure potential that the microwave background confirms, and the vacuum is measured to carry real non-zero energy. The first-person registration of temporal flow is routed out. The support for the mechanism is genuine, but indirect for the specific claim that our universe originated this way, since the pre-geometric state is no more directly observable than the origin it replaces. *Kernel:* [LOCK], $\det(R) = 0.9185$, $\lambda = -0.9584$, $\kappa(G) = 1.75$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9185$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural and conditional grade, on the spontaneous-symmetry-breaking mechanism, a direct simulation of a pre-geometric-to-geometric transition, and the confirmed non-empty vacuum, as a coherent and better-motivated account of the early universe than an undefined singularity. The grade is conditional because the origin itself is not directly observable and the corroboration is indirect; the seal rests on the external physics, not on the framework that motivates it.

Volume II · Demonstrative Tier

Nine propositions in a second domain rotation. The clearer cases of the set: theorem-grade seals in physics and logic, corroboration seals in psychology and social science, and two breaks where the

warrant either inverts under symmetric comparison or dissolves into a common cause.

37 · The Correspondence Theory of Truth

Logic and Foundations · Demonstrative

Proposition. A proposition is true if and only if it corresponds to the way the world actually is.

The proposition carries a formal core and a substantive thesis that certify at different grades. The formal core is the disquotation schema, that "snow is white" is true exactly when snow is white, which model theory makes precise as a specifiable mapping from a sentence to the world-state it describes; that schema is a theorem of standard semantics and seals. The substantive thesis, that truth consists in this correspondence rather than in coherence or usefulness, is supported at structural grade: every functioning measurement practice in science is a correspondence test in action, calibration and replication and standardization all checking whether a proposition matches an independent world-state, and the registration record of belief-correction tracks the same. The one rival the formal core does not exclude is deflationism, which accepts the disquotation schema while declining to read a robust relation into it, so the seal is on correspondence as the operative notion of truth, not a refutation of every alternative. The first-person experience of discovering an error is routed out as warrant. *Kernel:* [LOCK], $\det(R) = 0.9908$, $\lambda = -0.9954$, $\kappa(G) = 1.21$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9908$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the disquotation schema and structural on the substantive correspondence thesis. Deflationism shares the formal core; the coherentist and pragmatist rivals, which the practice of measurement runs against, do not.

38 · Attachment Theory

Psychology · Demonstrative

Proposition. The quality of the early caregiver bond systematically shapes lifelong relational patterns.

The claim seals on independent empirical convergence. The theoretical mechanism is the internal working model, a cognitive template of caregiver availability built from repeated early experience that then functions as a default expectation in later relationships; this is a model rather than a formal necessity, so the load is carried by the evidence, which is unusually strong. Four independent programs converge: a standardized laboratory assessment replicated across thirty-two countries with consistent classifications, a longitudinal study following children to adulthood in which early classification predicted adult

relationship quality and parenting at twenty-year follow-up, and an independent narrative-based adult instrument that concords with the childhood measure at high rate. *Kernel*: [LOCK], $\det(R) = 0.9607$, $\lambda = -0.9801$, $\kappa(G) = 1.48$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9607$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, corroboration-grade across four independent programs spanning cross-cultural, longitudinal, and adult-narrative measures. The seal is on a real and replicated influence and is scope-fenced: early attachment shapes later patterns, it does not deterministically fix them, and the prediction is empirical rather than formally entailed.

39 · The Broken Windows Theory of Crime

Social Science · Demonstrative

Proposition. Visible signs of disorder cause serious crime by signaling that norms do not hold.

This is the same shape as the earlier poverty-to-crime case: a real correlation with a common cause and an unsupported causal pathway. The formal model is coherent as a signaling story but conflates two distinct claims, that disorder correlates with crime and that disorder produces it, and it never specifies how observing a broken window leads a person to commit assault. The empirical axis settles it against the causal reading. The crime decline that accompanied disorder-focused policing in one city occurred equally in cities that never adopted the policy, which points to common confounders, and the decisive controlled study found that once concentrated disadvantage is held constant, disorder does not independently predict crime. Disorder and serious crime are both products of concentrated disadvantage; the link between them is that shared cause, not a pathway from one to the other. *Kernel*: [X], $\det(R) = 0$, $\kappa(G) = 1.8 \times 10^{16}$, $N = 24$.

Triaxial. G4 CAUSAL fails: disorder-crime link confounded, no clean mechanism. Two axes fall into one, the warrant volume collapses. *Kernel* [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, a common-cause confound with the causal pathway unspecified and refuted under control. A weaker claim holds, that disorder and crime co-occur as markers of the same underlying disadvantage; the claim that disorder causes crime does not.

40 · Gravitational Waves Exist and Carry Energy

Cosmology · Demonstrative

Proposition. Disturbances in spacetime curvature propagate as waves at the speed of light and carry energy.

This seals at theorem grade on a formal derivation, an indirect confirmation that preceded detection by four decades, and three

independent direct detections. The field equations of general relativity, taken in the weak-field limit, yield a transverse wave propagating at the speed of light and carrying energy by a derivable radiation formula, and the orbital decay of a binary pulsar matched that predicted energy loss to within a fraction of a percent long before any wave was caught directly. The direct confirmation is three independent interferometer programs on different continents, with different engineering, locations, and funding, recording waveforms that match the numerical-relativity predictions to measurement precision. The registration record is the catalog of detected events. *Kernel*: [LOCK], $\det(R) = 0.9855$, $\lambda = -0.9927$, $\kappa(G) = 1.27$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9855$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the general-relativistic derivation and corroboration-grade across an indirect binary-pulsar test and three independent direct detectors.

41 · Wave-Particle Duality

Physics · Demonstrative

Proposition. Quantum entities exhibit both wave-like and particle-like behavior.

The claim seals at theorem grade, with one clarification the formalism supplies. Quantum mechanics represents an entity as a single mathematical object, a state evolving by a wave equation whose squared amplitude gives the probability of a definite measured outcome, so the wave character and the particle character are not two contradictory natures but complementary aspects of one object read in different measurement contexts. The empirical axis is three independent lineages using incompatible instruments, single particles building an interference pattern one detection at a time, neutron interferometry confirming the wave nature of a massive particle, and interference observed for large molecules, each showing the same dual behavior. *Kernel*: [LOCK], $\det(R) = 0.9982$, $\lambda = -0.9991$, $\kappa(G) = 1.09$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9982$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the quantum formalism and corroboration-grade across three independent experimental lineages. The complementarity reading dissolves the apparent contradiction: one object, two contexts, not two conflicting natures.

42 · Gödel's Incompleteness Theorems

Logic and Foundations · Demonstrative

Proposition. Any consistent formal system expressing basic arithmetic contains true statements it cannot prove.

This is a pure mathematical theorem and seals at the strongest grade the method issues. The construction encodes statements about provability as arithmetic and builds a sentence asserting its own unprovability, which is true and unprovable if the system is consistent, and the second theorem shows that a consistent system cannot prove its own consistency. The proof has been verified by several independent proof-checking systems built on entirely different formal architectures, and the prediction that concrete undecidable arithmetic statements exist was later confirmed by independently discovered examples shown to be true but unprovable in elementary arithmetic. *Kernel:* [LOCK], $\det(R) = 0.9828$, $\lambda = -0.9914$, $\kappa(G) = 1.30$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9828$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, with independent machine verification and concrete confirming instances. The method acknowledges the limitative theorems at full strength here, consistent with how it treats formal-system ceilings throughout rather than claiming to escape them.

43 · The Pygmalion Effect

Psychology · Demonstrative

Proposition. The expectations authority figures hold about individuals systematically influence their performance.

The claim seals on a specified mechanism and independent replication, at a fenced effect size. The mechanism is the self-fulfilling prophecy operating through observable channels: an evaluator who expects more provides more feedback, harder material, and warmer engagement, and the recipient receives differential inputs and performs accordingly, a causal chain with identifiable links rather than any unmediated transmission. The original demonstration was flagged for researcher-expectancy contamination, so the seal rests on what came after: a meta-analysis across dozens of studies confirming a moderate effect, larger for disadvantaged populations, and independent confirmation in military and management settings with different populations and measures. *Kernel:* [LOCK], $\det(R) = 0.9896$, $\lambda = -0.9948$, $\kappa(G) = 1.22$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9896$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, corroboration-grade across independent programs after the contaminated original is set aside, on a

real and moderate expectancy effect. Scope-fenced: a moderate effect, not a deterministic shaping.

44 · Nuclear Energy Is Categorically More Dangerous Than Fossil Fuels

Geopolitics · Demonstrative

Proposition. Nuclear energy poses greater risk to human health and safety than fossil fuels.

The warrant for this claim is fear-framing and the extrapolation of a few salient accidents to a categorical judgment, and the symmetric comparison the formal axis requires inverts it. Risk must be measured as deaths per unit of energy generated, not as a count of dramatic events, and on that measure three independent mortality analyses agree that nuclear is orders of magnitude safer than fossil fuels, on the order of hundredths of a death per terawatt-hour against roughly twenty-five for coal, even when the worst accident is included, while fossil-fuel air pollution kills millions every year. The claim asserts the reverse of what the independent record shows. *Kernel:* [X], $\det(R) = 0$, $\kappa(G) = 3.7 \times 10^{16}$, $N = 24$.

Triaxial. G8 CSCG fails: inverted by deaths-per-unit-energy data. Two axes fall into one, the warrant volume collapses. Kernel [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, empirical inversion. The supporting warrant is narrative fear-framing and single-event extrapolation; the symmetric per-energy comparison from three independent analyses shows nuclear is far safer, the opposite of the claim.

45 · The Flynn Effect

Social Science · Demonstrative

Proposition. Average IQ scores rose substantially across populations over the twentieth century.

The claim seals on multi-national independent replication and carries a formal corollary that follows deductively. The empirical axis is independent national datasets, military and conscription and testing archives across many countries collected for purposes unrelated to the research, culminating in a meta-analysis of hundreds of studies spanning a century and thirty-one countries, all confirming a rise of roughly three points per decade. The formal corollary is a clean deduction: gene-pool change requires thousands of years, the observed rise is far faster, so the scores cannot be tracking a fixed genetic capacity, which is direct evidence against the strong hereditarian reading and consistent with the earlier finding that broke the claim of IQ as a fixed natural entity. *Kernel:* [LOCK], $\det(R) = 0.9757$, $\lambda = -0.9878$, $\kappa(G) = 1.37$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9757$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed, corroboration-grade across multi-national archival datasets, on the empirical rise, with the deductive corollary that the rise is too fast for genetic change and so reflects substantial environmental influence.

Volume II · Analytical Tier

Nine harder propositions. Three genuine formal results about the structure of knowledge seal at theorem grade, two replication-crisis casualties and two refuted claims break, and two non-isolable causal claims go under-determined.

46 · Hume's Problem of Induction

Logic and Foundations · Analytical

Proposition. Inferring universal laws from finite observation has no rational justification that escapes circularity.

The formal axis carries it as an exhaustive dilemma. A justification for induction must be deductive or inductive: deduction cannot derive a contingent universal from any finite set of observations, which is provable, and an inductive defense, that induction has worked so it will keep working, uses induction to license itself, which is circular. No third option exists. The empirical record of philosophy of science confirms the gap rather than closing it: falsificationism removes induction only by giving up positive prediction, which all science uses, and Bayesian confirmation relocates the problem into the justification of priors, itself arbitrary or inductive. This is a genuine structural result, the companion to the case that broke falsifiability as a sole criterion. *Kernel:* $[LOCK]$, $\det(R) = 0.9898$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9898$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed, theorem-grade on the exhaustive dilemma. Induction lacks a non-circular justification, a fact about the structure of justification rather than a defect to be fixed.

47 · Ego Depletion

Psychology · Analytical

Proposition. Willpower functions as a finite metabolic resource depleted by acts of self-control.

The strength model makes precise predictions about sequential tasks and glucose restoration, and the original literature ran to hundreds of confirming studies. The empirical axis breaks the claim, because the decisive test is the rigorous one. The largest pre-registered multi-lab replication, twenty-three labs and over two thousand participants, found a near-zero effect with a confidence interval crossing zero, a second coordinated

replication returned null, and the glucose mechanism was independently refuted. *Kernel:* $[X]$, $\det(R) = 0$, $N = 24$.

Triaxial. G8 CSCG fails: the effect fails large pre-registered replication. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, failure under pre-registered replication. The accumulated original literature did not survive coordinated direct replication at the claimed magnitude, and the proposed metabolic mechanism was refuted.

48 · Nuclear Deterrence Prevents Great Power War

Geopolitics · Analytical

Proposition. Mutual nuclear possession is the primary cause of the absence of great-power war since 1945.

The deterrence model is a coherent equilibrium: if each side can assure the other's destruction, neither rationally strikes first, and the Long Peace is consistent with it. The empirical axis cannot isolate it as the primary cause, because the same period brought decolonization, deep economic interdependence, dense international institutions, and the spread of democracy, each an independent candidate, across a single unrepeatable run. The documented near-misses, several occasions on which escalation was averted by narrow individual decisions, also show deterrence operating closer to failure than its theory allows. *Kernel:* $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, the causal claim not isolable from co-occurring candidate causes across a single historical run. The deterrence effect is real in the model; its primacy among the causes of the Long Peace cannot be separated out.

49 · The Measurement Problem

Physics · Analytical

Proposition. Standard quantum mechanics lacks a complete description of measurement.

The formal axis establishes it directly. The Schrödinger equation is linear and deterministic, so a measuring apparatus treated as a quantum system entangles with what it measures and the whole evolves into a superposition of outcomes, while the Born rule asserts that exactly one outcome is observed with a given probability. The two rules are inconsistent as stated, and that inconsistency is the problem. The empirical register is the ninety-year failure of the physics community to converge on an interpretation: every interpretation exists precisely to address this gap, which is itself the evidence that the gap is real and unresolved. *Kernel:* $[LOCK]$, $\det(R) = 0.9995$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9995$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\epsilon]$.

Verdict: $[\epsilon]$ Sealed, theorem-grade on the formal inconsistency of unitary evolution with definite-outcome measurement. The gap is genuine; the seal is on its existence, not on any interpretation that proposes to fill it.

50 · Cosmic Inflation

Cosmology · Analytical

Proposition. The early universe underwent a brief epoch of exponential expansion.

Inflation seals at conditional grade. The formal axis is a real achievement: a brief epoch of exponential expansion resolves three independent puzzles of standard cosmology, the uniformity of the microwave background across regions that could not have been in contact, the observed spatial flatness, and the absence of predicted relics, none of which standard cosmology explains without fine-tuning. The empirical axis supports it without nailing it down. The near-scale-invariant spectrum of primordial fluctuations that inflation predicts was confirmed by an independent satellite, and the seeding of large-scale structure is quantitatively consistent, but these observations are consistent with inflation rather than unique to it, and the definitive signature, primordial gravitational waves, remains unconfirmed after one prominent claim was withdrawn as dust. *Kernel:* [LOCK], $\det(R) = 0.8683$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.8683$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\epsilon]$.

Verdict: $[\epsilon]$ Sealed at conditional grade, on three independent solved problems and a confirmed spectral prediction. The grade is conditional because the supporting observations are consistent but not unique and the decisive primordial-wave signature has not been found.

51 · "Immigration Causes Crime"

Social Science · Analytical

Proposition. Immigration into a host country increases its overall crime rate.

No formal criminological theory predicts that immigration as such raises crime; the theories that link the two route through specific conditions, economic marginalization or absent social capital, that may co-occur with immigration but are not immigration itself. The empirical axis inverts the claim. Independent meta-analyses across dozens of studies find a majority of negative or null associations and no consistent positive one, with the small

positive effects that appear confined to property crime and attributable to poverty conditions. The warrant for the proposition is fear-framing; the independent record runs the other way. *Kernel:* [X], $\det(R) = 0$, $N = 24$.

Triaxial. G8 CSCG fails: collides with the criminological record. Two axes fall into one, the warrant volume collapses. Kernel [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, contradicted by the meta-analytic record. The association is mostly negative or null, the warrant is narrative, and the claimed direction is not what the independent evidence shows.

52 · Occam's Razor as a Universal Principle

Logic and Foundations · Analytical

Proposition. The simplest explanation is always to be preferred over more complex alternatives.

Parsimony has a genuine formal justification: simpler hypotheses carry higher prior probability under a complexity prior, and Bayesian model comparison penalizes complexity through the evidence integral. That licenses the razor as a prior, the weaker claim, which survives. It does not license the word always, which converts a probabilistic preference into an absolute rule, and the history of science refutes that conversion directly. The correct theory has repeatedly been the more complex one, special relativity adding a postulate, general relativity replacing a force with curved spacetime, the standard model elaborating structure, each buying greater empirical reach at the cost of formal simplicity. *Kernel:* [X], $\det(R) = 0$, $N = 24$.

Triaxial. G9 CSEG fails: the universal claim exceeds a defeasible heuristic. Two axes fall into one, the warrant volume collapses. Kernel [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, overreach from a probabilistic prior to an absolute rule. Parsimony as a prior survives; the claim that the simplest explanation is always to be preferred is refuted by the cases where complexity was correct.

53 · The Oedipus Complex as a Universal Stage

Psychology · Analytical

Proposition. Every child universally passes through an Oedipal developmental stage.

The claim requires a universal sequence with specific content, a fixed resolution mechanism, and a permanent structural legacy, all holding across cultures. The empirical axis refutes the universality. In a matrilineal society where the father is not the authority figure, the anxieties the theory attributes to rivalry with the father attach instead to the maternal uncle, which shows the pattern tracks social structure rather than a universal biological stage, and cross-cultural psychopathology research finds no universal Oedipal predictor of adult outcomes. *Kernel:* [X], $\det(R) = 0$, $N = 24$.

Triaxial. G8 CSCG fails: cross-cultural counterevidence interferes. Two axes fall into one, the warrant volume collapses. Kernel $[X] \det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, cross-cultural counterevidence refuting universality. The configuration is culturally contingent, not a universal developmental stage.

54 · Social Capital Theory

Social Science · Analytical

Proposition. Social trust and civic network density causally produce economic prosperity.

The mechanism is coherent and multiple: civic networks build norms of reciprocity, repeated interaction lowers transaction costs, and generalized trust reduces the need for costly enforcement, all reducing friction in coordination. The cross-national correlation between trust and prosperity is robust, and the regional and historical studies reinforce it. The causal direction is the open question. Prosperity may build trust as readily as trust builds prosperity, the two plausibly co-evolve, and the studies do not separate the directions, so the correlation is established while the causal arrow the proposition asserts is not. *Kernel:* $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel $[?] \det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, robust correlation with the causal direction unresolved. Trust and prosperity move together; that trust causally produces prosperity, rather than the reverse or a mutual loop, is not established.

Volume II · Superior Tier

Nine cases where the contested part is institutional, interpretive, or beyond measurement. One two-claim split seals a documented core and fences the stronger version, two formal results seal, two replication or inference failures break, and four claims about the inaccessible or the non-isolable go under-determined.

55 · The CIA-Contra-Cocaine Connection

Geopolitics · Superior

Proposition. US intelligence officials had knowledge of, and in some cases facilitated, cocaine trafficking by the Nicaraguan Contras.

The proposition splits along the line between knowledge and direction. That officials had knowledge of, and in places obstructed investigation into, trafficking by the Contras is documented in two independent official sources: a US Senate committee report concluded that the networks were involved in trafficking and that government agencies took actions obstructing drug investigations, and the agency's own later inspector-general

report admitted that its officers were aware of the trafficking and did not act. That much seals, and on the official record, not on the journalism that first raised it. *Kernel (knowledge and obstruction):* $[LOCK]$, $\det(R) = 0.9881$, $N = 24$.

The stronger claim, that the agency actively directed or orchestrated the trafficking as an instrument of policy, is not established by those reports and goes beyond what the documents show. *Kernel (active orchestration):* $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9881$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at corroboration grade on knowledge and obstruction, documented in two independent official reports; [?] Under-determined on active orchestration. The deciding input for the stronger claim would be documentary evidence of direction rather than awareness.

56 · The Many-Worlds Interpretation

Physics · Superior

Proposition. Every quantum measurement branches the universe, with all outcomes occurring in separate worlds.

Applying the Schrödinger equation universally, with no collapse, yields all measurement outcomes as separate branches of one wavefunction, and the construction is internally consistent. It is also empirically undecidable by construction. Every test that confirms quantum mechanics confirms Many-Worlds and its rivals equally, and no possible measurement distinguishes it from collapse interpretations, so the empirical axis can never be populated to favor it. A formal problem compounds this, the difficulty of recovering the usual probabilities when every outcome occurs. *Kernel:* $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel $[?] \det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, empirically undecidable by construction with an unresolved probability problem. It is a consistent interpretation that no experiment could select over its rivals.

57 · The Stanford Prison Experiment

Psychology · Superior

Proposition. Assigned social roles fully determine behavior, overriding individual character.

The strong situationist reading requires the role to be a sufficient cause, individual character becoming negligible under situational force. The empirical axis refutes it. Reanalysis of the original

found that specific guards, not all, drove the abuse, that their behavior was actively encouraged by the experimenters rather than arising from the role, and that several guards refused to abuse at all, each fact contradicting full determination, while a pre-registered independent replication found the opposite outcome. *Kernel*: $[X]$, $\det(R) = 0$, $N = 24$.

Triaxial. G8 CSCG fails: demand characteristics collapse the demonstration. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, contradicted by reanalysis and a failed pre-registered replication. A weaker claim, that situations influence behavior, holds; the claim that roles fully determine it does not.

58 · The Fermi Paradox as Evidence Life Is Rare

Cosmology · Superior

Proposition. The absence of detected signals is significant evidence that intelligent life is absent or rare.

The argument rests on a formal product of estimated rates whose terms are not independently estimable, most of them guesses, so the apparent paradox is not well-posed. The empirical claim is the breaking point. The silence is real, but the systematically searched volume corresponds to roughly a thousand nearby stars against some hundred billion trillion in the universe, a sampled fraction so small that non-detection carries essentially no weight about how common life is. An absence of signal across that fraction is not significant evidence of absence. *Kernel*: $[X]$, $\det(R) = 0$, $N = 24$.

Triaxial. G9 CSEG fails: inference from silence exceeds its support. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, an unwarranted evidential inference from a vanishingly small search fraction. The underlying question of how rare life is remains genuinely open; the claim that the silence is significant evidence does not hold.

59 · NATO Expansion as the Primary Cause of the 2022 Invasion

Geopolitics · Superior

Proposition. NATO's eastward expansion is the primary cause of Russia's 2022 invasion of Ukraine.

The realist model is coherent, expansion raising threat perception and prompting a military response, and the documentary record shows genuine and longstanding Russian objection to expansion. Primacy is what cannot be established. Russia did not invade during the earlier rounds of expansion, which complicates a monocausal reading, the 2022 decision is entangled with domestic and ideological factors and the specific context of that moment, and the single case offers no way to isolate the weight of the

expansion variable against the others. *Kernel*: $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, expansion a real factor whose status as the primary cause cannot be isolated from co-occurring factors and the timing of prior non-responses. The method names a contributing cause and declines to rank it first on this evidence.

60 · The Münchhausen Trilemma

Logic and Foundations · Superior

Proposition. Every attempt at complete justification meets one of three equally unsatisfactory ends.

The formal axis carries it as an exhaustion argument. Any justification needs a further justification, which can only continue without end, double back on what it was meant to support, or stop at something posited without further ground, an infinite regress, a circle, or an axiom. The trichotomy is exhaustive; no fourth structure exists. The empirical record is that every major epistemological tradition terminates in one of the three horns, rationalism at axioms, empiricism at circularity or regress, coherentism in the circle, foundationalism at the axiom. *Kernel*: $[LOCK]$, $\det(R) = 0.9984$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9984$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the exhaustive trichotomy. Complete justification is structurally unavailable, the companion result to the problem of induction.

61 · Loop Quantum Gravity

Physics · Superior

Proposition. Spacetime is fundamentally discrete at the Planck scale.

Quantizing general relativity in the relevant variables yields quantum states of geometry whose area and volume operators have discrete spectra, so a minimal spatial scale is a formal consequence of the construction rather than an assumption. The construction is genuine; the claim that spacetime is in fact discrete at that scale is what the empirical axis cannot reach. The scale lies some twenty orders of magnitude below accelerator energies, and the available indirect probe, the arrival timing of high-energy photons across cosmological distances, found no predicted effect, which constrains the cruder versions of the theory without confirming any. *Kernel*: $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, a formal consequence of an unconfirmed framework with the relevant scale beyond measurement.** The discreteness follows from the theory; whether the theory describes the world is not settled, and the indirect tests constrain rather than confirm.

62 · The Multiverse

Cosmology · Superior

Proposition. Physical reality extends beyond the observable horizon and contains multiple universes.

Several independent frameworks generate other universes as a by-product, eternal inflation nucleating bubbles, the string landscape supplying vast numbers of vacua, the universal wavefunction branching. The claim that such universes exist is empirically inaccessible by definition: nothing beyond the cosmological horizon can send a signal here, so no instrument can reach the others, and the one proposed indirect signature, imprints of a collision between bubbles in the microwave background, was searched for and not found. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, empirically inaccessible by definition with null indirect searches.** The frameworks that imply a multiverse are themselves unconfirmed, and the claim sits beyond the reach of measurement.

63 · Universal Basic Income Improves Well-Being

Social Science · Superior

Proposition. A guaranteed unconditional income improves recipient well-being.

The well-being claim seals at conditional grade. Welfare economics frames competing effects, a possible reduction in labor supply against a security effect that lowers financial stress and may raise risk-taking, and the empirical axis on well-being is consistent across independent randomized pilots: an unconditional income raised mental health and life satisfaction in a Nordic trial, improved food security and psychological well-being in a large cash-transfer program in Kenya with no measured drop in labor supply, and improved outcomes in a US guaranteed-income pilot. *Kernel:* [LOCK], $\det(R) = 0.8993$, $N = 24$. The grade is conditional because every one of these is a time-limited pilot rather than a permanent universal program, so the macro-scale

effects, on labor markets, prices, and fiscal sustainability, are untested.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.8993$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at conditional grade on the well-being improvement, consistent across independent randomized pilots.** The improvement is real at pilot scale; the effects of a permanent, population-wide program are a separate and untested question.

Volume II · Supreme Tier

Nine cases closing Volume II. The dominant cause of warming and the building blocks of life seal on the physics and chemistry, two genuine philosophical results seal, the religious framing of a national conflict and the strong claim of central-bank independence break, and the simulation hypothesis, trauma primacy, and solipsism's conclusion resist the seal.

64 · Anthropogenic Climate Change

Geopolitics · Supreme

Proposition. Human greenhouse-gas emission is the dominant cause of the observed rise in global temperature.

The mechanism is derivable physics: greenhouse gases absorb outgoing long-wave radiation and re-emit it in all directions, so rising atmospheric concentration increases the opacity of the atmosphere to that radiation and traps more energy, the same physics used to design infrared instruments. The empirical axis is six independent streams: direct atmospheric concentration measurement, ice cores from multiple independent drilling teams spanning hundreds of thousands of years, and satellite temperature records produced by groups originally skeptical of warming who independently confirmed the trend, among others. The attribution to human emission as the dominant driver rests on the isotopic and spectral fingerprint, not on the bare correlation. *Kernel:* [LOCK], $\det(R) = 0.9994$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9994$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed, theorem-grade on the radiative mechanism and corroboration-grade across six independent evidence streams, including confirmation from originally skeptical groups.**

65 · The Gettier Problem

Logic and Foundations · Supreme

Proposition. Justified true belief is insufficient for knowledge.

The formal axis carries it by construction. A short paper produced cases in which a belief is both justified and true yet plainly not knowledge, because the justification connects to the truth only by luck, and the structure is reproducible at will. The empirical record of the field is sixty years of failed repair: every proposed fourth condition, no-false-lemmas, reliability, causal, safety, generates fresh counterexamples that defeat it, which confirms the gap is structural rather than a missing detail. *Kernel:* [LOCK], $\det(R) = 0.9482$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9482$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the counterexample and the sixty-year record of failed repairs. Justified true belief is demonstrably insufficient; what to add is the open question, not whether the gap is real.

66 · Psychedelic-Assisted Therapy

Psychology · Supreme

Proposition. Psilocybin- and MDMA-assisted therapy produce clinically significant, durable reductions in treatment-resistant depression and PTSD.

The claim seals at corroboration grade. The mechanism is grounded, a serotonergic action that loosens the over-rigid default-mode architecture of depression and opens a window of plasticity. The empirical axis is independent clinical programs including multi-site phase-three trials, with large response rates against control and a comparison favoring the psychedelic against a standard antidepressant. The fence is methodological: these therapies are hard to blind, since participants can tell whether they received an active dose, which inflates apparent effects, so the seal is on a real and replicated benefit whose exact magnitude carries that caveat. *Kernel:* [LOCK], $\det(R) = 0.9110$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9110$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, corroboration-grade across independent clinical programs including phase-three trials. Scope-fenced by the functional-unblinding problem, which qualifies the magnitude without erasing the effect.

67 · The Israel-Palestine Conflict as Primarily Religious

Geopolitics · Supreme

Proposition. The conflict is fundamentally driven by religious difference between Judaism and Islam.

The historical record runs against the religious framing. The founding movements on both sides were secular and nationalist, not religious, and the structure of the dispute is competing nationalisms over the same land. Independent historians writing from opposed national and political perspectives converge on origins in competing nationalism and land displacement rather than religious difference, the principal partition line was drawn demographically and politically, and the major wars were territorial and national in character. *Kernel:* [X], $\det(R) = 0$, $N = 24$.

Triaxial. G9 CSEG fails: the monocausal framing exceeds a multi-driver conflict. Two axes fall into one, the warrant volume collapses. Kernel [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, contradicted by convergent historical scholarship across perspectives. Religion is one dimension and has grown more salient over time; the claim that the conflict is primarily religious is not what the record shows.

68 · The Simulation Hypothesis

Cosmology · Supreme

Proposition. Physical reality is a computational simulation run by a more advanced civilization.

The trilemma is a valid exhaustive argument, that civilizations go extinct before computational maturity, or mature ones do not run ancestor simulations, or simulated minds vastly outnumber unsimulated ones and we are probably among them. It does not select a horn. The empirical axis cannot be populated: no test distinguishes a simulated universe from an unsimulated one, and the one proposed signature, lattice artifacts in the high-energy cosmic-ray spectrum, was not found, which constrains a narrow version without reaching the claim. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, empirically undecidable with the trilemma valid but non-selecting. No measurement distinguishes the simulated from the unsimulated, so the claim sits outside the method's reach.

69 · Solipsism

Logic and Foundations · Supreme

Proposition. Only one's own mind is certain to exist; all external reality is uncertain.

The proposition has a defensible kernel and an overreaching conclusion. The kernel is the cogito: the act of doubting confirms that thinking is occurring, so first-person thought is certain, and

that much seals. *Kernel (first-person certainty):* [LOCK], $\det(R) = 0.9916$, $N = 24$.

The solipsist conclusion, that therefore only one's own mind is certain and external reality is unknowable, is the overreach, and it is self-undermining: the private-language argument shows that formulating the position at all requires a public, shared language whose criteria presuppose other minds, and no one sustains the position in practice. The undermining is strong but not the clean performative contradiction that closes radical skepticism, so the conclusion does not break outright; it fails to seal. *Kernel (solipsist conclusion):* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9916$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the cogito's first-person certainty; [?] Under-determined on the solipsist conclusion, self-undermining and unsustained but not decisively refuted. The certain kernel does not extend to the claim that only that kernel is certain.

70 · The Federal Reserve's Operational Independence

Social Science · Supreme

Proposition. The Federal Reserve operates with genuine independence from short-term political pressure.

The Fed has genuine instrument independence, the freedom to choose its tools without per-decision approval, and that narrow claim holds. The proposition's stronger claim, genuine independence from short-term political pressure, is what the evidence undercuts. Its mandate is set by Congress and can be changed by Congress, and systematic research documents that policy shifts respond to political pressure when threats to its independence are credible, with partisan patterns in policy inconsistent with a purely technocratic driver and at least one documented episode of direct political influence on a chairman. *Kernel:* [X], $\det(R) = 0$, $N = 24$.

Triaxial. G9 CSEG fails: the strong-independence claim exceeds structural constraint. Two axes fall into one, the warrant volume collapses. Kernel [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken on the strong claim of genuine independence from political pressure, which the documented responsiveness contradicts. The surviving qualified claim is real instrument independence and more insulation than most agencies, not immunity.

71 · Trauma's Embodiment

Psychology · Supreme

Proposition. Physiological dysregulation of the autonomic nervous system is the primary architecture of trauma.

That trauma has a genuine physiological dimension is established: imaging shows that during trauma activation the brain shifts toward sensorimotor and emotional processing and away from language and executive regions, a real and measurable dysregulation. The proposition's strong claim, that physiological dysregulation is the primary architecture over cognitive and memory-based models, is what the method will not seal. The primacy is not isolable from cognitive accounts, and the specific frameworks the thesis leans on, the polyvagal model and body-based completion therapies, are contested in the literature and thin on rigorous controlled confirmation, while the clinical phenomenology that carries much of the case is not admissible as primary warrant. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the physiological dimension real but its primacy over cognitive architectures unestablished and its specific frameworks contested. The deciding input would be controlled evidence isolating the physiological architecture as primary.

72 · The Origin of Life

Cosmology · Supreme

Proposition. The origin of life from non-living chemistry is a physically plausible natural process.

The plausibility claim seals at conditional grade. Prebiotic chemistry has demonstrated that the canonical amino acids form from simple precursors under early-Earth conditions and that a synthetic route to nucleotides exists without pre-formed sugars and bases, and the empirical axis adds that these building blocks occur independently in nature, in meteorites carrying scores of amino acids and in alkaline hydrothermal systems that generate the proton gradients and catalysts life uses. What is demonstrated is the plausibility of the chemistry, step by step; what is not demonstrated is the complete pathway from that chemistry to a self-replicating living system. *Kernel:* [LOCK], $\det(R) = 0.9638$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9638$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at conditional grade on physical plausibility, the component steps demonstrated and the precursors found independently. The complete chemistry-to-life pathway remains undemonstrated, the open remainder rather than a defeat of plausibility.

Volume III · Tier I

Nine established results across the disciplines. Five formal and mathematical results seal at theorem grade, three well-replicated empirical findings seal at corroboration grade, and the strong innateness claim for universal grammar goes under-determined on genuinely split evidence.

73 · Radical Skepticism Is Structurally Self-Defeating

Epistemology · Tier I

Proposition. The claim that nothing can be known with certainty defeats itself.

The claim that nothing can be known with certainty is itself asserted as a certainty, so it generates a statement and its denial at once, a performative contradiction that follows from the grammar of the skeptic's own claim rather than from any rhetorical move, and it is identified independently in formal semantics and in phenomenology. The empirical floor is that reliable perceptual tracking and error-correction operate below the level at which the doubt can even be stated, and no organism able to formulate the argument could have reached that point without knowledge-yielding mechanisms. *Kernel: [LOCK], $\det(R) = 0.9930$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9930$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the self-refutation. The position defeats itself in the act of assertion.

74 · Cantor's Diagonal Argument

Mathematics and Logic · Tier I

Proposition. The real numbers cannot be placed in one-to-one correspondence with the natural numbers.

The proof assumes any enumeration of the reals and constructs, by altering the diagonal, a real that differs from every listed entry, so no enumeration is complete and the reals are uncountable. It has been verified without modification in several independent proof-checkers, and the constructivist objections build a different formal system rather than defeating the argument in classical logic. The same diagonal method drives the halting result, giving it an operational echo in uncomputability. *Kernel: [LOCK], $\det(R) = 1.0000$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 1.0000$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade with independent machine verification. Uncountable infinities exist as a proven consequence of the diagonal construction.

75 · Neuroplasticity

Neuroscience · Tier I

Proposition. The adult brain undergoes structural and functional reorganization in response to experience.

The formal axis makes structural change a requirement of learning: strengthening of co-active connections follows from long-term potentiation, and any computational model that learns must modify its weights, so learning without structural change has no coherent account. The empirical axis is four independent platforms converging, enlarged hippocampal structure in drivers who navigate for a living, cortical remapping in Braille readers, functional reorganization after stroke, and increased dendritic branching in enriched animals, measured by structural imaging, functional imaging, histology, and behavior. *Kernel: [LOCK], $\det(R) = 0.9988$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9988$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the learning formalism and corroboration-grade across four independent measurement platforms.

76 · Cognitive Dissonance

Psychology · Tier I

Proposition. Holding two inconsistent cognitions drives measurable attitude change.

A coherence-seeking cognitive system must reduce inconsistency, and the prediction is exact: larger inconsistency drives stronger attitude change. The empirical axis is the classic reversal, in which those paid less to endorse a dull task came to rate it more favorably than those paid more, replicated across incentive sizes and cultures, with a stable moderate-to-large effect and an independent biological channel in the brain region that activates during induced dissonance. *Kernel: [LOCK], $\det(R) = 0.9902$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9902$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, corroboration-grade across cross-cultural replication, a stable effect size, and a neural correlate.

77 · Universal Grammar as an Innate Module

Linguistics · Tier I

Proposition. All human languages share deep structure caused by an innate, species-specific grammar module.

The poverty-of-the-stimulus argument implies an information source beyond the input, and the modular parameter framework gives it form, supported by critical-period effects, modality-independent structure in sign languages, and a gene whose mutation produces specific grammatical deficits. The counterevidence is equally serious: a language reported to lack recursion, typological surveys documenting vast structural variation, and statistical-learning studies showing infants acquiring syntactic patterns without innate templates. The empirical picture is genuinely split, and the strong claim of an innate species-specific grammar module is not settled. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] Under-determined, the evidence genuinely split between innate-module support and learning-based counterevidence. A weaker innate learning bias is more defensible than the strong module the proposition asserts.

78 · Comparative Advantage Does Not Guarantee Gains for All

Economics · Tier I

Proposition. Trade produces aggregate gains but does not guarantee welfare gains for every participant.

Two theorems settle the proposition inside one framework. The classical result proves aggregate gains from trade, and a companion theorem derives the distributional consequence, that liberalization raises returns to the abundant factor and lowers them to the scarce one, so aggregate gain and harm to some are both formal outputs, not an empirical accident. The empirical axis confirms the harm: persistent earnings and employment losses concentrated in specific manufacturing communities exposed to import competition, not offset by retraining or mobility as the simpler theory assumed, with the pattern repeated across other liberalization episodes. *Kernel:* [LOCK], $\det(R) = 0.9981$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9981$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] Sealed, theorem-grade on the joint derivation of aggregate gain and distributional harm, empirically confirmed. The gains are real and so is the uncompensated harm.

79 · Durkheim's Social Integration Thesis

Sociology · Tier I

Proposition. Variation in social integration predicts differential suicide rates.

The taxonomy is derived from a matrix of integration and regulation and predicts a counterintuitive pattern, elevated risk at both extremes, that individual psychology alone does not yield. The empirical axis is five independent epidemiological streams across a century and a half: a denominational difference holding under economic controls, elevation during military demobilization, a measurable rise following widowhood, cross-national correlation with social-capital indices, and a modern clinical theory that operationalizes the same variables. *Kernel:* [LOCK], $\det(R) = 0.9747$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9747$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] Sealed, corroboration-grade across five independent epidemiological streams spanning a hundred and fifty years.

80 · The Halting Problem Is Undecidable

Computer Science and Logic · Tier I

Proposition. No general algorithm decides, for an arbitrary program and input, whether the program halts.

Assuming a general halting decider, one builds a program that halts exactly when the decider says it runs forever and runs forever when it says it halts, a contradiction, so no such decider exists. The proof is machine-verified and is structurally the same result as the diagonal argument and the first incompleteness theorem. The empirical echo is that termination, equivalence, and deadlock analysis are undecidable in practice, a boundary the verification industry meets routinely. *Kernel:* [LOCK], $\det(R) = 0.9901$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9901$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] Sealed, theorem-grade with machine verification, one member of the family of diagonal undecidability results.

81 · Moral Intuitions Are Causally Prior to Reasoning

Ethics · Tier I

Proposition. For most people in most contexts, moral reasoning is post-hoc rationalization of intuitive response.

The dual-process account predicts that fast automatic pattern-matching returns a verdict before deliberation finishes, and the Bayesian framing makes intuitions priors that weight incoming arguments, placing them causally upstream of reasoning. The empirical axis is moral dumbfounding, in which people condemn harmless taboo violations strongly while failing to produce supporting arguments, neuroimaging showing emotional regions activating before deliberative ones, and cross-cultural replication of the architecture. The proposition's hedge, for most people in most contexts, matches the evidence. *Kernel:* [LOCK], $\det(R) = 0.9908$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9908$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, corroboration-grade across dumbfounding, neuroimaging timing, and cross-cultural replication, on the appropriately hedged claim.

Volume III · Tier II

Nine theory-laden cases. Two physics claims and the structural insight of intersectionality seal at conditional grade, and the strong versions of five influential theories break or go under-determined: completeness for reliabilism, uniqueness for Rawls, resolution for decoherence, and the strong-form efficient market hypothesis outright.

82 · Reliabilism Is a Complete Theory of Justified Belief

Epistemology · Tier II

Proposition. A belief is justified if and only if it is produced by a reliable cognitive process.

The formal defeat is the generality problem: a process can be described at any grain, vision, vision at noon, vision in poor light, and the reliability verdict changes with the description, so without a principled way to fix the reference class the central predicate is indeterminate, and no accepted solution exists. The empirical axis sharpens it: processes statistically reliable in their class still produce false beliefs in specific applications, certain heuristics and in-group biases among them, and the data does not fix which class is the process. *Kernel:* [X], $\det(R) = 0$, $N = 24$.

Triaxial. G9 CSEG fails: the completeness claim exceeds the theory's reach. Two axes fall into one, the warrant volume collapses. Kernel [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, the completeness claim defeated by the unsolved generality problem. The reliabilist core, that reliability

bears on justification, survives; the claim to be a complete theory does not.

83 · Terror Management Theory

Psychology · Tier II

Proposition. Reminders of mortality cause measurable increases in worldview defense.

Reminders of mortality do produce measurable worldview defense, and a meta-analysis of hundreds of studies finds the effect real at small-to-medium size. Its attribution specifically to mortality is what cannot be isolated. The standard manipulation is confounded with negative affect in general, and rival mechanisms, uncertainty and threat to meaning, produce similar effects without any mortality-specific component, so the signal is genuine but not cleanly traceable to mortality processing. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the effect real but its mortality-specific attribution confounded with general negative affect and matched by rival mechanisms. The deciding input would be a manipulation isolating mortality from affect.

84 · Intersectionality

Sociology · Tier II

Proposition. Inequalities of race, gender, and class interact multiplicatively to produce unique forms of disadvantage.

The structural insight is sound and seals at conditional grade. Where disadvantage runs along several axes that interact, a linear decomposition into separate single-axis effects is formally invalid, and the legal critique that motivated the concept identified exactly this failure in single-category anti-discrimination law. The empirical axis confirms non-additive interaction in specific domains: earnings at the intersection of race and gender that do not match additive predictions, and interaction effects in cardiovascular and maternal-health outcomes. The fence is that many studies testing additive against interactive models lack the power to distinguish them, so the universal claim that all such inequalities combine multiplicatively is not established at that strength. *Kernel:* [LOCK], $\det(R) = 0.9504$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9504$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at conditional grade on the structural insight that multi-axis disadvantage can interact non-

additively, demonstrated in earnings and health. The universal multiplicative claim is fenced as underpowered, not sealed.

85 · The Efficient Market Hypothesis, Strong Form

Economics · Tier II

Proposition. All information, public, private, and insider, is already fully reflected in asset prices.

It fails formally and empirically at once. Formally it is self-undermining: if all information were already priced, no one would be paid to gather it, but without gatherers prices could not come to reflect it, a proven internal contradiction. Empirically it is refuted directly, by documented above-market profits from inside information, regulatory cases confirming private information generates excess returns before incorporation, drift in prices after announcements, and momentum and value anomalies persisting across decades. *Kernel:* $[X], \det(R) = 0, N = 24$.

Triaxial. G8 CSCG fails: persistent anomalies refute the strong form. Two axes fall into one, the warrant volume collapses. Kernel $[X] \det(R) \rightarrow 0. \vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, self-contradictory in form and refuted in fact. The weaker claim that public information is largely priced is defensible; the strong claim that insider information is already priced is not.

86 · Rawls's Original Position

Political Philosophy · Tier II

Proposition. Deliberation behind the veil of ignorance uniquely generates just principles.

The uniqueness fails. The derivation depends on choosing the maximin rule, maximize the worst outcome, which requires extreme risk aversion that the rationality assumption alone does not supply; a competing result shows that expected-utility maximizers behind the same veil should choose differently, so the setup does not single out one answer. The empirical axis agrees: subjects placed in veil-of-ignorance scenarios do not consistently choose maximin, preferring a guaranteed floor with maximized average, and justice intuitions vary across cultures rather than converging. *Kernel:* $[X], \det(R) = 0, N = 24$.

Triaxial. G9 CSEG fails: the uniqueness claim exceeds the derivation. Two axes fall into one, the warrant volume collapses. Kernel $[X] \det(R) \rightarrow 0. \vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, the uniqueness claim defeated by an unjustified risk attitude and by experimental non-convergence. The original position is a valuable construction; that it generates uniquely just principles is not sustained.

87 · Hawking Radiation

Cosmology · Tier II

Proposition. Black holes emit thermal radiation through quantum effects and gradually lose mass.

The formal warrant is unusually strong. Applying quantum field theory in curved spacetime to a black hole yields a thermal emission spectrum fixed by the mass with no free parameters, and three independent derivations by entirely different routes converge on the same formula. Direct empirical confirmation for astrophysical black holes is absent and effectively unreachable, since the predicted temperature for a stellar-mass hole lies far below the microwave background, but laboratory analog systems, acoustic and optical, have observed the predicted effect and confirm the mathematical structure of the mechanism. *Kernel:* $[LOCK], \det(R) = 0.8997, N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK] \det(R) = 0.8997, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at conditional grade on three convergent parameter-free derivations and analog-system confirmation of the mechanism. The grade is conditional because direct astrophysical detection is absent and likely permanently out of reach.

88 · Decoherence Resolves the Measurement Problem

Quantum Foundations · Tier II

Proposition. Environmental decoherence explains why definite measurement outcomes occur.

Decoherence is real and exactly derivable: environmental entanglement suppresses interference between macroscopic alternatives almost instantly, and the suppression is confirmed across eraser experiments, ion traps, and qubits. It does not resolve the measurement problem, and the reason is precise. Suppressing interference turns a coherent superposition into a mixture of alternatives; it does not select one alternative as the outcome that occurs, which is exactly what the measurement problem asks. The observation that one definite result is registered fits decoherence under several interpretations equally, so decoherence explains a different thing than the problem it is said to solve. *Kernel:* $[X], \det(R) = 0, N = 24$.

Triaxial. G12 ADEG fails: relocates the measurement problem rather than closing it. Two axes fall into one, the warrant volume collapses. Kernel $[X] \det(R) \rightarrow 0. \vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, decoherence addressing interference suppression rather than the definite-outcome question that constitutes the problem. The phenomenon is genuine and important; the claim that it resolves the measurement problem is not.

89 · The Default Mode Network

Neuroscience · Tier II

Proposition. The default mode network is the neural substrate of self-referential processing.

The network is reliably involved in self-referential processing, confirmed across hundreds of imaging studies in self-judgment, autobiographical memory, future thinking, and theory of mind. The stronger claim, that it is the substrate of self-referential processing specifically, fails on functional specificity. The same regions activate in narrative comprehension, moral reasoning, and egocentric navigation, so the network is recruited broadly for internally directed, context-rich processing rather than uniquely for self-reference. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, the network involved in self-referential processing but not specific to it.** The deciding input would be a dissociation isolating self-reference from other internally directed processing.

90 · Functionalism

Philosophy of Mind · Tier II

Proposition. Mental states are defined by their functional roles over physical substrates.

The multiple-realizability argument is valid: the same functional state can be instantiated in different physical substrates, and computation is definable independently of implementation, so a functional account of cognitive states, beliefs and desires defined by their causal roles, is coherent and largely workable. The strong claim that minds simply are functional states, substrate-independent down to qualitative experience, meets the same explanatory gap the hard problem names: the thought experiments of an inverted or absent experience under identical function, and the qualia question generally, are not settled by the functional role. The method holds that gap open here as it does there. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, functionalism workable for cognitive states but the qualitative-experience component meeting the explanatory gap the method does not seal.** The functional account does not, on the available warrant, extend to experience.

Volume III · Tier III

Nine cases at the edge of theory. Two physics results seal, the second at theorem grade, and the remainder split between broken arguments, the ontological proof and the social contract among them, and genuinely contested metaphysics held under-determined.

91 · Black Hole Information Is Preserved

Cosmology · Tier III

Proposition. Information about matter falling into a black hole is preserved; unitarity holds.

The case for preservation has strengthened into a conditional seal. Quantum mechanics requires unitarity while classical gravity implies loss inside the singularity, so a quantum theory of gravity must break one, and the holographic duality supplies the resolution: a gravitational system in the relevant spacetime is dual to a manifestly unitary boundary theory, and the recent island calculations recover the unitary curve that information preservation requires. The empirical axis is indirect, with heavy-ion collision data matching bounds from the same duality, validating the framework without testing a full evaporation. *Kernel:* [LOCK], $\det(R) = 0.8926$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.8926$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at conditional grade, on a holographic duality with a unitary boundary theory and the island recovery of the unitary curve.** The grade is conditional because the result is framework-specific and the empirical support is indirect.

92 · The Quantum Zeno Effect

Quantum Foundations · Tier III

Proposition. Continuous or very frequent measurement arrests a quantum state's transition.

The derivation is exact: a quantum state's survival probability decays quadratically at short times, and sufficiently frequent measurement keeps resetting the system before the decay reaches the regime where transition probability accumulates, with the relevant timescale fixed by the energy spread of the state. The empirical axis is confirmation across independent platforms, trapped ions, optical cavities, superconducting qubits, and cold atoms, by separate teams, with the complementary acceleration effect also observed where the theory predicts it. *Kernel:* [LOCK], $\det(R) = 0.9921$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK]

$\det(R) = 0.9921$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed, theorem-grade on the derivation and corroboration-grade across multiple independent experimental platforms.

93 · Milgram Obedience as Universal Compliance

Sociology · Tier III

Proposition. Ordinary people universally comply with authority instructing them to harm.

The empirical axis breaks the universal claim the way it broke the prison study. The original is contaminated: the authority figure repeatedly departed from the script to pressure resistant subjects in non-standard ways, obedience rates swung dramatically across conditions in a manner indicating experimenter effects, cross-cultural replications returned much lower rates, and many subjects saw through the deception. *Kernel:* $[X]$, $\det(R) = 0$, $N = 24$.

Triaxial. G9 CSEG fails: the universal-compliance claim exceeds the variation data. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, contaminated procedure, dramatic rate variation, and cross-cultural non-replication refuting universality. A weaker claim, that authority can raise harmful compliance in some settings, holds; universal compliance does not.

94 · The Ontological Argument

Philosophy and Theology · Tier III

Proposition. God's existence follows by definition from the concept of the greatest conceivable being.

The argument infers existence from the premise that a being existing in reality is greater than one existing only in conception, and the inference commits a formal category error: existence is not a first-order predicate that adds to a concept, but a second-order claim about whether a concept is instantiated, so it cannot be packed into a definition. The modal reformulation shifts the weight to a premise that a maximally great being is possible, which already presupposes the conclusion. The empirical axis is empty by design. *Kernel:* $[X]$, $\det(R) = 0$, $N = 24$.

Triaxial. G11 OMA fails: existence derived from concept, a category error. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, a category error in treating existence as a defining predicate, with the modal version begging the question. The method breaks the argument, not the theological conclusion: that this proof fails says nothing about whether its conclusion is true.

95 · Social Contract Theory

Political Philosophy · Tier III

Proposition. Political authority derives its legitimacy from an actual or hypothetical agreement of the governed.

On the literal reading the proposition is refuted: no founding agreement exists anywhere in the historical record, and political entities arise from conquest, migration, inheritance, or seizure rather than from consent, with stateless societies showing no pre-political individuals contracting into authority. The hypothetical reframing avoids the empirical refutation only by giving up the mechanism, since a hypothetical contract is a decision procedure rather than an agreement, and consent that no one actually gave cannot generate the obligation the theory needs. *Kernel:* $[X]$, $\det(R) = 0$, $N = 24$.

Triaxial. G9 CSEG fails: the descriptive-contract claim exceeds any actual consent. Two axes fall into one, the warrant volume collapses. *Kernel* $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken, the literal claim empirically refuted and the hypothetical version not consent at all. Social contract reasoning retains value as a normative test of principles; the claim that authority derives from consent does not hold in either reading.

96 · Moral Realism

Ethics · Tier III

Proposition. Objective moral facts exist independently of any mind.

The question is genuinely contested, and the method reaches the same verdict it reached for the existence of abstract objects. The arguments for realism are real but not decisive: that mathematics is also non-physical yet treated as objective, and that moral statements function as genuine propositions in inference. The empirical axis underdetermines the question, with some pan-cultural prohibitions alongside enormous variation, fitting realist and anti-realist readings equally. *Kernel:* $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, the arguments inconclusive and the cross-cultural data fitting both readings, a genuinely open metaphysics rather than a settled fact. As with abstract objects, the deciding input would be a consideration that breaks the symmetry, which the standard arguments do not supply.

97 · Embodied Cognition

Cognitive Science · Tier III

Proposition. Bodily states constitutively shape abstract thought.

The evidence has split under scrutiny. A weaker neural version has some support, with language comprehension recruiting motor regions associated with the described actions. The signature priming results that made the strong case, that incidental warmth or head movement shifts judgment, largely failed to replicate in the large pre-registered studies of the past decade, so the constitutive claim is left without its main empirical support. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, the constitutive claim undercut by replication failure while a weaker sensorimotor-involvement claim survives.** The deciding input would be robust pre-registered evidence that bodily states shape, rather than merely accompany, abstract processing.

98 · The Fiscal Multiplier

Economics · Tier III

Proposition. The fiscal multiplier consistently exceeds one across economies.

The claim is refuted by treating the multiplier as what it is, a regime-dependent quantity rather than a constant. The formal frameworks already span zero to well above one depending on conditions, and the better-identified natural experiments resolve the picture: multipliers are larger in recessions and in slack economies with monetary accommodation, and near zero at the zero lower bound under tight policy. A quantity that ranges from near zero to above one with the regime does not consistently exceed one. *Kernel:* [X], $\det(R) = 0$, $N = 24$.

Triaxial. G7 DUAL fails: regime-dependent, the value shifts with conditions. Two axes fall into one, the warrant volume collapses. *Kernel* [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow$ [X].

Verdict: [X] **Broken, the multiplier regime-dependent rather than a stable parameter above one.** The conditional claim, that it exceeds one in a slack economy with accommodation, holds; the unconditional claim does not.

99 · Elite Theory

Political Science · Tier III

Proposition. Political outcomes in modern democracies are primarily determined by narrow elites.

The thesis has formal grounding in several classical traditions and a microfoundation in the lower collective-action costs of small organized groups. The empirical support is genuine but contested. A prominent analysis of many policy outcomes found elite preferences prevailing when they conflicted with the median voter, but a reanalysis disputes the statistical interpretation on the

ground that elite and popular preferences usually coincide, and capture rates vary across countries. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, real evidence of disproportionate elite influence against a disputed interpretation and variable cross-national rates.** The weaker claim of disproportionate influence is defensible; that outcomes are primarily elite-determined is not settled on this evidence.

Volume III · Tier IV

Nine cases closing Volume III. The genuine structural results in epistemology and the rule-following argument seal, two robust empirical regularities seal, and the speculative metaphysics resists: sufficient reason, personal identity, and a cyclic cosmology go under-determined, while panpsychism breaks for relocating the problem it claims to solve.

100 · The Principle of Sufficient Reason

Metaphysics · Tier IV

Proposition. Every fact, event, and entity has a complete explanatory ground.

Science proceeds as if every fact has a sufficient reason, treating each successful explanation as a confirmation. The principle cannot be established that way, since the absence of a confirmed brute fact in the record does not show that none exists, and quantum mechanics supplies a genuine candidate: the specific outcome of an individual quantum event, the moment a particular atom decays, has no sufficient reason for that outcome rather than another, only a probability, even though the state is fully specified. As a universal metaphysical truth the principle is a regulative assumption, not an established fact, and it faces a live counterexample. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, a powerful regulative assumption that science relies on but cannot confirm, with individual quantum outcomes a standing candidate counterexample.** The deciding input would be a resolution of whether quantum indeterminacy is genuine brute contingency or hidden determination.

101 · Personal Identity as Psychological Continuity

Metaphysics · Tier IV

Proposition. What makes a person the same over time is psychological continuity, not physical continuity.

Psychological continuity is clearly relevant to identity: clinical cases show that the loss of memory and continuity changes who a person is in a way the loss of mere matter does not. It is not sufficient, and the branching problem is the reason. If continuity alone constituted identity, then a process splitting one psychological stream into two would make both successors identical to the original, which is impossible, since they are not identical to each other. The question of what constitutes identity over time is genuinely open among competing accounts, psychological, physical, narrative, and the deflationary view that identity is not what matters. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] Under-determined, psychological continuity relevant but not sufficient, the branching problem defeating sufficiency and the constitutive question open among rival theories. The deciding input would be a principled resolution of the fission case.

102 · Penrose's Conformal Cyclic Cosmology

Cosmology · Tier IV

Proposition. The universe undergoes an infinite sequence of aeons, each rescaled conformally into the next.

The mathematics of the model is defined, but two things leave it unconfirmed. It requires that all massive particles eventually decay, including protons, which has never been observed, and the claimed observational signature, anomalous low-variance spots in the microwave background from a previous aeon, failed to reproduce its statistical significance when independent groups reanalyzed the same data with different pipelines. The model is neither supported nor refuted, a defined speculative proposal whose evidence did not survive replication. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] Under-determined, the claimed signature failing independent replication and a key premise unobserved. The deciding input would be a confirmed observational signature or the detection of the required particle decay.

103 · Wittgenstein's Private Language Argument

Philosophy of Language · Tier IV

Proposition. A language whose terms are defined purely by reference to private sensation is incoherent.

A word is a rule for use, and a rule requires a difference between using it correctly and merely seeming to, which a purely private language cannot supply, since its only check is the speaker's impression of remembering the use, and an impression that cannot be wrong is no check at all. Without a public criterion there is no correctness, so a language whose terms are fixed entirely by private ostension is incoherent. The empirical axis corroborates: language acquisition requires social interaction, input without interaction does not produce it, and no child has developed a symbolic system in isolation. *Kernel:* [LOCK], $\det(R) = 0.9979$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9979$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] Sealed at structural grade on the rule-following argument and corroboration-grade on the social requirement for language. This is the consideration that undermined solipsism, here sealed as a result in its own right.

104 · The Problem of the Criterion

Epistemology · Tier IV

Proposition. The problem of the criterion cannot be resolved without circularity.

To identify which beliefs are knowledge one needs the criteria for knowledge, and to establish the criteria one needs cases of knowledge to test them against, so the two are mutually dependent and no non-circular path exists. Every escape is one of two unsatisfactory moves: accept some knowledge without criteria, an arbitrary starting point, or accept criteria without grounding them in cases, an ungrounded rule. The empirical echo is that every calibration procedure presupposes which states are accurate enough to calibrate against, and no method validates itself from outside. *Kernel:* [LOCK], $\det(R) = 0.9889$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9889$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] Sealed, theorem-grade on the mutual dependence of criteria and cases, a member of the family with the regress trilemma and the problem of induction.

105 · The Thermodynamic Arrow of Time

Physics · Tier IV

Proposition. The thermodynamic arrow is the empirically grounded macroscopic temporal asymmetry.

The fundamental equations of physics are symmetric under time reversal, with only the marginal exception of a confirmed but second-order microscopic asymmetry that does not generate a macroscopic direction. The macroscopic arrow comes instead from entropy increase against a low-entropy past, and the empirical support is comprehensive: every observed process at human scales increases entropy, laboratory and cosmological evidence agree, and the early universe sits in the required low-entropy state. *Kernel*: [LOCK], $\det(R) = 0.9260$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9260$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade on T-symmetric fundamental laws plus a multiply-confirmed entropic macroscopic arrow. The deeper question of why the initial state was low-entropy is a separate open problem the seal does not close.

106 · Trolley Variants and Moral Inconsistency

Ethics · Tier IV

Proposition. Systematic variation across structurally equivalent trolley cases reveals that moral intuitions are structurally inconsistent.

Systematic variation in moral judgment between cases with identical outcome structure is robust: people refuse to push one person to save five while accepting a switch that does the same, the difference tracking personal contact and using a person as a means. Imaging shows the personal version recruiting emotional regions more strongly, responses to it run slower, and the asymmetry replicates across dozens of countries, so it is a cross-cultural feature of moral cognition rather than a local artifact. *Kernel*: [LOCK], $\det(R) = 0.9871$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9871$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, corroboration-grade on the cross-cultural variation with a dual-process basis. Fenced: whether to call this inconsistency depends on the standard, a consequentialist seeing inconsistency where a deontologist sees principled sensitivity to a real distinction; the empirical pattern is what seals.

107 · The Matthew Effect

Sociology · Tier IV

Proposition. Cumulative advantage is a self-reinforcing structural property of resource distribution.

Cumulative advantage is self-reinforcing by a derivable mechanism: when new links attach in proportion to existing ones, the result is a power-law distribution with structural inequality as a mathematical consequence rather than an accident. The empirical axis confirms it across four independent domains, disproportionate citation to already-credited scientists, power-law wealth distributions in every measured society, scale-free structure in technological and social networks, and a controlled experiment in which identical items diverged sharply in popularity once early advantage was allowed to compound. *Kernel*: [LOCK], $\det(R) = 0.9910$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9910$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the preferential-attachment mechanism and corroboration-grade across four independent domains including a controlled experiment.

108 · Panpsychism Resolves the Hard Problem

Philosophy of Mind · Tier IV

Proposition. Attributing phenomenal character to all matter resolves the hard problem of consciousness.

Panpsychism attributes some phenomenal character to every physical entity, so that complex consciousness becomes a combination of proto-conscious parts rather than an emergence from non-conscious matter, which appears to dissolve the emergence problem. It does not resolve the hard problem; it relocates it. The combination problem, how micro-experiences could combine into a single unified experience, is as intractable as the emergence problem it replaces, so the difficulty is moved rather than removed. And no test distinguishes panpsychism from materialism, since it makes no measurable prediction the other does not. *Kernel*: [X], $\det(R) = 0$, $N = 24$.

Triaxial. G12 ADEG fails: relocates the hard problem into the combination problem. Two axes fall into one, the warrant volume collapses. *Kernel* [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, the resolution claim trading the emergence problem for an equally hard combination problem and adding no empirical discrimination. Panpsychism relocates the hard problem rather than resolving it, the same move that broke the decoherence claim.

Volume IV · Tier III

Nine quantum and physical results, almost all proven theorems with operational or multi-instrument confirmation. They seal at theorem grade, the calibrated outcome for established results, with the area law noting its indirect empirical confirmation and underdetermination joining the structural epistemology family.

109 · The No-Cloning Theorem

Quantum Mechanics · Tier III

Proposition. No physical operation can perfectly copy an arbitrary unknown quantum state.

The proof is short and exact. A cloning operation applied to a superposition, by linearity, produces a superposition of cloned basis states, not the product the genuine clone would require, and the two coincide only for basis states, so no operation copies an arbitrary unknown state. The result is operationally confirmed in quantum key distribution, where it guarantees that eavesdropping disturbs the channel and is detected, exactly as the theorem requires. *Kernel:* [LOCK], $\det(R) = 0.9931$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9931$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade with operational confirmation in quantum cryptography.

110 · The No-Communication Theorem

Quantum Mechanics · Tier III

Proposition. Quantum entanglement cannot enable superluminal signaling.

Whatever measurement one party performs on an entangled pair, the other party's local state, obtained by tracing out the first, is the same maximally mixed state regardless of basis or outcome, so no choice on one side changes the local statistics on the other and entanglement cannot carry a signal. Every working quantum-cryptography protocol assumes this and functions, and loophole-free Bell tests confirm nonlocal correlation with no detectable signaling. *Kernel:* [LOCK], $\det(R) = 0.9876$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9876$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade with operational confirmation. The correlations are nonlocal; signaling is not.

111 · The Heisenberg Uncertainty Principle as Ontological

Quantum Mechanics · Tier III

Proposition. The uncertainty principle is a constraint on states, not an artifact of measurement disturbance.

The uncertainty relation follows from the non-commutativity of the operators alone and applies to any state, including one that has never met a measuring device, so it is a constraint on preparation

rather than a disturbance from observation. The two were separated experimentally: the original measurement-disturbance formula can be violated under weak measurement, confirmed with neutrons and photons, while the preparation relation remains inviolate. *Kernel:* [LOCK], $\det(R) = 0.9964$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9964$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade with the ontological preparation relation confirmed inviolate as the disturbance interpretation was separated off.

112 · CPT Symmetry

Quantum Field Theory · Tier III

Proposition. CPT symmetry is necessary in any local Lorentz-invariant quantum field theory.

The combined symmetry follows rigorously from Lorentz invariance, locality, and unitarity, so any local Lorentz-invariant field theory respects it. The empirical record confirms the structure precisely: the individual symmetries of charge and parity are violated in several particle systems while the combined symmetry remains exact, and matter-antimatter comparisons confirm it to high precision. *Kernel:* [LOCK], $\det(R) = 0.9994$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9994$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade with confirmation that individual symmetries break while the combined one holds exactly.

113 · Neutrino Oscillations Imply Mass

Astrophysics · Tier III

Proposition. The observed oscillation of neutrinos between flavors requires non-zero rest mass.

Oscillation between flavors arises from a mismatch between the states produced in weak interactions and the states that propagate, and the oscillation formula requires a non-zero mass-squared difference, so massless neutrinos could not oscillate. Four independent detector programs, atmospheric, solar, reactor, and accelerator, confirm the oscillation with the predicted dependence on distance and energy. *Kernel:* [LOCK], $\det(R) = 0.9997$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9997$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the derivation and corroboration-grade across four independent detector lineages.

114 · The Cosmic Microwave Background

Cosmology · Tier III

Proposition. The microwave background is direct physical evidence of a hot dense early universe.

A relic radiation field was predicted from early-universe theory years before its discovery near the predicted temperature, with two sharp predictions: a near-perfect blackbody spectrum from thermal equilibrium and near-isotropy with tiny seeded anisotropies. Successive instruments confirmed the most perfect blackbody ever measured, then the predicted acoustic-peak structure, then a percent-level cosmological model, across independent space and ground platforms. *Kernel:* $[LOCK]$, $\det(R) = 0.9991$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9991$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, a prediction confirmed and then characterized to extraordinary precision by multiple independent instruments.

115 · The Pauli Exclusion Principle

Quantum Mechanics · Tier III

Proposition. The exclusion principle has no classical analog and cannot be derived from classical physics.

Exclusion follows from the antisymmetry of the wavefunction for identical fermions, which forces the amplitude to zero when two share a state, and the antisymmetry itself follows from the spin-statistics theorem for half-integer spin, with no classical counterpart. Its empirical reach is the structure of matter: atomic shells and chemistry, the band structure of solids, and the degeneracy pressure that supports white dwarfs and neutron stars against gravity. *Kernel:* $[LOCK]$, $\det(R) = 0.9989$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9989$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the spin-statistics derivation and corroboration-grade across chemistry, solid-state physics, and stellar structure.

116 · The Bekenstein-Hawking Area Law

Cosmology · Tier III

Proposition. Black hole entropy is proportional to one quarter of the event-horizon area.

Three independent formal routes converge on an entropy proportional to one quarter of the horizon area, and two microscopic theories that share almost nothing, a string-theoretic state count for certain black holes and a loop-quantum count, both reproduce the same formula. That convergence from unrelated directions is unusually strong theoretical warrant. The empirical confirmation is indirect, since black-hole entropy is not directly measured. *Kernel:* $[LOCK]$, $\det(R) = 0.9992$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9992$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the convergence of independent semiclassical and microscopic derivations. The empirical confirmation is indirect, which the seal notes without lowering the grade given the strength of the convergence.

117 · Underdetermination of Theory by Evidence

Epistemology · Tier III

Proposition. Finite evidence is in principle compatible with multiple incompatible theories.

A hypothesis yields predictions only with auxiliary assumptions, so a failed prediction does not single out the hypothesis as false, and for any theory with given consequences one can construct empirically equivalent alternatives. The historical record shows the structure concretely: theories empirically equivalent on all data of their time, separated only later by novel predictions, and interpretations of quantum mechanics that are provably indistinguishable by any experiment. *Kernel:* $[LOCK]$, $\det(R) = 0.9920$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9920$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the Duhem-Quine structure with concrete historical instances, a companion to the other structural results on the limits of inference.

118 · Steady-State Theory Is Refuted

Cosmology · Tier III

Proposition. The Steady-State model of an unchanging universe is empirically refuted.

The Steady-State model holds the universe statistically identical at all epochs, which forbids relic radiation, requires fixed source populations across redshift, and rules out epoch-varying abundances. Four independent lines contradict it: the microwave background exists where the model predicts none, quasars are far more abundant at early epochs than locally, the primordial helium fraction matches hot-early-universe synthesis with no Steady-State mechanism, and radio-source counts vary with redshift. The proposition is that the model is refuted, and it is. *Kernel: [LOCK], $\det(R) = 0.9719, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9719, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on four independent empirical refutations. The claim audited is the refutation itself, which holds.

119 · MOND Is Insufficient as a Complete Dark-Matter Replacement

Astrophysics · Tier III

Proposition. Modified Newtonian dynamics is structurally insufficient to replace dark matter entirely.

Modified dynamics reproduces galaxy rotation curves and the Tully-Fisher relation as a mathematical consequence of a single modification, which is a real success at galactic scale. It fails as a complete replacement. The Bullet Cluster separates the lensing mass from the baryonic gas by a wide margin, and a theory in which gravity tracks visible matter cannot produce that separation; clusters require additional unseen mass even after the modification; and the original form has no complete relativistic extension. *Kernel: [LOCK], $\det(R) = 0.9936, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9936, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the success at galactic scale genuine and the insufficiency at cluster scale decisive. MOND captures a real regularity but cannot replace dark matter entirely, which is exactly what the proposition claims.

120 · Variable Speed of Light Is Not a Viable Alternative to Inflation

Cosmology · Tier III

Proposition. Variable-speed-of-light cosmology is not a viable alternative to inflation.

A varying speed of light requires the permittivity or permeability of the vacuum to vary, which would shift atomic spectra and the

fine-structure constant in ways not seen, and it sits in tension with local Lorentz invariance. The claimed fine-structure variation that motivated the program was not confirmed by independent spectra, and the model does not naturally reproduce the nearly scale-invariant fluctuation spectrum that inflation predicts and that the microwave background confirms. *Kernel: [LOCK], $\det(R) = 0.9280, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9280, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at conditional grade. The model is strongly disfavored on several independent counts, not strictly impossible, so the seal records that it is not currently viable rather than that it is ruled out forever.

121 · Zero-Point Energy Cannot Be Extracted as Unlimited Free Energy

Quantum Field Theory · Tier III

Proposition. The vacuum zero-point energy cannot be extracted as unlimited free energy.

The vacuum energy is the ground state of the field, the lowest available energy, and extracting work requires a transition to something lower, which by definition does not exist. The argument is thermodynamic and tight. The empirical record agrees: every claimed extraction device has failed controlled testing or proved to be measurement error, and the Casimir force, measured to one percent, is conservative, returning no net energy over a cycle. *Kernel: [LOCK], $\det(R) = 0.9865, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9865, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the ground-state argument and corroboration-grade on the uniform failure of extraction claims. The vacuum is real; the free lunch is not.

122 · Superdeterminism Cannot Resolve Measurement Without Unfalsifiability

Quantum Foundations · Tier III

Proposition. Superdeterminism cannot resolve the measurement problem without becoming unfalsifiable.

Superdeterminism escapes the Bell result by denying that the experimenter's setting is independent of the hidden variables, correlating choice with system in advance. The cost is structural. Cosmic Bell tests sourced settings from quasar light billions of years old, and the position absorbs this simply by extending the required pre-correlation to those epochs, a move available against

any conceivable test, since no experiment can reach a horizon predating all correlation. The escape is purchased with unfalsifiability. *Kernel*: [LOCK], $\det(R) = 0.9916$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9916$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade. The proposition is that the resolution requires giving up falsifiability, and the structure of the evasion confirms exactly that.

123 · The Anthropic Principle Is Not a Scientific Explanation of Fine-Tuning

Epistemology · Tier III

Proposition. The anthropic principle is not a scientific explanation of physical fine-tuning.

An explanation answers why a constant takes its value rather than another, by mechanism or derivation. The anthropic observation answers a different question, why observers find themselves where observation is possible, which is a selection effect, not a cause. The distinction is formal. The one celebrated anthropic prediction required an independent assumption, a multiverse with a distribution of values, to do its work; the selection observation alone predicts nothing. *Kernel*: [LOCK], $\det(R) = 0.9825$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9825$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade on the distinction between a selection effect and a causal derivation. The principle constrains what we can observe; it does not explain why the constants are what they are.

124 · Orchestrated Objective Reduction Is Not a Physical Theory of Consciousness

Philosophy of Mind · Tier III

Proposition. The Penrose-Hameroff orchestrated-objective-reduction proposal is not a physical theory of consciousness.

The proposal rests on an undeived collapse mechanism, a gravity-induced reduction not obtained from any established theory of quantum gravity, reached by an extrapolation from an incompleteness result to non-computable cognition to quantum gravity that no step justifies. The empirical side is bare: no quantum coherence has been measured in neural microtubules at body temperature, anesthetic action is better explained conventionally, and the measured correlates of consciousness are macroscopic. *Kernel*: [LOCK], $\det(R) = 0.9912$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9912$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the proposition that this is not an established physical theory holding on both the underived mechanism and the absent empirical support. It is a conjecture with a mechanism it has not derived and a signature no one has found.

125 · Verlinde's Emergent Gravity Cannot Replace General Relativity

Physics · Tier III

Proposition. Verlinde's entropic emergent gravity cannot replace general relativity as a foundational framework.

The entropic-force derivation recovers the Newtonian law and the field equations as emergent, but from a holographic-screen construction not obtained from a confirmed microscopic theory, and it assumes results, a thermal relation and an entropy formula, that it then uses to derive gravity. As a foundational replacement it cannot match the incumbent: it has no completed relativistic wave equation to reproduce the confirmed polarization of gravitational waves, and has not been shown to recover the precisely measured perihelion precession. *Kernel*: [LOCK], $\det(R) = 0.9870$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9870$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the framework incomplete where the incumbent makes confirmed precise predictions. Suggestive as a perspective, it cannot replace the theory whose predictions it has yet to reproduce.

126 · The Standard Model Is Demonstrably Incomplete

Particle Physics · Tier III

Proposition. The Standard Model of particle physics is demonstrably incomplete.

Four confirmed phenomena lie outside the model. Neutrino oscillations require masses the model sets to zero. Galactic and cosmological dynamics require a dark-matter particle the model does not contain. The matter excess over antimatter requires far more symmetry violation than the model supplies. And gravity has no place in it at all. Each is independently established, and the preferred extensions have not appeared at accessible energies. *Kernel*: [LOCK], $\det(R) = 0.9658$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9658$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on four independent confirmed phenomena the model cannot accommodate. Its incompleteness is not a matter of taste; it is forced by measurement.

Volume IV · Tier IV

Nine cases at the top tier of Volume IV. Three are open problems of physics correctly identified as genuine, three are no-go or equivalence results, and three concern interpretations of quantum mechanics, each sealing on the precise form of its claim.

127 · The GZK Cutoff

Astrophysics · Tier IV

Proposition. The GZK cutoff is a physical consequence of ultra-high-energy cosmic rays interacting with the microwave background.

Above a threshold energy, cosmic-ray protons interact with microwave-background photons to produce a resonance that bleeds off their energy, and the threshold follows from known masses and the background temperature with no free parameters, near five times ten to the nineteenth electron-volts. Three independent observatories, using different detection methods, observed the predicted spectral suppression at the predicted energy. *Kernel:* [LOCK], $\det(R) = 0.9965$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9965$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, a parameter-free prediction from established physics confirmed by three independent experiments.

128 · The Frauchiger-Renner No-Go Result

Quantum Mechanics · Tier IV

Proposition. The extended Wigner's-friend scenario shows quantum mechanics admits no consistent observer-independent ontology.

The extended Wigner's-friend scenario proves that three assumptions cannot all hold: that quantum mechanics applies universally, that agents may reason with it, and that different observers' outcomes are jointly consistent in an observer-independent way. At least one must be abandoned. A six-photon implementation observed correlations violating an inequality derived from the scenario, confirming the joint incompatibility in a physical system. *Kernel:* [LOCK], $\det(R) = 0.9616$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9616$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the proven incompatibility, with experimental confirmation. The theorem fixes that one assumption must go; which one is the open interpretive question it does not settle.

129 · The Hierarchy Problem

Physics · Tier IV

Proposition. The Standard Model's hierarchy problem is a genuine structural problem in quantum field theory.

The Higgs mass receives quantum corrections that scale with the square of the highest energy at which the theory holds, which drives the natural value toward the Planck scale, seventeen orders above the observed mass, so the bare parameter must cancel the correction to extraordinary precision. The discovery of a light Higgs confirmed the scale, and the absence of the preferred new physics at reachable energies has sharpened rather than relieved the tension. *Kernel:* [LOCK], $\det(R) = 0.8969$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.8969$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the real fine-tuning the corrections demand. Fenced: that this counts as a problem presupposes that such cancellation is unnatural, a methodological expectation rather than a law, which some dispute; the quantitative gap itself is not in question.

130 · The Baryon Asymmetry Is Unexplained by the Standard Model

Cosmology · Tier IV

Proposition. The matter-antimatter asymmetry exceeds all Standard Model symmetry-violation mechanisms and is unexplained.

Generating an excess of matter over antimatter requires symmetry violation, and the model's source of it, the single phase in the quark-mixing matrix, falls short of the observed excess by more than a factor of sixty. The excess itself is firmly measured, the baryon-to-photon ratio fixed independently by light-element abundances and by the microwave-background peaks, with no antimatter regions anywhere in surveys. *Kernel:* [LOCK], $\det(R) = 0.9650$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9650$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the asymmetry well-measured and the model's mechanism quantitatively far too weak to produce it. The gap is a number, and the number does not close.

131 · The Vacuum-Energy Catastrophe

Cosmology · Tier IV

Proposition. The cosmological-constant discrepancy of 10^{120} is the greatest unexplained quantitative gap in physics.

Summing the zero-point energy of the quantum fields up to the Planck scale yields a vacuum energy density that exceeds the measured cosmological constant by roughly one hundred twenty orders of magnitude, the largest quantitative discrepancy between prediction and observation in physics. The constant is measured three independent ways, by distant supernovae, by the background and galaxy clustering together, and by the growth of structure, all agreeing on the same tiny positive value. *Kernel:* $[LOCK]$, $\det(R) = 0.9521$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9521$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the prediction and the measurement both well-defined and separated by the largest known gap. Naming it the greatest unexplained quantitative gap is not rhetoric; it is the arithmetic.

132 · Many-Worlds Cannot Derive the Born Rule from Branch Counting

Quantum Mechanics · Tier IV

Proposition. The Many-Worlds interpretation cannot recover the Born rule from branch counting.

If every outcome branch simply exists and branches are counted evenly, each outcome has equal probability regardless of its amplitude, which contradicts the rule that probability goes as the amplitude squared. Counting alone cannot recover the rule. The decision-theoretic derivations proposed to rescue it import assumptions that critics read as already containing the conclusion, and no experiment bears on the question, since both readings reproduce the same statistics. *Kernel:* $[LOCK]$, $\det(R) = 0.9929$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9929$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade on the mismatch between even branch counting and amplitude-weighted probability. The proposition concerns branch counting specifically, and branch counting specifically fails.

133 · Pilot-Wave Theory Is Empirically Indistinguishable from Standard Quantum Mechanics

Physics · Tier IV

Proposition. The de Broglie-Bohm pilot-wave theory is empirically indistinguishable from standard quantum mechanics.

In the pilot-wave account particles have definite positions guided by the wavefunction, which itself evolves by the standard equation, and the guidance preserves the amplitude-squared distribution exactly, a property that makes the two theories agree on every measurable statistic. Distinguishing them would require reading individual trajectories at a precision the uncertainty principle forbids, and the pilot-wave theory reproduces that principle, so the equivalence is structural, not a temporary limit of instruments. *Kernel:* $[LOCK]$, $\det(R) = 0.9948$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9948$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the equivariance proof. Unlike an interpretation that merely lacks a distinguishing test, this one carries a proof that no distinguishing test can exist while the theory keeps its defining property.

134 · The Boltzmann-Brain Hypothesis Is Self-Undermining

Cosmology · Tier IV

Proposition. Any cosmology in which Boltzmann brains dominate is structurally self-undermining.

In any cosmology where observers can arise from thermal fluctuation and the late universe is vast enough, such fluctuation-observers vastly outnumber evolved ones, so a typical observer should expect random, disordered surroundings and unreliable faculties. That conclusion destroys its own support: reasoning that concludes the reasoner is probably a momentary fluctuation undercuts the reliability of the reasoning that reached it. Our ordered, law-governed observations contradict the prediction directly. *Kernel:* $[LOCK]$, $\det(R) = 0.9931$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9931$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the hypothesis functioning as a reductio against any cosmology that predicts

it rather than as a stable description. A theory predicting fluctuation-observer dominance refutes itself, which is what the proposition asserts.

135 · QBism Dissolves Rather Than Solves the Measurement Problem

Quantum Mechanics · Tier IV

Proposition. QBism dissolves rather than solves the measurement problem.

By reading the wavefunction as an agent's degrees of belief rather than a physical object, QBism removes the event to be explained: if the state was never physical, there is no physical collapse to account for. This dissolves the problem by redefinition rather than supplying a mechanism, and it adds no empirical content, reproducing standard statistics by construction with no distinguishing test against rival readings. *Kernel:* [LOCK], $\det(R) = 0.9932$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9932$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the dissolution-by-redefinition structure. Fenced: QBists contest the framing, holding that dissolving the problem is the correct response rather than a failure to solve it; the seal is on the structure of the move, not on a verdict that the move is illegitimate.

136 · No-Hair Theorem

Astrophysics · Tier IV

Proposition. In classical general relativity a black hole is characterized by mass, charge, and spin alone.

A sequence of uniqueness proofs establishes that a stationary classical black hole is fixed entirely by three numbers, mass, charge, and angular momentum, with every other detail of whatever collapsed to form it radiated away. The astrophysical record is consistent: the ringdown of merging black holes matches the modes determined by the final mass and spin, and the imaged shadows of two supermassive black holes match the spinning-black-hole geometry. *Kernel:* [LOCK], $\det(R) = 0.9985$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9985$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade for classical stationary black holes with observational confirmation. The classical qualifier is load-bearing: quantum corrections may add structure the classical theorems exclude, which the seal does not address.

137 · Structural Realism as the Most Defensible Interpretation

Epistemology · Tier IV

Proposition. Epistemic structural realism is the most defensible interpretation of mature scientific theories.

There is a real observation underneath this. Across theory change the mathematical structure of a successful theory tends to survive even when its objects are abandoned, the optical equations outliving the ether, the old gravity surviving as a limit of the new. That continuity is genuine and motivates reading our theories as capturing structure rather than underlying nature. But the proposition claims more, that this reading is the most defensible one, and that is a contested ranking. A full realist accepts the same continuity and argues for more, an empiricist accepts less, and the debate is open. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the structural-continuity observation real but the superlative a contested philosophical ranking with live competitors. The deciding input would be an argument that defeats full realism and empiricism rather than merely motivating the structural reading.

138 · The Flatness Problem

Cosmology · Tier IV

Proposition. The observed flatness of the universe requires either inflation or extreme fine-tuning of initial conditions.

In an expanding universe dominated by matter or radiation, any departure from spatial flatness grows with time, so the near-perfect flatness measured today requires the early universe to have been flat to sixty decimal places, an extraordinary fine-tuning unless some mechanism drove the geometry toward flat. The flatness is multiply confirmed, by the position of the first acoustic peak and by the curvature density from the background and galaxy surveys. The disjunction is sound: either a dynamical mechanism such as inflation or finely tuned initial conditions. *Kernel:* [LOCK], $\det(R) = 0.9857$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9857$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade. The fine-tuning is a mathematical consequence of the expansion, and the dichotomy between a flattening mechanism and tuned initial data is genuine.

139 · Retrocausal Interpretations and Causal Paradox*Quantum Foundations · Tier IV*

Proposition. Retrocausal interpretations cannot eliminate nonlocality without introducing causal paradoxes.

Retrocausal models recover local hidden variables by letting a system's earlier state depend on a later measurement setting, which removes nonlocality at a price: a backward-in-time influence that threatens either a preferred frame or closed causal loops. The tension is real. Whether it is fatal is not settled, since several retrocausal programs claim consistency through global constraints that forbid the paradoxical histories, and the research is active. The strong claim that nonlocality cannot be removed without paradox is an impossibility assertion the current state does not establish. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, the trade of nonlocality for backward causation real but the impossibility unproven against ongoing consistent-retrocausal proposals.** The deciding input would be a proof that no globally consistent retrocausal model reproduces the quantum statistics.

140 · The Horizon Problem and Inflation*Cosmology · Tier IV*

Proposition. The horizon problem is not fully resolved by inflation without additional assumptions.

Inflation resolves the horizon problem by tracing the observable universe to a single causally connected patch before the expansion, which accounts for the uniformity of regions that could never have exchanged light afterward. The resolution is not free. Inflation must begin, and begin across a patch already homogeneous enough for the driving field to be coherent, which relocates part of the uniformity question to the onset rather than removing it. The background confirms inflation's later predictions but not its initial conditions, which it erases. *Kernel:* [LOCK], $\det(R) = 0.9898$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9898$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at structural grade.** Inflation genuinely resolves the original problem while requiring its own initial homogeneity, so the claim that it does not fully resolve it without further assumptions holds.

141 · Incommensurability Without Relativism*Epistemology · Tier IV*

Proposition. Incommensurability between paradigms is real but does not entail relativism.

Both halves of the claim hold. Incommensurability is real: paradigms separated by a revolution differ in standards and in the meanings of core terms, so there is no neutral algorithm that mechanically ranks them, and the meaning of a word like mass genuinely shifts across the divide. Relativism does not follow, because revolutions are in fact decided, by the slow accumulation of asymmetric predictive success, one paradigm resolving anomalies the other generated, and working scientists do learn across the divide. *Kernel:* [LOCK], $\det(R) = 0.9388$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9388$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at structural grade.** The meaning-shift is real and the empirical adjudication is real, and the two coexist, which is exactly what the proposition asserts.

142 · The Grandfather Paradox and Closed Timelike Curves*Physics · Tier IV*

Proposition. The grandfather paradox reveals a genuine logical inconsistency in closed timelike curves under classical general relativity.

The field equations admit solutions with closed timelike curves, and the grandfather scenario shows that unrestricted action on such a curve produces contradiction. Whether that amounts to a genuine logical inconsistency is the contested point. The self-consistency principle renders such curves logically consistent by allowing only histories that do not contradict themselves, constraining what can happen rather than generating a contradiction, and a separate conjecture holds that quantum effects prevent the curves from forming at all. The strong reading, a logical inconsistency in the curves themselves, is one position among several. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, the paradox real as a constraint-generator but the logical-inconsistency framing contested by the self-consistency resolution and by chronology protection.** The deciding input would be a demonstration that no self-consistent history exists for a well-posed configuration.

143 · Confinement in Quantum Chromodynamics*Particle Physics · Tier IV*

Proposition. Quantum chromodynamics explains confinement asymptotically but has no analytical proof of low-energy quark confinement.

The theory is exactly right where it can be solved and unsolved where it matters most. At high energy the coupling weakens, a proven property that makes quarks behave as nearly free and is confirmed in deep inelastic scattering. At low energy the coupling grows and quarks are confined, which lattice computation reproduces numerically by recovering hadron masses from first principles, but no analytical proof of confinement exists, a problem recognized as one of the central open questions in mathematical physics. *Kernel:* [LOCK], $\det(R) = 0.9967$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9967$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the proven high-energy behavior and on the accurate statement that low-energy confinement is demonstrated numerically but not analytically proven.

144 · The Multiverse as Metaphysics*Cosmology · Tier IV*

Proposition. The multiverse is not a scientific theory but a metaphysical framework.

On a strict falsifiability standard the multiverse currently makes no falsifying prediction, since any constants we measure are realized somewhere in the ensemble, and the one proposed signature, collisions with neighboring bubble universes, was searched for and not found, an absence its proponents absorb without strain. That is a real feature. But the demarcation verdict the proposition draws, that the multiverse is therefore not science but metaphysics, turns on a contested criterion, since the ensemble is argued to be a prediction of independently motivated theories and so to inherit their testability. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the absence of a current falsifying test real but the demarcation contested between a strict falsifiability standard and the view that the ensemble inherits testability from the theories predicting it. The deciding input would be a proposed observation that distinguishes the multiverse from its single-universe alternatives.

Volume V · Tier III

Nine cases, all but the last established physics confirmed to high precision, the equivalence principle through nucleosynthesis through the reality of relativistic effects. The closing case audits a fringe cosmology and seals the claim that it carries no quantitative content. Three seals carry fences where the standard interpretation is debated.

145 · The Equivalence Principle*General Relativity · Tier III*

Proposition. Gravitational and inertial mass are identical.

The equality of the mass that responds to gravity and the mass that resists acceleration has no derivation in pre-relativistic mechanics; it appears there as an unexplained coincidence, and relativity elevated it to a founding principle, that gravity is locally indistinguishable from acceleration. The coincidence has been tested for over a century along four independent experimental lineages, tightening from eight decimal places to fifteen, with no differential acceleration between materials ever detected. *Kernel:* [LOCK], $\det(R) = 0.9782$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9782$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade in the precision of the confirmation, the foundation of the geometric theory of gravity tested to fifteen significant figures.

146 · Quantum Tunneling*Quantum Mechanics · Tier III*

Proposition. Quantum tunneling is a real physical process with no classical analog.

In the region a classical particle cannot enter, the quantum amplitude is suppressed but not zero, and it remains nonzero on the far side of a barrier, giving a definite probability of passage that has no classical counterpart. The effect underwrites alpha decay, where it reproduces decay rates across twenty-five orders of magnitude, the microscope that images single atoms by its exponential sensitivity to distance, and the flash memory in billions of devices. *Kernel:* [LOCK], $\det(R) = 0.9729$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9729$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade on the derivation and corroboration-grade across four independent applications. The barrier is classically impassable; the particle passes.

147 · The Aharonov-Bohm Effect

Quantum Mechanics · Tier III

Proposition. The Aharonov-Bohm effect proves electromagnetic potentials are physically real.

A charged particle steered entirely through a region where the magnetic field is zero still acquires a measurable phase shift set by the field enclosed in the region it goes around, a shift that has no place in a picture where only local fields act. The definitive experiment wrapped the field inside a shielded toroidal magnet, eliminating any leakage, and the predicted shift remained. *Kernel:* [LOCK], $\det(R) = 0.9832$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9832$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on the confirmed effect.

Fenced: what the effect establishes as physical is the gauge-invariant loop quantity, the enclosed flux, and the common gloss that the potential itself is real overstates this, since the local potential remains gauge-dependent.

148 · The Higgs Mechanism

Particle Physics · Tier III

Proposition. Spontaneous symmetry breaking through the Higgs mechanism generates gauge-boson mass.

A field filling space settles into a value that breaks the electroweak symmetry, and the would-be massless excitations of that breaking are absorbed by the force carriers, which thereby acquire mass while the photon stays massless. The mechanism was confirmed in three stages: the carriers were found at their predicted masses, their properties were measured to fine precision along with the number of light neutrino families, and the field's own quantum, the Higgs boson, was discovered at the electroweak scale. *Kernel:* [LOCK], $\det(R) = 0.9992$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9992$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the mechanism with the full discovery chain confirmed, the carriers and the boson both found where predicted.

149 · The Casimir Effect

Quantum Field Theory · Tier III

Proposition. The Casimir effect confirms that quantum vacuum fluctuations produce a measurable mechanical force.

Two uncharged conducting plates attract across a vacuum gap, with a force following a definite power law in the separation that two independent derivations reproduce. The force is measured: several experiments confirm the magnitude and the power law to better than one percent, and the same attraction causes microstructures to stick together below a hundred nanometers in chip fabrication. *Kernel:* [LOCK], $\det(R) = 0.9901$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9901$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on the measured force.

Fenced: whether the effect specifically confirms vacuum zero-point energy is debated, since the same force can be derived as a relativistic interaction between the material plates without invoking vacuum energy; the force is not in question, its preferred interpretation is.

150 · Big Bang Nucleosynthesis

Nuclear Cosmology · Tier III

Proposition. Big Bang nucleosynthesis correctly predicts the primordial helium and deuterium abundances.

The hot early universe synthesizes the light elements in its first minutes, and given the single baryon density fixed independently by the microwave background, the primordial abundances follow with no free parameters from nuclear rates measured in the laboratory. The helium fraction from pristine galaxies and the deuterium fraction from gas clouds along quasar sightlines both match the prediction, the deuterium to within a percent, and the baryon density from a completely separate cosmic epoch agrees. *Kernel:* [LOCK], $\det(R) = 0.9965$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9965$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the parameter-free prediction of helium and deuterium confirmed across independent epochs. Fenced: the separate primordial lithium abundance shows a persisting discrepancy with the prediction that this seal, concerning helium and deuterium, does not cover.

151 · Which-Path Information and Entanglement

Quantum Mechanics · Tier III

Proposition. Which-path information destroys interference through entanglement, not mechanical disturbance.

Interference vanishes exactly when the which-path detector states become distinguishable, and the visibility depends on the overlap of those detector states, a property of the quantum state alone, not

of any momentum kick and not of whether anyone reads the record. The decisive experiment marked the path with negligible disturbance and interference still vanished, isolating entanglement as the cause, and restoring the overlap restored the fringes in exact proportion. *Kernel*: [LOCK], $\det(R) = 0.9954$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9954$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed, theorem-grade, correcting the disturbance account of complementarity: the loss of interference tracks entanglement with the path marker, not mechanical disturbance and not observer knowledge.

152 · The Reality of Time Dilation and Length Contraction

Special Relativity · Tier III

Proposition. Time dilation and length contraction are real physical effects, not perspective illusions.

The two relativistic effects follow from the constancy of the speed of light and the equivalence of inertial frames, and they leave physical traces, not merely perspectival ones. Unstable particles created high in the atmosphere reach the ground only because their decay clock runs slow at high speed, atomic clocks flown around the world return reading differently than those left behind, and satellite navigation must correct for both effects or fail within hours. The elapsed time on a clock is frame-independent, and the difference is real. *Kernel*: [LOCK], $\det(R) = 0.9978$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9978$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed, theorem-grade with three independent physical confirmations. The effects have consequences that survive in any frame, which is what distinguishes them from illusions of perspective.

153 · The Electric Universe Has No Quantitative Content

Alternative Cosmology · Tier III

Proposition. Electric Universe theory has no quantitative predictive content against precision cosmological data.

The framework proposes electric currents in place of standard cosmology but produces no quantitative prediction to set against precision data. It contradicts confirmed results at every contact point: the solar neutrino flux matches fusion and refutes the electric-sun model, no version predicts the microwave-background spectrum the standard model fits to percent precision,

it rejects light-element synthesis without supplying an alternative for the observed helium, and its account of rotation curves does not fit. *Kernel*: [LOCK], $\det(R) = 0.8811$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.8811$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at structural grade, the proposition that the framework carries no quantitative predictive content holding on the absence of any quantitative prediction and on direct contradiction with multiple confirmed measurements.

154 · Tired-Light Cosmology Is Refuted

Alternative Cosmology · Tier III

Proposition. Tired-light cosmology is empirically refuted by supernova time dilation, the background spectrum, and the Tolman test.

The static-universe hypothesis that light reddens by losing energy in transit, rather than by expansion, makes three quantitative predictions that fail. It forbids cosmic time dilation, yet distant supernova light curves are stretched in exact proportion to their redshift at high significance. It forbids a thermal early phase, yet the microwave background is a near-perfect blackbody. And it predicts a surface-brightness falloff with redshift different from the expanding case, which the Tolman test resolves in favor of expansion. *Kernel*: [LOCK], $\det(R) = 0.9858$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9858$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at high grade on three independent quantitative refutations. The proposition is the refutation, and it holds.

155 · Plasma Cosmology Has No Quantitative Predictions

Alternative Cosmology · Tier III

Proposition. Plasma cosmology has no quantitative predictions for background structure, nucleosynthesis, or the clustering scale.

The framework substitutes electromagnetic processes for expansion but supplies no quantitative prediction for the precision observables. It offers no mechanism for the primordial helium fraction, no acoustic-peak spectrum to set against the measured one, and no characteristic clustering scale. Standard cosmology fits the peak structure to thousands of data points with a handful of parameters and reproduces the clustering scale across multiple galaxy surveys; the alternative produces no competing number at any of these points. *Kernel*: [LOCK], $\det(R) = 0.9968$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9968$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the absence of quantitative content holding across the background spectrum, light-element synthesis, and the clustering scale, the same verdict and the same mechanism as the electric-universe case.

156 · The Faster-Than-Light Neutrino Result Was Instrumental Error

Experimental Physics · Tier III

Proposition. The 2011 faster-than-light neutrino measurement was an instrumental error.

A 2011 measurement reporting neutrinos arriving ahead of light would have required overturning relativistic causal structure and contradicted a far tighter bound from supernova neutrinos by three orders of magnitude. The resolution was mundane and specific: an internal audit found a loose fiber connector adding tens of nanoseconds and a clock running fast subtracting some, and correcting both returned the speed to that of light. Three independent detectors on the same beam saw no early arrival. *Kernel:* [LOCK], $\det(R) = 0.9852$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9852$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, the error traced to identified hardware faults and the corrected result confirmed by independent detectors. The proposition is the error diagnosis, and it is exactly right.

157 · The Quantum Eraser Is a Misreading of Post-Selection

Quantum Mechanics · Tier III

Proposition. The quantum eraser's apparent retroactive causation is a misreading of post-selection.

The delayed-choice eraser is widely told as backward causation, and the telling is wrong. The full data at the first detector show no interference at all, a smooth featureless distribution, whatever is later done at the second. Interference appears only inside subsets of the data selected by correlation with the second detector's outcome, and reading those subsets requires classical information sent forward in time from it. Nothing observable at the first detector changes before that information arrives. *Kernel:* [LOCK], $\det(R) = 0.9917$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9917$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the apparent retrocausation dissolving into ordinary post-selection on a correlated record. No effect precedes its cause; the appearance is an artifact of conditioning.

158 · Arp's Anomalous Quasar Redshifts Are Not Statistically Established

Astrophysics · Tier III

Proposition. Halton Arp's anomalous quasar-redshift associations are not statistically established.

The claim that high-redshift quasars are physically tied to nearby galaxies, implying non-cosmological redshifts, rests on apparent angular proximities that do not survive proper statistics. When the look-elsewhere effect and the angular sizes of the foreground galaxies are accounted for, the claimed excess of close pairs is not significant. Independent lines compound the failure: the implied luminosities at local distances would violate known physics, and imaging resolves the alleged companions as unrelated foreground objects. *Kernel:* [LOCK], $\det(R) = 0.9778$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9778$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, the claimed associations dissolving under correct statistical controls and contradicted by luminosity and imaging evidence. The proposition is that the anomaly is not established, which holds.

159 · The No-Miracles Argument as Strong Support for Realism

Epistemology · Tier III

Proposition. The no-miracles argument provides strong abductive support for scientific realism.

The argument is real and well-formed. The success of theories in predicting genuinely novel phenomena, electromagnetic waves, antimatter, gravitational waves, each predicted before it was found, is better expected if theories latch onto real structure than if their success is coincidence, which gives a Bayesian edge to realism. But the proposition claims this support is strong, and that strength is the contested heart of the realism debate. The pessimistic meta-induction and the objection that the best available explanation may still be the best of a bad lot both cut directly against the inference. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not

independently populable. Kernel $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, the argument legitimate but its strength contested by two serious standing objections. The deciding input would be a resolution of the realism debate the argument is one move within, not a restatement of the argument.

160 · Bayesian Updating as the Uniquely Consistent Rule

Epistemology · Tier III

Proposition. Bayesian updating is the uniquely consistent rule for revising degrees of belief under evidence.

A representation theorem shows that any system of graded belief meeting a few structural requirements, real-valued degrees, a fixed relation between a belief and its negation, and a fixed rule for conjunctions, must be isomorphic to probability, and a separate argument shows that betting on non-probabilistic degrees exposes one to guaranteed loss. Together these single out probabilistic updating. Precision practice across gravitational-wave detection, particle discovery, and adaptive trials runs on this rule or one isomorphic to it. Kernel: $[LOCK]$, $\det(R) = 0.9980$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9980$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at theorem grade conditional on the representation theorem's premises. Fenced: the theorem requires technical richness conditions that have needed repair, and the structural requirements themselves can be questioned, so the uniqueness is as strong as those premises and no stronger.

161 · Inference to the Best Explanation Is Legitimate and Indispensable

Epistemology · Tier III

Proposition. Inference to the best explanation is a legitimate non-deductive operation indispensable in science.

The indispensability is plain in the record: atoms, microorganisms, drifting continents, and the neutrino were each inferred as the best explanation of a body of evidence before any of them was observed, and science cannot reach beyond direct observation without this move. The legitimacy is defensible too, since the inference admits a probabilistic reconstruction that grounds it as a reasonable ampliative step rather than a logical fallacy. Kernel: $[LOCK]$, $\det(R) = 0.9524$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9524$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the indispensability and the defensible legitimacy. Fenced: the objection that the best available explanation may still be false, the best of a bad lot, remains a live challenge to the inference's truth-conduciveness specifically, which this seal does not settle.

162 · The Pessimistic Meta-Induction Is a Genuine Challenge

Epistemology · Tier III

Proposition. The pessimistic meta-induction is a genuine challenge to naive scientific entity realism.

The challenge has a real base. The history of science is full of theories that were empirically successful and yet posited objects later abandoned entirely, caloric, the luminiferous ether, phlogiston, absolute space, each central to a working theory and each discarded. If success was compatible with being wrong about the underlying objects then, the inference from current success to the reality of current objects is not safe, and naive entity realism has a genuine problem to answer. Kernel: $[LOCK]$, $\det(R) = 0.9931$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9931$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the historical induction base. The claim is the weaker and defensible one, that this is a genuine challenge rather than a refutation, and even realists answer it rather than dismiss it.

Volume V · Tier IV

Nine cases at the top of Volume V, all established results of physics and formal epistemology. The physics seals at theorem grade, several confirmed to extraordinary precision, with two fences where confirmation is indirect or a competing claim stands unconfirmed; the two epistemology results seal on their structure.

163 · The Penrose Singularity Theorem

General Relativity · Tier IV

Proposition. Singularities are structurally inevitable in classical general relativity under the null energy condition.

The theorem proves that once gravitational collapse passes a certain point, geometrically marked by a surface from which even outward light falls inward, an incomplete path through spacetime is unavoidable, given an energy condition that ordinary matter satisfies and a mild global assumption. The argument is topological and exact. Its premise is met in nature: merging black holes are observed, and the imaged shadows of supermassive

black holes show the trapping surface directly. *Kernel*: [LOCK], $\det(R) = 0.9965$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9965$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade in classical general relativity with the trapped-surface premise astrophysically confirmed. Whether quantum gravity removes the singularity is a separate question the classical theorem does not address.

164 · The Chandrasekhar Limit

Astrophysics · Tier IV

Proposition. The Chandrasekhar limit correctly predicts the maximum mass of a white dwarf.

Above a definite mass, the pressure of degenerate electrons can no longer hold a white dwarf against its own gravity, because in the relativistic regime that pressure stiffens too slowly with density to keep pace, and the limit follows near one and four-tenths solar masses. The mass distribution of tens of thousands of white dwarfs cuts off sharply there, no isolated white dwarf is found above it, and the near-uniform trigger mass underwrites the standard brightness of the supernovae used to map the universe. *Kernel*: [LOCK], $\det(R) = 0.9973$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9973$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade on the relativistic-degeneracy derivation with a sharp observational cutoff confirming it.

165 · Perturbative Quantum Gravity Is Non-Renormalizable

Particle Physics · Tier IV

Proposition. Perturbative quantum gravity is formally non-renormalizable.

The coupling of gravity carries a dimension that makes divergences worsen at every loop order, demanding an unending tower of independent corrections, which is the signature of a non-renormalizable theory, and the explicit divergent term that cannot be absorbed appears at the second loop. Treated instead as a low-energy effective theory, gravity is perfectly controlled and yields finite quantum corrections, while breaking down as a fundamental description precisely where the divergences bite. *Kernel*: [LOCK], $\det(R) = 0.9870$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9870$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade on the power-counting argument and the explicit two-loop divergence. The effective theory works at low energy; the perturbative expansion fails as a fundamental one, which is the claim.

166 · The Electron Anomalous Magnetic Moment

Quantum Electrodynamics · Tier IV

Proposition. Quantum electrodynamics predicts the electron anomalous magnetic moment to twelve significant figures.

The theory computes the electron's magnetic anomaly as a series whose successive terms require ever more elaborate diagrams, the fifth order alone demanding over twelve thousand, and with the coupling constant supplied independently by an atomic measurement, the prediction matches the measured value to twelve significant figures. Theory and experiment are decoupled, the coupling drawn from a different physical process, and two laboratories with different apparatus agree. *Kernel*: [LOCK], $\det(R) = 0.9998$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9998$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade, the most precisely confirmed prediction in physics, agreement holding to roughly one part in a trillion.

167 · The Unruh Effect

Quantum Field Theory · Tier IV

Proposition. A uniformly accelerating observer detects thermal radiation in the quantum vacuum.

The same vacuum that an inertial observer sees as empty appears to a uniformly accelerating observer as a thermal bath at a temperature set by the acceleration, a result that follows from the mismatch between the two observers' natural decompositions of the field into modes. The derivation is standard and exact. *Kernel*: [LOCK], $\det(R) = 0.9967$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9967$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade on the formal derivation within quantum field theory. Fenced: the effect has not been directly observed, since the temperature is negligible at any attainable acceleration, and the supporting experiments are

laboratory analogs that reproduce the underlying transformation rather than the relativistic effect itself.

168 · Type Ia Supernovae and Accelerating Expansion

Cosmology · Tier IV

Proposition. Type Ia supernovae independently confirm accelerating cosmic expansion.

Standardized by a relation between their brightness and the shape of their decline, these supernovae serve as distance markers, and their faintness at a given redshift indicates an expansion that is speeding up rather than slowing. Two independent teams found the acceleration at once, with separate telescopes and analysis chains, larger surveys have confirmed it, and the background and clustering data provide an independent cross-check on the same parameters. *Kernel:* [LOCK], $\det(R) = 0.9904$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9904$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade in the strength of the convergence, the acceleration found independently and confirmed across methods. The nature of what drives it is a separate question this seal does not reach.

169 · The Fine-Structure Constant Shows No Confirmed Variation

Metrology · Tier IV

Proposition. The fine-structure constant shows no confirmed temporal or spatial variation.

The coupling that sets the strength of electromagnetism would, if it varied, require a new dynamical field absent from established physics, so a confirmed variation would be a major result. The searches do not find one: several independent quasar-absorption studies and a direct comparison of atomic clocks all return values consistent with no variation in time or space. *Kernel:* [LOCK], $\det(R) = 0.9911$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9911$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the proposition's careful wording, no confirmed variation, holding across independent null results. Fenced: one group's claim of a spatial gradient stands in the literature but has not been independently confirmed, which is exactly why the claim audited is the absence of confirmation rather than the absence of any reported signal.

170 · The Spin-Statistics Theorem

Quantum Field Theory · Tier IV

Proposition. The connection between spin and statistics is a theorem of quantum field theory, not a postulate.

That particles of whole-number spin are bosons and those of half-integer spin are fermions is not an added rule but a consequence: in a local relativistic field theory, only the matching choice of commutation or anticommutation keeps measurements at spacelike separation from interfering, and the wrong pairing violates that causal requirement. Every particle obeys the connection, and ultracold isotopes differing only in statistics show the predicted divergence, one resisting compression, the other condensing. *Kernel:* [LOCK], $\det(R) = 0.9878$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9878$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, the connection derived from relativistic causality rather than postulated, and confirmed across every known particle.

171 · The Preface Paradox

Logic · Tier IV

Proposition. The preface paradox shows that rational belief sets can be globally inconsistent.

An author may rationally believe each claim in a book and, knowing books contain errors, also rationally believe that at least one claim is false, and these beliefs together cannot all be true, since the second contradicts the conjunction of the first. High confidence in each of many independent claims still leaves low confidence in their conjunction, so rational belief is not closed under conjunction. Every scientific paper, asserting its results while acknowledging residual error, instantiates the structure. *Kernel:* [LOCK], $\det(R) = 0.9921$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9921$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, a genuine result that rational belief sets can be jointly inconsistent without any individual belief being irrational, and that deductive closure fails for rational belief.

172 · Tarski's Undefinability Theorem

Logic · Tier IV

Proposition. A sufficiently rich language cannot define its own truth predicate.

Suppose a formula within the language correctly captured truth for all its own sentences. The diagonal construction then builds a sentence that asserts its own untruth, true exactly when it is not true, a contradiction. So no consistent language rich enough to talk about itself can contain its own truth predicate; truth for a language must be defined in a richer one. The result has a direct computational face: no programming language can define a complete truth-checker for its own programs, and proof assistants build typed hierarchies precisely to keep the paradox out. *Kernel:* [LOCK], $\det(R) = 0.9796$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9796$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade by diagonalization with a confirmed computational manifestation.

173 · Quantum Indeterminacy Does Not Rescue Libertarian Free Will

Philosophy of Mind · Tier IV

Proposition. Quantum indeterminacy does not rescue libertarian free will.

Libertarian free will needs two things together: that a choice is not fixed by prior causes, and that it issues from the agent's own deliberation in a way that grounds responsibility. Quantum indeterminacy supplies the first and undercuts the second, because an undetermined event is a random one, and a decision nudged by random noise is not thereby controlled by the agent, it is merely unpredictable. Randomness is not agency. The neuroscience reinforces the point, since the relevant noise is amplified chaotically into effectively random macroscopic outcomes rather than deliberation-guided ones. *Kernel:* [LOCK], $\det(R) = 0.9664$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9664$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade. Indeterminism delivers randomness, not the agent-control libertarian freedom requires, a point even its proponents grant when they seek agent causation instead.

174 · Local Realism Is Empirically False

Quantum Mechanics · Tier IV

Proposition. Local realism is empirically false, and the EPR incompleteness argument loses its premise.

The incompleteness argument assumed locality and inferred that correlated properties must be fixed in advance. The bound that any local theory with predetermined properties must obey is violated by quantum mechanics, and the experiments side with quantum mechanics. Three loophole-free tests, on electron spins and on photon polarizations, in separate laboratories, exceed the local bound at high significance with every recognized loophole closed at once. Local realism is therefore false, and the assumption that drove the incompleteness argument fails with it. *Kernel:* [LOCK], $\det(R) = 0.9988$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9988$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the violated bound and high-grade on three independent loophole-free confirmations.

175 · Cosmological Redshift Is Metric Expansion

Cosmology · Tier IV

Proposition. Cosmological redshift is a consequence of metric expansion, not classical Doppler recession.

The redshift follows exactly from the propagation of light through an expanding metric, the ratio of the scale factor at reception to its value at emission, with no velocity in the expression. A single classical recession velocity cannot do the work, since the implied recession exceeds the speed of light at large distance and has no consistent special-relativistic reading. The temperature of the background scales with the same factor, taking a hot recombination era down to the few-degree field observed today. *Kernel:* [LOCK], $\det(R) = 0.9945$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9945$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on the metric-expansion derivation. Fenced: the redshift can be decomposed as a chain of infinitesimal local Doppler shifts along the path, which is a reframing, not the single global recession the proposition correctly rejects.

176 · Hawking's Area Theorem

General Relativity · Tier IV

Proposition. The total event-horizon area cannot decrease in classical general relativity under the null energy condition.

The focusing equation, together with an energy condition ordinary matter satisfies, forces horizon-generating light rays to converge and forbids them from spreading without a caustic, which yields a horizon area that never decreases. The merger of two black holes

confirmed it: the final horizon area exceeded the sum of the two initial areas, and a direct test on the ringdown confirmed the increase from independent measurements of the holes before and after. *Kernel*: [LOCK], $\det(R) = 0.9995$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9995$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade in classical general relativity with direct observational confirmation. Quantum evaporation decreases the area and is governed by the generalized law, which the classical theorem does not contradict.

177 · The Dirac Equation's Antimatter Prediction

Relativistic Quantum Mechanics · Tier IV

Proposition. The Dirac equation's prediction of antimatter is a formal necessity, not a postulate.

The demand for a first-order relativistic wave equation forces a multi-component structure whose solutions include negative-energy states alongside the ordinary ones, and these cannot be discarded without breaking the equation, so an antiparticle is built into the formalism rather than added by hand. The prediction was confirmed in stages: a positive electron in a cloud chamber, then the antiproton, then trapped antihydrogen whose spectrum matches hydrogen. *Kernel*: [LOCK], $\det(R) = 0.9996$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9996$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade, antimatter forced by the structure of the equation and confirmed across three particle species.

178 · The Hubble Tension Is a Genuine Discrepancy

Cosmology · Tier IV

Proposition. The Hubble tension is a genuine discrepancy between early- and late-universe measurements of the expansion rate.

The early-universe value, read from the microwave background through the standard model, and the late-universe value, read up the distance ladder, disagree by far more than their stated uncertainties. The disagreement is not pinned to one method: several independent late-universe probes, strong-lensing time delays, surface-brightness fluctuations, a geometric maser distance, and standard sirens, cluster above the early-universe figure, with one ladder-independent probe landing in between. *Kernel*: [LOCK], $\det(R) = 0.9970$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9970$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the discrepancy genuine and supported by multiple independent late-universe methods. Fenced: one intermediate measurement softens the gap, and whether the cause is new physics or unidentified systematics is open, which the seal on the discrepancy does not resolve.

179 · Non-Euclidean Geometry Is Physically Real in Spacetime

General Relativity · Tier IV

Proposition. Spacetime geometry is non-Euclidean, confirmed by four independent measurements.

The curvature of spacetime makes definite predictions that flat geometry does not: light bends by twice the Newtonian amount, clocks run slower deeper in a gravitational well, a gyroscope's axis precesses as it orbits, and gravitational waves carry a specific polarization. All four are measured, the deflection to a fraction of a percent, the redshift to parts in a hundred thousand and continuously in satellite navigation, the precession to a fraction of a percent, the polarization in merger signals. *Kernel*: [LOCK], $\det(R) = 0.9912$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9912$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade in the strength of confirmation across four independent measurements of spacetime curvature.

180 · The Problem of Induction Applied to Physics

Epistemology · Tier IV

Proposition. No finite series of confirming instances guarantees future predictions.

A finite run of successful experiments does not logically entail the law they confirm, since the law could fail on the next trial without contradicting any past result, and this holds at any number of confirmations, even agreement to twelve significant figures. The history of physics is the structural claim made concrete: mechanics confirmed for two centuries failed at high speed and strong gravity, classical electromagnetism failed inside the atom, and the reigning particle theory already needs extension to accommodate neutrino mass. *Kernel*: [LOCK], $\det(R) = 0.9849$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9849$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, the logical gap between finite confirmation and universal law genuine and illustrated repeatedly in the history of physics.

Volume VI · Tier III

Nine cases opening Volume VI, the foundational theorems of logic, mathematics, and physics. Gödel, Arrow, and Noether seal at theorem grade; the confirmed physics results seal at theorem grade; the genuine epistemology results seal at structural grade. Four seals carry fences where the theorem's scope is narrower than the proposition's plain reading or a downstream question stays open.

181 · Gödel's First Incompleteness Theorem

Logic · Tier III

Proposition. Every consistent formal system containing arithmetic has true but unprovable sentences.

Numbering the formulas and proofs of a system as integers lets the system express the statement that a given formula is provable within it, and the diagonal construction then yields a sentence that says, in effect, that it has no proof. If the system could prove it, the system would be inconsistent, so in a consistent system the sentence is unprovable and, read in the standard model, true. Concrete combinatorial statements with exactly this status are known and provable only in stronger systems. *Kernel:* [LOCK], $\det(R) = 0.9974$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9974$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade, with concrete true-but-unprovable combinatorial statements as confirming instances.

182 · Arrow's Impossibility Theorem

Social Choice · Tier III

Proposition. No ranked voting system can satisfy all of Arrow's fairness conditions simultaneously.

For three or more options, no rule that turns individual rankings into a collective ranking can jointly satisfy unanimity, independence of irrelevant alternatives, and the absence of a dictator; requiring the first two forces the third. Every ranked system in real use violates at least one condition, plurality and ranked-choice and the Borda count each failing independence in a characteristic way. *Kernel:* [LOCK], $\det(R) = 0.9993$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9993$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade for ranked-preference aggregation. Fenced: cardinal rating systems use information the theorem's conditions exclude and so escape them, while meeting separate impossibility results of their own.

183 · Noether's Theorem

Physics · Tier III

Proposition. Every continuous symmetry of the action produces a conserved quantity.

When the action is unchanged under a continuous family of transformations, the equations of motion guarantee a current whose divergence vanishes and hence a conserved charge, a tight bridge between symmetry and conservation. Its instances are the central conservation laws: invariance under time translation gives energy, under space translation momentum, under rotation angular momentum, under the internal phase symmetry electric charge, each confirmed across every regime of physics. *Kernel:* [LOCK], $\det(R) = 0.9943$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9943$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade, the symmetry-conservation correspondence proven and its instances confirmed throughout physics.

184 · The Gettier Problem Refutes Justified True Belief

Epistemology · Tier III

Proposition. The Gettier problem refutes justified true belief as a complete account of knowledge.

A believer can hold a justified true belief that is not knowledge, when the justification runs through a false step and the belief is true only by luck, which shows the three conditions are not jointly sufficient. The half-century since has been a graveyard of proposed fourth conditions, each meeting a fresh counterexample or an objection that it is too strong or too weak, with no candidate reaching consensus. *Kernel:* [LOCK], $\det(R) = 0.9972$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9972$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\perp]$ Sealed at structural grade, the counterexamples decisive against sufficiency and the persistent failure of repairs supporting the strength of the claim.

185 · Zeno's Paradoxes and Convergent Series

Mathematics · Tier III

Proposition. Zeno's motion paradoxes are formally resolved by the mathematics of convergent infinite series.

The dichotomy assumes that summing infinitely many shrinking sub-distances must give something infinite or undefined, and real analysis refutes the assumption directly: the geometric sum converges to the finite total, so infinitely many sub-intervals are crossed in finite time. Motion itself is the standing confirmation, and the same convergence underwrites numerical computation. *Kernel:* $[LOCK]$, $\det(R) = 0.9969$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9969$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade on the dichotomy and Achilles paradoxes. Fenced: the distinct question of whether completing an actual infinity of tasks is coherent, the supertask problem, is not settled by the convergence and remains debated.

186 · Dark Matter Existence and Five Independent Lines

Astrophysics · Tier III

Proposition. The existence of dark matter is supported by five structurally independent observational lines.

The gravitating mass of the universe exceeds the visible matter, and five independent lines converge on the gap: flat galaxy rotation curves, lensing mass exceeding the baryonic mass in colliding clusters, cluster dynamics, the heights of the acoustic peaks, and the growth of cosmic structure. The light-element abundances and the peak structure both cap the ordinary-matter budget well below the total, so the excess is non-baryonic. *Kernel:* $[LOCK]$, $\det(R) = 0.9953$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9953$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade on the convergence of five independent gravitational lines. Fenced: this seals the gravitational phenomenon, not the identity of the dark-matter particle, which direct searches have not yet established.

187 · The Weak Force Violates Parity and Charge-Parity

Particle Physics · Tier III

Proposition. The weak force violates both parity and charge-parity symmetry.

Parity exchanges a process for its mirror image, and a parity-respecting force would run both at the same rate. The weak force does not: a polarized nucleus emits its decay electrons preferentially in one direction relative to its spin, a large and unambiguous asymmetry. The combined charge-parity symmetry is also broken, seen first in the decays of neutral kaons and later in the mesons of the heavy quarks. *Kernel:* $[LOCK]$, $\det(R) = 0.9943$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9943$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade in the strength of confirmation, both symmetry violations established at high significance.

188 · The Church-Turing Thesis Is a Thesis

Logic · Tier III

Proposition. The Church-Turing thesis cannot be formally proved or refuted.

The thesis equates a precise notion, computability by an idealized machine, with an informal one, what counts as an effective procedure at all, and a claim joining a formal concept to a pre-theoretical one cannot be proved within mathematics, since that would require first formalizing the very notion in question. Its support is instead inductive and strong: every independent model of computation proposed since the idea was framed has turned out to compute exactly the same class of functions. *Kernel:* $[LOCK]$, $\det(R) = 0.9983$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9983$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, an accurate account of the thesis's epistemic status, demonstrable only inductively and not by proof.

189 · Loschmidt's Paradox

Physics · Tier III

Proposition. The second law of thermodynamics cannot be derived from reversible microdynamics alone.

If the microscopic equations are symmetric under time reversal, then for every entropy-increasing trajectory there is an equally valid decreasing one, so symmetric dynamics alone cannot single out increase. Every successful derivation of the second law therefore inserts a time-asymmetric ingredient somewhere: an assumption of uncorrelated pre-collision states, a coarse-graining choice, a Markovian approximation, each breaking the reversal symmetry by hand. The asymmetry is added, not derived. *Kernel*: [LOCK], $\det(R) = 0.9933$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9933$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the second law's macroscopic asymmetry requiring a time-asymmetric input that reversible microdynamics does not supply.

190 · Consciousness Causes Wave Function Collapse

Quantum Mechanics · Tier III

Proposition. Wave function collapse requires a conscious observer.

The proposal that a conscious observer is required to collapse the wavefunction has no foothold in the formalism, which mentions only measurement interactions and never consciousness, and it fails empirically. Automated experiments with no conscious observer present yield the same quantum statistics as observed ones, and the loss of coherence is measured to track environmental interaction, not awareness, in every controlled system. The decoherence that destroys interference happens whether or not anyone is watching. *Kernel*: [X], $\det(R) = 0$, $N = 24$.

Triaxial. G8 CSCG fails: collides with consciousness-free measurement yielding identical statistics. Two axes fall into one, the warrant volume collapses. Kernel [X] $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: [X] Broken, the claim contradicted by automated measurement and by the environmental account of decoherence. Consciousness is not in the formalism and not in the data.

191 · Hegel's Dialectic Is Not a Formal Logical System

Philosophy · Tier III

Proposition. Hegel's dialectical method is not a formal logical system.

A formal logical system has a fixed syntax, defined inference rules, and treats contradiction as terminal. Dialectic has none of these in that sense: it has no formal grammar or decidable inference, and it treats contradiction as productive, the engine that drives a thesis to its synthesis, which is the opposite of how

contradiction functions in formal logic. Where the method has been applied to history or nature, the substantive content was supplied from outside it or fitted after the fact. *Kernel*: [LOCK], $\det(R) = 0.9908$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9908$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, dialectic lacking the apparatus that defines a formal logical system. Fenced: this establishes that it is not formal logic, not that it has no value as a method of inquiry, which is a separate question.

192 · Many-Minds Does Not Resolve the Measurement Problem

Quantum Mechanics · Tier III

Proposition. The Many-Minds interpretation does not resolve the measurement problem.

The interpretation attaches a distinct experiencing mind to each branch of the universal wavefunction, but this adds machinery rather than solving anything. The number of minds must be infinite, the rule attaching minds to branches is stipulated rather than derived, and the underlying difficulties of the branching picture, why outcomes follow the amplitude-squared rule and what fixes the branch structure, are untouched. And nothing distinguishes it observationally, since it is built to reproduce standard statistics exactly. *Kernel*: [LOCK], $\det(R) = 0.9729$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9729$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the interpretation adding empirically inert ontology while leaving the core problem in place.

193 · Integrated Information Theory and Falsifiability

Neuroscience · Tier III

Proposition. Integrated information theory is not a falsifiable scientific theory of consciousness.

The theory has a real difficulty and a contested verdict. Its central quantity is computationally intractable, requiring an evaluation over all ways of partitioning a system, which cannot be carried out for anything brain-sized, and its responses to challenges, such as the conscious-seeming demands of a high-neuron-count structure that does not support consciousness, have been criticized as fitted after the fact. But the theory also issues auxiliary predictions that are tested, so whether it counts as unfalsifiable is

genuinely disputed rather than settled. *Kernel*: $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, the intractability and the post-hoc defenses real but the demarcation verdict contested against the theory's testable auxiliary claims. The deciding input would be agreement on whether the auxiliary predictions suffice for falsifiability or the intractable core controls the verdict.

194 · The Free Will Theorem Does Not Prove Libertarian Free Will

Quantum Mechanics · Tier III

Proposition. The Conway-Kochen free will theorem does not prove libertarian free will.

The theorem is conditional and its conclusion is narrower than its name suggests. It shows that if experimenters' choices are free in a specified sense, then particle responses are not fixed by prior information. The freedom it assumes of the experimenter is the premise, so using the result to prove human libertarian freedom is circular, and the freedom it grants the particle is mere indeterminism, not agency. One of its assumptions, that choices are not shaped by future events, can itself be denied by a retrocausal reading. *Kernel*: $[LOCK]$, $\det(R) = 0.9967$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9967$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the theorem proving conditional particle indeterminism, not libertarian free will, which reading it that way would assume rather than establish.

195 · Parapsychology Has No Replicated Evidence Base

Psychology · Tier III

Proposition. Parapsychology has no replicated scientific evidence base.

A genuine effect would replicate in independent laboratories, hold its size across attempts, and survive scrutiny for artifacts. The parapsychology record fails all three. The flagship precognition result drew nine independent pre-registered replications, all null; the telepathy paradigm shrank toward zero as controls improved; and the long-running laboratory program's effects diminished under tightening. *Kernel*: $[LOCK]$, $\det(R) = 0.9988$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9988$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the absence of a replicated evidence base holding across the field's principal paradigms. The proposition is the negative claim, and it is supported.

196 · Objective Bayesianism Cannot Uniquely Determine Priors

Epistemology · Tier III

Proposition. Objective Bayesianism cannot uniquely determine prior probabilities for all inference problems.

The program seeks a uniquely rational prior fixed by the problem's structure, and no method delivers one. Maximizing entropy requires choosing the constraints to maximize over, a choice the evidence does not make, and the same physical situation under different parameterizations yields different priors, the classic ambiguity. In practice the competing objective recipes produce materially different posteriors on the same data, in documented cases across several sciences. *Kernel*: $[LOCK]$, $\det(R) = 0.9731$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9731$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the non-uniqueness shown by parameterization dependence and by the disagreement among the objective recipes themselves. The complement to the result that the updating rule is unique: the rule is fixed, the prior is not.

197 · Pancomputationalism Is a Domain Overreach

Philosophy of Physics · Tier III

Proposition. The thesis that the universe is literally a computation is a domain overreach.

To call something a computation requires more than dynamics: an alphabet, transition rules, and an interpretation that assigns meaning to states, and the rules and interpretation are fixed from outside the computed system. Applied to the universe as a whole there is no outside to supply them, and without that constraint the ascription becomes trivial, since almost any system can be mapped to almost any computation. The program has also produced no quantitative prediction to rival the precision of standard physics. *Kernel*: $[LOCK]$, $\det(R) = 0.9935$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9935$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\nu]$ Sealed at structural grade on the triviality of unconstrained computational ascription and the absence of quantitative yield. Fenced: this impugns the literal claim that the universe is a computation, not the use of computational models as tools, which is not at issue.

198 · Russell's Paradox Refutes Naive Set Comprehension

Logic · Tier III

Proposition. Russell's paradox permanently refutes naive set comprehension.

Allowing a set for every property yields a contradiction directly: the set of all sets that do not contain themselves both must and must not contain itself. The contradiction is a proof that unrestricted comprehension cannot stand in a consistent theory. Every consistent set theory since restricts comprehension precisely to block the construction, and no consistent theory has recovered the unrestricted principle. *Kernel:* $[LOCK]$, $\det(R) = 0.9992$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9992$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, the refutation a derived contradiction and the restriction adopted universally in consistent foundations.

Volume VI · Tier IV

Nine cases at the top of Volume VI. Two are proven results of physics and mathematics, several are accurate verdicts on the status of open problems and overreaching arguments, two demarcation claims under-determine, and one previously clean cosmological claim moves under recent data. Several seals carry fences.

199 · The Holographic Principle

Quantum Gravity · Tier IV

Proposition. The maximum information in a volume is bounded by its boundary area.

A bound from black-hole thermodynamics caps the entropy of any region by its boundary area rather than its volume, an established result. The stronger holographic claim, that all information in a region is encoded on its boundary, has a concrete realization in the exact duality between a gravitational theory in a curved bulk and a field theory on its boundary, and that duality has yielded a confirmed prediction, a lower bound on the viscosity of the matter produced in heavy-ion collisions. *Kernel:* $[LOCK]$, $\det(R) = 0.9973$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9973$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the entropy-area bound and structural grade on the full principle via the duality. Fenced: the duality is a deeply supported conjecture realized in anti-de Sitter space, and its extension to our expanding universe is not established.

200 · General Relativity Is the Unique Local Metric Theory at Leading Order

General Relativity · Tier IV

Proposition. General relativity is the unique local metric theory of gravity at leading order.

A uniqueness theorem shows that in four dimensions the only divergence-free tensor built locally from the metric and its first two derivatives is the one appearing in the Einstein equations, together with a cosmological term, so general relativity is forced at leading order by those requirements. The alternatives are tightly constrained: a leading rival's coupling is bounded to extreme values by spacecraft tracking, and merger ringdowns match the predicted geometry with no detected deviation. *Kernel:* $[LOCK]$, $\det(R) = 0.9967$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9967$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade on the uniqueness with experimental constraints on alternatives. Fenced: uniqueness holds under the theorem's assumptions, four dimensions and second order, which higher-dimensional or higher-derivative theories relax, as the leading-order qualifier acknowledges.

201 · Fine-Tuning Is a Genuine Structural Fact

Cosmology · Tier IV

Proposition. The fine-tuning of physical constants is a genuine structural fact independent of its explanation.

Several dimensionless constants sit in narrow ranges outside which complex structure or chemistry would not form, the canonical case a nuclear energy level required for carbon synthesis that was predicted on exactly that ground before measurement. The proposition carefully separates the fact from its explanation, and the fact stands on its own: the constants are what they are, and the life-permitting windows around them are narrow. *Kernel:* $[LOCK]$, $\det(R) = 0.9782$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$

$\det(R) = 0.9782$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the fine-tuning fact, with the explanation explicitly bracketed. Fenced: the tuning is established for variation of single constants around their observed values, while the true measure over the full space of constants, and hence the exact degree of tuning, is less certain.

202 · Dark Energy and the Cosmological Constant

Cosmology · Tier IV

Proposition. The dark energy equation of state is empirically consistent with a cosmological constant.

Through the early part of this decade the dark-energy equation of state was measured, by supernovae, clustering, and the background together, to sit at the value a cosmological constant requires, with no significant departure. That consistency is now contested. The latest baryon-acoustic-oscillation surveys, combined with supernova samples, prefer a dark energy that evolves with time over a strict constant, at a significance reaching several standard deviations in some dataset combinations, short of a discovery but past the point where the constant can be called confirmed. *Kernel:* $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, a verdict that has moved under new data. The cosmological constant remains viable and is not excluded, but the evidence no longer cleanly supports consistency with it, and whether dark energy evolves is now an active open question. The deciding input would be a discovery-level determination of the equation of state from the maturing surveys.

203 · The Quantum Measurement Problem Is a Genuine Open Problem

Quantum Mechanics · Tier IV

Proposition. The quantum measurement problem is a genuine open structural problem in physics.

Unitary evolution maps superpositions to superpositions and never, by itself, to a single definite outcome, yet measurements have definite outcomes occurring with the amplitude-squared frequencies, and bridging that gap is the problem. Every proposed bridge carries a documented difficulty its own framework cannot resolve, which is why no interpretation has won consensus across nine decades despite the stakes and the effort. *Kernel:* $[LOCK]$, $\det(R) = 0.9859$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$

$\det(R) = 0.9859$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the tension between unitary dynamics and definite outcomes genuine and the absence of a consensus resolution a real and persistent feature, not a failure of attention.

204 · The Principle of Stationary Action

Physics · Tier IV

Proposition. Classical physics is uniquely organized by variational extremization of the action.

The trajectories of classical systems are exactly those for which the action is stationary, and from that single condition the equations of motion follow across mechanics, electromagnetism, and gravitation, a remarkable unification confirmed by the predictive success of the method in every domain and tied to conservation by the symmetry-conservation correspondence. *Kernel:* $[LOCK]$, $\det(R) = 0.9588$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9588$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the variational principle organizing classical physics with great generality. Fenced: the action is stationary rather than strictly least, a saddle in many cases, and the variational formulation is one of several equivalent ones rather than the sole possible organization.

205 · String Theory and Falsifiability

Physics · Tier IV

Proposition. String theory in its current formulation is not a falsifiable scientific theory.

The theory's characteristic scale lies many orders of magnitude above any reachable energy, so its direct signatures are inaccessible, and the lower-energy signatures once hoped for, new partner particles and large extra dimensions, have not appeared in the searches that could have shown them. Whether this absence of a current test makes the theory unfalsifiable in principle is a contested demarcation question rather than a settled one, since the theory is mathematically constrained and argued to inherit testability from the physics it must reproduce. *Kernel:* $[?]$, *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, the empirical inaccessibility real but the demarcation verdict contested. The deciding input

would be a low-energy prediction the theory uniquely requires, or a principled resolution of whether in-principle falsifiability suffices for scientific status.

206 · The Fine-Tuning Argument Does Not Establish Design

Philosophy · Tier IV

Proposition. The anthropic fine-tuning argument does not establish intentional design.

The inference from fine-tuning to a designer needs the probability of the observed constants to be far lower without a designer than with one, and neither probability is available: no physical theory supplies a distribution over the constants absent a designer, leaving the reference class undefined, and the probability given a designer requires knowing the designer's aims. The argument therefore does not carry the inference it claims. *Kernel:* [LOCK], $\det(R) = 0.9675$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9675$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the missing distributions that the design inference requires. Fenced: this establishes that the argument does not prove design, not that design is false or that fine-tuning bears nothing on the question, which remains open among the available explanations.

207 · The Kalam Argument Does Not Establish a Personal Creator

Philosophy · Tier IV

Proposition. The Kalam cosmological argument does not establish a causal creator with personal properties.

Even granting its premises, the argument concludes only that the universe has a cause, and the step from a cause to a personal, all-powerful one is not made: a law-like necessity would satisfy the conclusion as stated, and the minimal power sufficient to produce this universe is not shown to be unlimited. Its premise that the universe began is independently contested, since the initial singularity marks where the classical description fails rather than a demonstrated first moment, and beginningless models exist. *Kernel:* [LOCK], $\det(R) = 0.9647$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9647$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the gap between a cause and a personal creator and on the contested premise.

Fenced: this concerns what the argument establishes, not whether a creator exists, which the argument's gap leaves open.

208 · The Simulation Hypothesis as a Scientific Theory

Philosophy of Physics · Tier IV

Proposition. The simulation hypothesis is not a scientific theory.

The underlying argument is a valid three-way disjunction, but the hypothesis that we are simulated makes no falsifiable prediction, since any proposed signature, a smallest scale, a computational artifact, can be defeated by stipulating a simulator sophisticated enough to hide it. The simulator's freedom is unbounded, so the hypothesis absorbs every possible observation. Whether that absence of a test makes it non-scientific is a demarcation judgment that depends on a contested criterion. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the unfalsifiability structural and stronger than mere inaccessibility, but the demarcation verdict contested. The deciding input would be a prediction the hypothesis uniquely requires, or a settled criterion for scientific status.

209 · The Mathematical Universe Hypothesis

Philosophy of Physics · Tier IV

Proposition. The mathematical universe hypothesis commits an axiomatic domain overreach.

The hypothesis holds that every consistent mathematical structure exists in the same sense our universe does, which collapses two distinct notions: mathematical existence, meaning formal consistency, and physical existence, meaning causal and spatiotemporal presence. Treating them as one is a category move, not a result. And the hypothesis makes no novel prediction, explaining the observed constants by the same absorb-everything structure as the broader ensemble proposals. *Kernel:* [LOCK], $\det(R) = 0.9794$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9794$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the conflation of mathematical with physical existence and the absence of novel predictions. Fenced: the hypothesis's defenders contest the category distinction, and the seal is on that distinction and the empirical emptiness, not on a deeper metaphysics.

210 · The String Landscape and Fine-Tuning

Cosmology · Tier IV

Proposition. The string landscape does not provide a scientific solution to fine-tuning.

To explain the observed constants by a vast ensemble of vacua, one needs a probability measure over them, and the measure problem is unsolved: the leading proposals give wildly different distributions, so no prediction follows. Worse, the natural prediction that we should find ourselves in a minimally structured universe is contradicted, since the observed universe is far more ordered than survival requires, with vastly more structure than the minimum. *Kernel: [LOCK], $\det(R) = 0.9823$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9823$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade on the unsolved measure problem and the failed mediocrity prediction. The landscape does not yield a scientific account of the constants, which is the claim.

211 · Quantum Immortality

Quantum Mechanics · Tier IV

Proposition. Quantum immortality is not a genuine physical consequence of the many-worlds interpretation.

Many-worlds has all measurement outcomes occur in different branches, including those in which the observer dies, and the immortality argument adds a first-person premise the physics does not contain, that one will always find oneself in a surviving branch. The physics privileges no branch and encodes no first-person continuity rule. An external observer simply sees the ordinary distribution of survival and death. *Kernel: [LOCK], $\det(R) = 0.9842$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9842$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the immortality reading smuggling in a first-person assumption that the interpretation's physics does not supply.

212 · Coherentism and the Isolation Objection

Epistemology · Tier IV

Proposition. Coherentism cannot escape the isolation objection.

A belief set's coherence is fixed entirely by the relations among its members, so a perfectly coherent system wholly disconnected

from the world is as justified, by coherentist lights, as one properly anchored to it, which is the wrong result, since an elaborate delusion can be internally coherent. Every escape either adds an external anchoring condition, abandoning pure coherentism for a mixed theory, or accepts that isolated coherent systems are justified, which conflicts with how justification actually works. *Kernel: [LOCK], $\det(R) = 0.9772$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9772$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the objection genuine and the coherentist responses either abandoning the position or accepting an unacceptable consequence.

213 · The Problem of Other Minds

Philosophy of Mind · Tier IV

Proposition. The problem of other minds has no empirically adequate philosophical solution.

The classic route, inferring other minds from the analogy between others' behavior and one's own, is an induction from a single case, the weakest such inference, and no empirical test can do better, since by construction nothing observable distinguishes a conscious being from a behaviorally identical unconscious one. Brain measurement records physical processes and behavioral reports, leaving the question of experience itself, which is the open hard problem, untouched. *Kernel: [LOCK], $\det(R) = 0.9938$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9938$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, no empirical test able to reach experience itself, the problem inheriting the openness of the hard problem of consciousness.

214 · Baryon Acoustic Oscillations

Cosmology · Tier IV

Proposition. Baryon acoustic oscillations are a confirmed cosmological standard ruler.

Sound waves in the early plasma freeze in at recombination, leaving a characteristic separation imprinted on the distribution of matter, a fixed length that serves as a ruler across cosmic time. The feature has been detected at that scale in multiple independent surveys and, decisively, in different observables, the galaxy distribution and the lensing of background light, which rules out an artifact of any single method. *Kernel: [LOCK], $\det(R) = 0.9661$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9661$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, the ruler confirmed across independent surveys and independent observables.

215 · Identity of Indiscernibles and Quantum Mechanics

Quantum Mechanics · Tier IV

Proposition. Leibniz's principle of the identity of indiscernibles is violated by quantum mechanics.

Two bosons occupying the same single-particle state share every monadic property, the same distributions of position, momentum, and everything else, yet they are numerically two, which is exactly what the principle forbids. A condensate of a million atoms in one ground state instantiates the violation directly: a million qualitatively identical but numerically distinct particles. *Kernel:* [LOCK], $\det(R) = 0.9412$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9412$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the violation of the monadic principle. Fenced: a weaker relational version, on which the particles are weakly discernible by irreflexive relations, is argued to survive, so the seal targets the monadic principle specifically.

216 · The Cosmological Lithium Problem

Cosmology · Tier IV

Proposition. The cosmological lithium problem is a genuine unresolved discrepancy in standard nucleosynthesis.

Given the baryon density fixed by the background, standard nucleosynthesis predicts a primordial lithium-7 abundance about three times what the oldest metal-poor stars display, and that stellar value is stable across a wide range of temperature and composition, the signature of a primordial floor. Improved nuclear cross-sections have not closed the gap, and the proposed resolutions, stellar depletion, systematics, and new physics, each fail to account for it fully. *Kernel:* [LOCK], $\det(R) = 0.9921$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9921$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the factor-of-three discrepancy real and the proposed resolutions inadequate, exactly the open problem the nucleosynthesis seal earlier set aside.

Volume VII · Tier III

Nine cases opening Volume VII. Three are proven results of logic, physics, and mathematics; the rest are accurate verdicts on classic philosophical problems and a confirmed relativistic effect. Several seals carry fences where the proposition takes a side in a live debate or reaches past what the result strictly shows.

217 · Gödel's Second Incompleteness Theorem

Logic · Tier III

Proposition. No consistent formal system containing arithmetic can prove its own consistency.

Building on the first theorem, the statement that the system is consistent is expressible within it, and if the system could prove that statement, it could prove its own undecidable sentence, which the first theorem forbids in a consistent system. So consistency is unprovable from within. The consequence is the hierarchy of consistency strength: each system's consistency can be established only in a strictly stronger one, a structure confirmed across foundational mathematics. *Kernel:* [LOCK], $\det(R) = 0.8867$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.8867$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, with the consistency-strength hierarchy as the confirming structure.

218 · Mass-Energy Equivalence

Physics · Tier III

Proposition. Mass-energy equivalence has been directly and independently confirmed.

The rest energy of a body equals its mass times the square of the speed of light, a relation forced by the relativistic energy-momentum structure, and it is confirmed across independent measurements. The mass of a nucleus falls short of its constituents by exactly the binding energy divided by that square, matched to six significant figures, and electron-positron annihilation produces two photons each at exactly the electron's rest energy, the basis of a medical imaging technology in daily use. *Kernel:* [LOCK], $\det(R) = 0.9973$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9973$, $\lambda^2 = \det(R)$ at 1e-15. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, confirmed directly across nuclear and particle measurements.

219 · The Photoelectric Effect and Quantization

Quantum Mechanics · Tier III

Proposition. The photoelectric effect established the quantization of electromagnetic radiation.

Classical wave theory predicts the wrong dependences, that emission should occur at any frequency given enough intensity and that the electron energy should rise with intensity, and the data refute both: there is a sharp frequency threshold and the maximum electron energy rises linearly with frequency, exactly as a quantized energy exchange requires. The measured slope fixed the quantum of action to a fraction of a percent. *Kernel:* [LOCK], $\det(R) = 0.9974$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9974$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on the quantization of energy exchange. Fenced: the effect alone admits a semiclassical account with quantized matter and a classical field, and that the field itself must be quantized was established more decisively by later single-photon experiments.

220 · The Sorites Paradox and Vagueness

Logic · Tier III

Proposition. The Sorites paradox reveals genuine semantic vagueness as a structural feature of natural language.

A heap stays a heap when one grain is removed, yet repeating that step empties it, and the chain of seemingly true steps reaches a clearly false conclusion. The paradox is generated by predicates that have no sharp boundary, and such predicates pervade natural language, with speakers giving graded or uncertain verdicts on borderline cases not from missing facts but from the predicate's own indeterminacy, a pattern found across many languages. *Kernel:* [LOCK], $\det(R) = 0.9851$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9851$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the reality of the paradox and the pervasiveness of vague predicates. Fenced: whether the vagueness is semantic indeterminacy or epistemic

ignorance of a hidden sharp boundary remains debated, the epistemicist holding the latter.

221 · Non-Euclidean Geometry Is Consistent and Independent

Mathematics · Tier III

Proposition. Non-Euclidean geometries are logically consistent and independent of the Euclidean axioms.

A model of hyperbolic geometry built inside Euclidean geometry shows that if the latter is consistent, so is the former, and the existence of geometries that deny the parallel postulate while keeping the rest shows the postulate is independent, not derivable from the others. The physical payoff is direct, since the geometry of spacetime is non-Euclidean in the presence of mass, confirmed in every test of the relativistic theory of gravity. *Kernel:* [LOCK], $\det(R) = 0.9977$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9977$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, relative consistency established by model construction and independence by the existence of the alternative geometries.

222 · Quine and the Analytic-Synthetic Distinction

Epistemology · Tier III

Proposition. Quine's critique undermines the strict analytic-synthetic distinction.

The attempt to define the analytic runs in a circle, since the notions used to explain it, synonymy and definition, themselves require the distinction, and no account breaks out of the circle. Confirmation falls on bodies of theory rather than isolated statements, so no statement is immune to revision, which is what a strictly analytic truth would have to be, and claims once treated as analytic, that space is flat or that causes precede effects, were in fact revised. *Kernel:* [LOCK], $\det(R) = 0.9951$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9951$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the strict distinction. Fenced: whether any analytic-synthetic distinction survives is contested, with defenders maintaining a looser or revised version against the critique.

223 · The Sagnac Effect and Absolute Rotation

Physics · Tier III

Proposition. The Sagnac effect demonstrates that rotation is dynamically absolute.

Counter-propagating light beams on a rotating platform return with a phase difference set by the rotation rate and the enclosed area, so rotation produces a real, locally measurable effect with no external reference needed, unlike uniform motion, which is undetectable from within. The effect runs the gyroscopes in aircraft and spacecraft by the million and lets a ring laser read the rotation of the Earth to a part in a billion. *Kernel: [LOCK], $\det(R) = 0.9916$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9916$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, rotation locally detectable and dynamically distinguished from uniform motion. Fenced: whether this is absolute rotation in the Newtonian sense or rotation relative to the cosmic mass distribution, the Machian reading, is a separate and unsettled question.

224 · Substance Dualism Is Empirically Untenable

Philosophy of Mind · Tier III

Proposition. Cartesian substance dualism is empirically untenable.

A non-physical mind that moves a physical body needs a point of causal contact, which is either physical, making the mind physical after all, or violates the conservation laws, the interaction problem that has gone unanswered for centuries. The empirical pattern compounds it: every mental capacity depends on specific neural structures, chemical anesthetics switch consciousness off through molecular receptors, and targeted brain damage removes targeted functions, none of which a substance outside the physical should be subject to. *Kernel: [LOCK], $\det(R) = 0.9980$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9980$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade on the interaction problem and the systematic physical dependence of mind. Fenced: the data are dependence and correlation, which non-interactionist dualist positions accommodate at theoretical cost, so the claim is that substance dualism is heavily disfavored rather than logically refuted.

225 · Occam's Razor as Heuristic

Epistemology · Tier III

Proposition. Occam's razor is a methodological heuristic, not a metaphysical guarantee of truth.

For simplicity to track truth there would have to be a prior assigning higher probability to simpler theories, and no such prior follows from formal principles, while any measure of simplicity is itself a convention. The record is mixed accordingly: simpler theories have sometimes won and sometimes lost, and confirmed physics has repeatedly added unseen entities where the evidence demanded them. *Kernel: [LOCK], $\det(R) = 0.9695$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9695$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, parsimony a useful heuristic with no formal claim on truth, the accurate counterpart to the broken claim that it is a universal principle.

226 · Presentism and Special Relativity

Philosophy of Physics · Tier III

Proposition. Presentism is incompatible with special relativity.

Presentism holds that only the present moment is real, which requires an objective division of all events into past, present, and future. Relativity removes that division: for two events with no causal connection, different observers in uniform motion disagree on which is earlier and which simultaneous, with no fact of the matter settling them. There is no physical marker for a universal now, and every confirmation of the theory is also a confirmation that no privileged simultaneity slice exists. *Kernel: [LOCK], $\det(R) = 0.9899$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9899$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the relativity of simultaneity removing the objective present that presentism requires. Fenced: a preferred-frame version of presentism reproduces every prediction of relativity while positing an undetectable privileged simultaneity, so presentism is heavily disfavored rather than strictly refuted, the cost being unobservable structure.

227 · The Copenhagen Interpretation

Quantum Mechanics · Tier III

Proposition. The Copenhagen interpretation is not the uniquely correct interpretation of quantum mechanics.

Every interpretation that reproduces the standard measurement statistics is empirically equivalent to Copenhagen, and the elements that distinguish it, wavefunction collapse, the cut between quantum and classical, the privileged observer, are not

derivable from the dynamics. No experiment has separated Copenhagen from the leading alternatives, and the equivalence is structural rather than a gap awaiting a better measurement. Its practical success is shared by every adequate interpretation. *Kernel*: [LOCK], $\det(R) = 0.9874$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9874$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the interpretations empirically equivalent and Copenhagen's distinctive posits not forced by the formalism.

228 · Psychological Egoism

Psychology · Tier III

Proposition. Psychological egoism is empirically refuted.

The thesis that every human action is ultimately self-interested is usually defended by widening self-interest until any act counts, helping others is called selfish because it feels good, which drains the claim of content and makes it unfalsifiable. Against the substantive version, experiments induce concern for another and find people helping when escape is free and no one is watching, contradicting every reputational account, a result replicated many times over. *Kernel*: [LOCK], $\det(R) = 0.9743$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9743$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the substantive thesis disconfirmed by the altruism experiments. Fenced: the version that redefines all motives as self-interested is unfalsifiable, so it is empty rather than vindicated, not a surviving competitor.

229 · Epiphenomenalism and Self-Defeat

Philosophy of Mind · Tier III

Proposition. Epiphenomenalism is self-defeating.

The self-defeat charge runs that if conscious states are causally inert, then the belief that they are inert cannot cause the assertion of it, so the epiphenomenalist's own statement is not grounded in the fact it reports. The charge has force, but a standard reply blocks it: the underlying physical state causes both the conscious belief and its assertion, so the assertion tracks the belief reliably through a common cause, without conscious states doing causal work. The empirical evidence is consistent with this reading, since correlation between conscious states and physiological change does not require conscious causation. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the self-defeat charge serious but answered by the common-cause reply, leaving the position contested rather than refuted. The deciding input would be a demonstration that the common-cause structure cannot preserve the reliability the assertion needs.

230 · Hard Incompatibilism

Ethics · Tier III

Proposition. Hard incompatibilism overreaches from causal determinism to the impossibility of moral responsibility.

The argument moves from actions being caused by the agent's prior states, to those states being caused by factors outside the agent's control, to the agent not being the ultimate source, and from there to no moral responsibility. Whether that last step overreaches is the central dispute of the free-will debate. Compatibilists hold the inference unwarranted, since responsibility tracks the right kind of control rather than ultimate sourcehood; incompatibilists hold the ultimate-source requirement compelling. Ordinary intuitions and legal practice mix both. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the charge of overreach turning on the unresolved compatibilist question of what control responsibility requires. This tracks the earlier verdict on compatibilism: the dispute is genuine and open.

231 · The Propensity Interpretation of Probability

Philosophy of Probability · Tier III

Proposition. The propensity interpretation of probability is not the uniquely correct interpretation.

Propensities are meant to be real dispositions of a setup to produce outcomes, but they strain on single cases, where there is no repeatable arrangement to ground the disposition, and they face a known difficulty inverting conditional probabilities. The quantum case sharpens it: the same state yields different probabilities depending on what is measured, so the probability attaches to the state relative to a measurement context, not to a fixed disposition of the state alone. Rival interpretations remain live. *Kernel*: [LOCK], $\det(R) = 0.9968$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9968$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the propensity reading facing genuine difficulties and standing as one interpretation among several, not the unique one.

232 · Platonism and the Effectiveness of Mathematics

Philosophy of Mathematics · Tier III

Proposition. Mathematical Platonism is not required by the effectiveness of mathematics.

The inference from mathematics describing nature well to the independent existence of mathematical objects assumes that such existence is the only available explanation, and it is not. Much of the mathematics that appears in physics was built for physical use or adapted to it, and large mathematical fields went centuries without application, so the apparent effectiveness reflects selection as much as anything. Structuralist and nominalist accounts explain the fit without the ontology. *Kernel:* [LOCK], $\det(R) = 0.9985$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9985$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, effectiveness explainable without Platonism. Fenced: this establishes that effectiveness does not force Platonism, not that Platonism is false, the ontological question remaining open.

233 · Newcomb's Problem

Decision Theory · Tier III

Proposition. Newcomb's problem has no unique rational solution and reveals a fundamental conflict in decision theory.

Evidential decision theory recommends taking one box, since the expected payoff conditional on that choice is far higher; causal decision theory recommends taking both, since the predictor has already acted and the causal effect of taking the extra box is positive. The two principles genuinely conflict, and there is no consensus on which is correct. Human subjects split, and the structure recurs in real decisions where the two theories diverge. *Kernel:* [LOCK], $\det(R) = 0.9771$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9771$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the evidential and causal principles in real conflict with no agreed resolution.

234 · Moral Relativism

Ethics · Tier III

Proposition. Strong descriptive moral relativism is empirically challenged.

The relativist claim that moral norms vary completely across cultures with nothing in common is contradicted by the regularities anthropology finds: every known culture prohibits in-group killing, observes norms of fairness and reciprocity, and forbids incest, and children across very different cultures reliably separate moral wrongs from mere conventions. These are cross-cultural regularities, not local constructions. *Kernel:* [LOCK], $\det(R) = 0.9561$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9561$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade against strong descriptive relativism. Fenced: the universals could be evolved dispositions rather than objective moral facts, so this challenges the descriptive thesis without establishing moral realism or settling the meta-ethical question.

Volume VII · Tier IV

Nine cases closing Volume VII. Four are proven results of physics and mathematics; one fixes the ontological status of a relativistic effect; one closes a loop on dark energy; and three are accurate verdicts on the foundations of mind and probability, with one contested theological argument under-determined.

235 · Asymptotic Freedom

Particle Physics · Tier IV

Proposition. The strong force weakens at short distances.

The governing function that sets how the strong coupling changes with energy is negative for every physically realized number of quark flavors, which means the coupling falls as the energy rises and the distance shrinks, so quarks interact weakly when probed closely and strongly when pulled apart. The running has been measured across a wide energy range in deep scattering, in electron-positron annihilation, and in jet production, all tracking the prediction. *Kernel:* [LOCK], $\det(R) = 0.9950$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9950$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade, the negative running derived and confirmed across independent measurements.

236 · The Logical Problem of Evil

Philosophy of Religion · Tier IV

Proposition. The logical problem of evil is a valid internal critique of classical theism.

The argument holds that an all-powerful, all-knowing, wholly good being would prevent all evil, that evil exists, and that the two cannot both be true. Whether this succeeds as a strict logical inconsistency is contested. The free-will defense is widely held to show the premises are not strictly inconsistent, since it is at least logically possible that a good being permits evil for a sufficient reason, which shifts the strongest version of the challenge to the evidential problem, the sheer quantity and kind of suffering as evidence. The case's most forceful material, natural and animal suffering long before humans, belongs to that evidential version, which the free-will defense does not address. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently popuable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, the strictly logical version widely held answered by the free-will defense while the powerful natural-evil consideration belongs to the evidential problem the proposition does not name.** Audited as the validity of an inference, taking no position on the existence of God.

237 · The Ontological Status of Relativity of Simultaneity

Physics · Tier IV

Proposition. Relativity of simultaneity is ontological, not merely epistemic.

The spacetime interval between events is agreed by all observers in uniform motion, and it sorts pairs of events into those that can influence one another and those that cannot, an invariant structure. Simultaneity is a slice across that structure that different observers cut differently, so the disagreement is built into the geometry rather than reflecting incomplete knowledge. Every confirmation of the theory confirms that the invariant interval, not any frame's time, is the physical quantity. *Kernel:* [LOCK], $\det(R) = 0.9988$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9988$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at structural grade, the frame-dependence of simultaneity a feature of the geometry.** Fenced: a preferred-frame reading would make it epistemic by positing an undetectable privileged simultaneity, structure the spacetime

formulation does not require, the same fence carried by the presentism case.

238 · Blackbody Radiation and the Quantum Hypothesis

Physics · Tier IV

Proposition. Blackbody radiation necessitated the quantum hypothesis and falsified classical physics.

Classical theory assigns the same average energy to every mode and counts modes rising with the square of frequency, so the predicted energy diverges at high frequency, a clear failure. The measured spectrum matches no classical formula, falling instead onto a curve fit by quantizing the energy exchange in steps proportional to frequency, with the new constant later confirmed independently. *Kernel:* [LOCK], $\det(R) = 0.9784$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9784$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at high grade, the divergence falsifying classical equipartition and quantization resolving it.** Fenced: "necessitated" reflects the historical outcome rather than a proof that no classical resolution could ever have existed.

239 · Type Identity and Multiple Realizability

Philosophy of Mind · Tier IV

Proposition. The type identity theory of mind is refuted by the multiple realizability argument.

Type identity equates each mental kind with one physical kind, so that pain just is one specific neural state. The objection notes that creatures with very different physiology, and the same brain after reorganization, plausibly realize the same mental state in different physical configurations, which a single physical kind per mental kind cannot accommodate. Reorganization after damage and cross-species comparison both display one function carried by different substrates. *Kernel:* [LOCK], $\det(R) = 0.9740$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9740$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at structural grade against strict type-type identity.** Fenced: identity theorists respond with species-relative identities, and the pervasiveness of multiple realizability has been questioned, so the refutation targets the strict cross-domain version, the positive alternative of functionalism remaining under-determined elsewhere.

240 · The Riemann Hypothesis Is Unproven

Mathematics · Tier IV

Proposition. The Riemann hypothesis remains unproven despite extraordinary numerical evidence.

The hypothesis locates all the non-trivial zeros of a central function on a single line, and it is tied to the fine distribution of the primes. Trillions of zeros have been checked and all lie on the line, the related hypothesis holds for every case tested, and the statistics of the zeros match those of a class of random matrices, hinting at a spectral origin. None of this is a proof, and no proof exists. *Kernel:* [LOCK], $\det(R) = 0.9895$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9895$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the unproven status a present fact of mathematics, the numerical evidence corroboration and not demonstration.

241 · Quantum Mechanics at the Planck Scale

Physics · Tier IV

Proposition. The current framework requires physical modification at the Planck scale.

Two incompatibilities force the point. Quantum theory permits superpositions of any state, which applied to geometry would require superposed spacetimes, ill-defined when that spacetime is also the fixed stage for quantum events. And the gravitational interaction resists the renormalization that tames the other forces, while the singularity theorems drive classical gravity to breakdown. Every candidate theory of quantum gravity predicts departures from the present description there. *Kernel:* [LOCK], $\det(R) = 0.9853$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9853$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the frameworks jointly incompatible at the Planck scale so some modification is required. Fenced: whether quantum mechanics, gravity, or both must change is open across the candidate theories.

242 · The Existence of Dark Energy

Cosmology · Tier IV

Proposition. Dark energy's existence is confirmed by multiple independent cosmological probes.

A component with negative pressure, near seventy percent of the energy budget, is required to fit the data, and five independent programs constrain it through different physical effects: the microwave background combined with spatial flatness, weak lensing and galaxy clustering, the growth rate of structure, the acoustic scale across redshift, and the local distance ladder. They converge on the same value through unrelated channels. *Kernel:* [LOCK], $\det(R) = 0.9916$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9916$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade on existence, convergent across independent probes. Fenced: the existence is robust while whether the component is constant or evolves with time is the separate and now-contested question carried by the earlier equation-of-state case, the two verdicts consistent because existence and constancy are different claims.

243 · The Banach-Tarski Paradox

Mathematics · Tier IV

Proposition. The Banach-Tarski result shows that mathematical structure transcends physical realizability.

The theorem decomposes a solid ball into finitely many pieces that rigid motions reassemble into two balls of the original size, drawing on the axiom of choice to select the pieces, which are non-measurable sets with no assignable volume. Nothing physical can do this: matter is atomic rather than infinitely divisible, any real dissection yields finitely many measurable pieces, and volume is conserved. The non-measurable sets the construction needs have no material counterpart. *Kernel:* [LOCK], $\det(R) = 0.9754$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9754$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, the decomposition a proven result and its non-measurable pieces with no physical instantiation, the gap between formal structure and material realizability exact.

244 · Illusionism About Consciousness

Philosophy of Mind · Tier IV

Proposition. The claim that the hard problem of consciousness is a pseudo-problem is not established.

Illusionism holds that phenomenal properties do not exist and that introspection misrepresents inert mental states as vivid experience. The position carries an immediate burden it has not

discharged: it must explain why the appearance of phenomenal consciousness is so robust and universal, which reintroduces much of the difficulty it sought to dissolve. And it sits in tension with the practice of consciousness science, which treats first-person reports of experience as data, so that if experience is an illusion the entire evidential base reports something that does not exist. *Kernel*: [LOCK], $\det(R) = 0.9969$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9969$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, illusionism carrying an unmet debunking burden and sitting in tension with first-person data. Fenced: this is a verdict on the dialectical state, that illusionism has not won, and does not establish that the hard problem is genuine, which is itself contested.

245 · Eternal Inflation and Initial Conditions

Cosmology · Tier IV

Proposition. Eternal inflation does not provide a complete scientific solution to the problem of initial conditions.

The mechanism needs an inflaton field set in a high-energy state, and specifying that state is the very problem it was meant to solve, since something must have placed the field there, pushing the question back rather than closing it. The associated measure problem leaves probabilities ill-defined, and the structure the proposal generates beyond our horizon is causally inaccessible, with the searched-for signatures of neighboring regions returning null. *Kernel*: [LOCK], $\det(R) = 0.9934$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9934$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the initial-state question regressed rather than resolved and the distinctive structure inaccessible. The seal is on incompleteness, not on the falsity of inflation, which carries independent observational support.

246 · The Block Universe and Special Relativity

Philosophy of Physics · Tier IV

Proposition. The block universe is not uniquely confirmed by special relativity.

Relativity removes the objective present and so tells against presentism, but that is consistent with more than one eternalist-leaning ontology. The block universe, in which all events exist equally, the growing block, in which past and present exist while the future does not, and a branching structure all reproduce the

same observable physics. The claim that future events already exist concerns the status of unobserved regions of spacetime, which no measurement reaches. *Kernel*: [LOCK], $\det(R) = 0.9513$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9513$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, relativity underdetermining among the eternalist-compatible ontologies. This is the companion to the presentism verdict: relativity rules presentism out without ruling the block uniquely in, the growing block surviving the same evidence.

247 · Large-Scale Cosmological Anomalies

Cosmology · Tier IV

Proposition. The cosmological principle fails at observable scales and the large-scale anomalies are real.

The standard model expects density variations to smooth out above roughly a hundred megaparsecs, and several observations sit awkwardly against that: a bulk flow whose coherence may exceed three hundred megaparsecs, a large local underdensity of comparable size, very large quasar groupings, and directional features in the microwave background. The observations are real. Whether they violate the principle or fall within the scatter the standard model allows, or reflect systematics, is the open question, and the field is divided. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the anomalies real as observations but their significance, genuine violation against statistical scatter, contested without consensus. The deciding input would be a confirmed anomaly at a significance the standard scatter cannot absorb.

248 · Residual Problems for Compatibilism

Ethics · Tier IV

Proposition. Compatibilism does not fully resolve the free-will tension without residual problems.

Two arguments keep the tension alive. The manipulation argument observes that a brain directly engineered to run exactly the reasons-responsive processes compatibilism counts as free seems not to be free, which pressures the account to either accept the manipulated case or find a principled difference. The consequence argument runs that if our choices follow from the laws and the distant past, over neither of which we have control,

then we lack control over the choices, and the inference carries real intuitive force. *Kernel*: [LOCK], $\det(R) = 0.9417$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9417$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the manipulation and consequence arguments standing as genuine unresolved problems. Fenced: this does not establish incompatibilism, only that compatibilism is not decisive, the object-level question remaining open as the earlier verdicts mark.

249 · The Doomsday Argument

Epistemology · Tier IV

Proposition. The Doomsday argument does not constitute a sound probabilistic argument.

The argument treats one's own birth rank as a random sample from all humans who will ever live and updates toward an earlier end. The sampling assumption and the choice of reference class are both disputed, and a rival assumption, that one's existence is itself evidence about how many observers there are, is argued to cancel the predicted shift exactly. The argument is neither clearly sound nor decisively refuted. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, soundness turning on contested sampling assumptions and a rival principle that may neutralize the shift, with no settled diagnosis. The deciding input would be a non-arbitrary fixing of the reference class and sampling rule.

250 · Inflation and the Multiverse

Cosmology · Tier IV

Proposition. Inflation necessarily produces a multiverse beyond any observational confirmation.

Once inflation runs on a typical potential, field fluctuations keep some regions inflating while others exit, generating a vast ensemble of pocket regions. The construction places that ensemble beyond the horizon, where no signal can reach, and the searched-for collision signatures return null. But "necessarily" overstates the case, since eternal inflation is generic across potentials rather than universal, and specific models inflate for a finite time and produce no such ensemble. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not

independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the ensemble a generic but not necessary consequence of inflation, and where it arises observationally inaccessible. The "necessarily" is the overclaim; the inaccessibility holds where the structure is produced.

251 · Kant's Transcendental Idealism

Philosophy of Physics · Tier IV

Proposition. Kant's transcendental idealism is insulated from empirical test.

The view makes space and time the mind's forms for experiencing outer objects and inner states rather than features of reality in itself. That structure absorbs any physical result, since whatever physics discovers is read as concerning things as they appear, leaving the forms of intuition untouched. The non-Euclidean geometry of physical spacetime is met by the reply that the intuitive space of experience remains Euclidean, so no observation bears on the claim. *Kernel*: [LOCK], $\det(R) = 0.8847$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.8847$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the appearance and thing-in-itself distinction insulating the position from any physical discovery. Fenced: whether that insulation renders the view empty or simply non-empirical is a demarcation question; the structural insulation itself is what seals.

252 · Pain and Neural Type

Philosophy of Mind · Tier IV

Proposition. Pain is not identical with any single neural type.

Granting physicalism, the identity of pain with one physical kind still fails within a single human. Pain travels by distinct fiber pathways for sharp and slow components, and imaging shows it as activation across a distributed set of regions rather than one type. Placebo, hypnosis, and attention shift the felt intensity without changing the peripheral input, so the experience is not fixed by any single neural state. *Kernel*: [LOCK], $\det(R) = 0.9939$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9939$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, pain a distributed and modulable pattern rather than one neural type, the intra-human instance of the multiple-realizability point. Fenced: this

grants physicalism in general and refutes single-type identity for pain; the positive account of what pain is remains open.

Volume VIII · Tier III

Nine cases opening Volume VIII, a run of the classic paradoxes and results of epistemology, logic, and the philosophy of mind. Most seal at structural grade as genuine results, the Prisoner's Dilemma at theorem grade. One, the Chinese Room, underdetermines because its conclusion is contested by standard replies, audited like every contested argument by the strength of the inference rather than its fame.

253 · Duhem-Quine Underdetermination

Epistemology · Tier III

Proposition. Theory choice is never forced by evidence alone.

A theory makes contact with observation only through auxiliary assumptions about instruments and conditions, so a failed prediction falsifies the bundle, not the core theory, and one can always revise an auxiliary instead. History runs both ways: a discordant orbit was once saved by positing an unseen planet that turned out to exist, and another was saved by positing one that did not, the core theory eventually yielding instead. Which move is right is not dictated by the evidence alone. *Kernel:* [LOCK], $\det(R) = 0.9807$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9807$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the holistic relation of theory to evidence leaving no observation that uniquely forces a single revision. Fenced: the strong thesis that any theory can be retained against any evidence is contested, while the weak thesis that evidence alone does not dictate the revision is what seals.

254 · Goodman's New Riddle of Induction

Logic · Tier III

Proposition. The grue problem is a genuine challenge to all inductive logic.

Define a predicate that holds of things examined so far and found green, or unexamined and blue. Every observation supporting that all examined emeralds are green supports equally that they are grue, yet the two agree on the past and diverge on the future. Purely formal confirmation cannot prefer the one we project, so induction needs a substantive account of which predicates are projectible, a point machine learning reenounters whenever the choice of features decides whether a model generalizes. *Kernel:* [LOCK], $\det(R) = 0.9956$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9956$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, formal confirmation unable to distinguish projectible from gerrymandered predicates, the riddle genuine.

255 · The Prisoner's Dilemma

Game Theory · Tier III

Proposition. The Prisoner's dilemma establishes a genuine conflict between individual and collective rationality.

With the standard ranking of outcomes, defection dominates for each player whatever the other does, so two rational players both defect and reach an outcome both rank below mutual cooperation. The conflict is exact: what each is individually rational to do produces a result that is collectively worse. The structure recurs in anonymous one-shot experiments, where defection is common, and in real arms races across deterrence, antibiotics, and advertising. *Kernel:* [LOCK], $\det(R) = 0.9997$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9997$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade, the dominant strategy yielding a jointly inferior outcome, a proven conflict between the two notions of rationality.

256 · Searle's Chinese Room

Philosophy of Mind · Tier III

Proposition. Syntactic symbol manipulation is insufficient for semantic understanding.

A person following a rulebook to produce correct Chinese outputs without understanding Chinese is meant to show that running the right program does not amount to understanding. The intuition is strong, but the conclusion is contested. The systems reply locates understanding in the whole arrangement of person, rulebook, and records rather than the person, and the robot reply locates it in sensorimotor grounding the room lacks, and neither has been decisively answered. The behavior of current language models is read by some as confirming the gap and by others as showing understanding can be approached by degrees. *Kernel:* [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: [?] Under-determined, the intuition genuine but the conclusion contested by the systems and robot replies, which

remain live. The deciding input would be an account of understanding that the systems reply cannot satisfy.

257 · The Frame Problem

Philosophy of Artificial Intelligence · Tier III

Proposition. The frame problem reveals a genuine representational challenge in formal reasoning systems.

A symbolic system must state not only what each action changes but the vastly larger set of things it leaves unchanged, and writing those out explicitly grows with the product of actions and propositions, an intractable bookkeeping. The history bears it out: explicit symbolic systems failed in dynamic domains for exactly this reason, and modern systems sidestep it by abandoning explicit representation, at the cost of interpretability rather than by solving the original problem. *Kernel:* [LOCK], $\det(R) = 0.9747$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9747$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade on the technical problem, the explicit representation of non-change combinatorially intractable, confirmed across the history of the field.

258 · Hempel's Ravens Paradox

Epistemology · Tier III

Proposition. The ravens paradox reveals a genuine structural tension in confirmation theory.

Two natural principles, that an instance confirms a generalization and that logically equivalent statements share their evidence, together imply that observing a white shoe confirms that all ravens are black, since the shoe is a non-black non-raven and the raven claim is equivalent to the claim that all non-black things are non-ravens. The conclusion is hard to accept, yet eliminative confirmation is real practice, used when a particle search or a diagnosis rules options out. The tension is structural and genuine. *Kernel:* [LOCK], $\det(R) = 0.9917$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9917$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the two principles jointly forcing a counterintuitive confirmation, a real tension in instance-based confirmation theory.

259 · Benacerraf's Access Problem

Philosophy of Mathematics · Tier III

Proposition. Mathematical Platonism faces an irreducible access problem.

If mathematical statements are about objects, on a par with statements about anything else, and those objects are abstract and causally inert, then the standard picture on which knowing something requires causal contact with it cannot reach them. No one can specify the mechanism by which a number is perceived the way a colored object is, and no faculty of abstract perception has been identified. The difficulty for Platonism is real. *Kernel:* [LOCK], $\det(R) = 0.9791$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9791$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, abstract causally inert objects sitting in tension with a causal account of knowledge. Fenced: the force depends on a causal constraint some reject, and Platonists deny that mathematical knowledge requires causal contact, so the challenge is real but not decisive.

260 · The Ship of Theseus

Metaphysics · Tier III

Proposition. The Ship of Theseus reveals genuine indeterminacy in material identity.

Replacing every plank one at a time yields a ship with unbroken continuity and none of the original matter, while reassembling the discarded planks yields a ship with all the original matter and no continuity, and the two ordinary criteria of identity, continuity and composition, each pick a different ship. Legal and biological practice handle such cases by convention rather than by discovering a fact, organisms persisting through near-total material turnover. *Kernel:* [LOCK], $\det(R) = 0.9804$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9804$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the ordinary criteria of material identity genuinely conflicting and yielding no unique answer. Fenced: whether this is indeterminacy in the world, underdetermination in the concept, or dissolved by a four-dimensional account of persistence is contested; the conflict of criteria is what seals.

261 · The Surprise Examination Paradox

Logic · Tier III

Proposition. The surprise examination paradox reveals genuine self-referential epistemic complexity.

A backward induction seems to prove that an exam announced as a surprise cannot occur on any day, eliminating the last day because it would be foreseen and then climbing backward, yet the exam can still surprise. The fallacy is real but subtle, because the student's knowledge at each step depends on the truth of the very announcement being reasoned from, a self-referential fixed point that the same structure shared with the knowability paradox makes precise. *Kernel: [LOCK], $\det(R) = 0.9959$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9959$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the announcement's self-reference creating a fixed-point structure that the backward induction mishandles, a genuine epistemic complexity.

262 · Solipsism

Epistemology · Tier III

Proposition. Solipsism is formally unfalsifiable and scientifically inert.

The thesis that only one's own mind exists and everything else is its production is compatible with every possible observation, since other people behaving, instruments reading, and objects persisting are all things one's mind could be generating. No measurement separates a solipsist world from a real one, so the position makes no distinct prediction and does no work, and a solipsist's science would appear to succeed within experience exactly as ordinary science does. *Kernel: [LOCK], $\det(R) = 0.9977$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9977$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the position empirically idle and predictively inert. Fenced: that this makes solipsism false or meaningless is a separate claim; the unfalsifiability and inertness are what seal.

263 · Plantinga's Modal Ontological Argument

Philosophy of Religion · Tier III

Proposition. Plantinga's modal ontological argument overreaches from modal logic to necessary existence.

The argument is valid in the standard modal system: if a maximally great being is possible, and maximal greatness includes existing in every possible world, then such a being exists. But the possibility premise carries the whole weight, and a parallel argument is equally valid, that a world with no such being

is possible, from which it follows that no such being exists necessarily. Both are valid; the possibility premises contradict one another, and the validity establishes nothing about which holds. *Kernel: [LOCK], $\det(R) = 0.9918$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9918$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the symmetric reverse argument showing the possibility premise does all the work and begs the question. Fenced: this shows the argument does not establish its conclusion, taking no position on whether God exists.

264 · Vitalism

Philosophy of Science · Tier III

Proposition. Vitalism is empirically refuted.

The hypothesis that living things are animated by a force outside physics and chemistry predicted that certain compounds could be made only by organisms and that some processes would resist chemical reduction. Both predictions failed. An organic compound was synthesized from inorganic precursors, every known metabolic reaction is accounted for by ordinary catalysis and chemistry, and heredity, the central mystery, turned out to be molecular information storage. No process anywhere has been found to require a vital force. *Kernel: [LOCK], $\det(R) = 0.9989$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9989$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, the reduction of life processes to chemistry complete and the vital force nowhere required.

265 · Mach's Principle as a Complete Theory of Inertia

Physics · Tier III

Proposition. The strong Machian theory, that inertia is determined entirely by distant masses, is empirically inadequate.

On the strong version, removing the distant matter would abolish inertia, so an isolated rotating body in an empty universe should show no centrifugal effects. The relativistic theory of gravity contradicts this: it admits empty solutions with well-defined inertial frames and no matter at all, and the frame-dragging it does predict is fixed by the local geometry rather than by the global matter distribution alone. *Kernel: [LOCK], $\det(R) = 0.9896$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9896$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade against the complete Machian theory, the matter-free inertial solutions decisive. Fenced: the theory does incorporate some Machian effects through frame-dragging, so it is the strong all-determining version that fails, not every Machian element, the open question carried by the rotation case.

266 · Strong Social Constructivism

Philosophy of Science · Tier III

Proposition. Strong social constructivism about natural science fails the independence test.

The strong position explains successful and failed theories by the same social causes and treats truth as irrelevant to the explanation, which turns on itself, since the position's own content would then be merely socially caused rather than correct. Against it stands the record of novel prediction: phenomena predicted from theory before they had any social relevance, and later confirmed, have no mechanism on a purely social account of theory content. Something must track the world. *Kernel:* $[LOCK]$, $\det(R) = 0.9743$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9743$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, the symmetry principle self-undermining and novel prediction inexplicable on a purely social account. Fenced: this targets the strong "only social" version; that science has social dimensions is uncontested.

267 · The Infinite Regress Objection to Foundationalism

Epistemology · Tier III

Proposition. The infinite regress objection does not defeat all forms of foundationalism.

The trilemma holds that any justified belief is justified by another, generating a regress, or by itself, circular, or by a stopping point accepted without justification, dogmatic. Foundationalism takes the stopping point, and the modest and reliability-based versions answer the charge of dogmatism by grounding basic beliefs in reliable processes rather than in arbitrary assertion or in further beliefs. The regress is not vicious where a different kind of justification applies at the base. *Kernel:* $[LOCK]$, $\det(R) = 0.9782$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9782$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, reliability-based foundations supplying non-dogmatic basic beliefs. Fenced: this does not establish foundationalism over coherentism or infinitism, only that it survives the regress objection, the broader debate open.

268 · Naive Color Realism

Philosophy of Perception · Tier III

Proposition. Naive realism about perceptual color properties is empirically refuted.

The view that objects bear the colors we see as mind-independent surface features fails on several counts. Different physical light mixtures produce the same perceived color, so no single wavelength corresponds to a color, and context decides the appearance, two physically identical patches looking different under engineered shading. Color experience also varies with the perceiver's receptors, the same surface yielding different experiences across people and species. Color is a relation to a visual system, not a bare surface property. *Kernel:* $[LOCK]$, $\det(R) = 0.9564$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9564$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade against the naive view, color response-dependent rather than a simple intrinsic property. Fenced: dispositional and relational realism about color, on which color is a disposition to cause experiences, survive; the seal targets the naive surface-property reading.

269 · The Verification Criterion of Meaning

Epistemology · Tier III

Proposition. The logical positivist verification criterion of meaning is self-undermining.

The criterion holds that a statement is meaningful only if it is analytic or empirically verifiable. Turned on itself, it fails: it is not a truth of logic or a definition, and it is not empirically verifiable, so by its own standard it is meaningless. Every attempt to repair it either admitted all statements indiscriminately or excluded the universal laws of physics, which are not directly verifiable, and the repair program never converged. *Kernel:* $[LOCK]$, $\det(R) = 0.9970$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9970$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\perp]$ Sealed at structural grade, the criterion failing its own test and the revision program failing to converge.

270 · Cartesian Skepticism

Epistemology · Tier III

Proposition. Cartesian skepticism is a genuine challenge that is frame-locked rather than empirically tractable.

The deceiving-demon scenario is built so that every possible observation is the same whether or not the demon exists, so no measurement bears on it, which is a real challenge to any empirical refutation of skepticism. The standard responses are non-empirical: that knowing one has hands is more certain than any skeptical premise, and that doubt requires a background of things held fixed, so doubting everything at once is self-undermining. *Kernel:* $[LOCK]$, $\det(R) = 0.9808$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9808$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the hypothesis empirically idle and the answers necessarily non-empirical. Fenced: whether those responses decisively defeat skepticism or set it aside is contested, so "answerable" overstates the consensus; the frame-locked diagnosis is what seals.

Volume VIII · Tier IV

Nine cases closing Volume VIII, weighted toward proven results of physics, information theory, and mathematics, with two genuine open puzzles. Where a theorem is established the seal carries theorem grade; the two puzzles seal at structural grade as real and unresolved, fenced on the assumptions their force depends on.

271 · The Kochen-Specker Theorem

Quantum Mechanics · Tier IV

Proposition. Quantum observables cannot all carry pre-assigned definite values.

For any quantum system of three or more dimensions, no assignment of definite values to all observables can respect the algebraic relations that the observables satisfy whenever they are measured together, so a non-contextual hidden-variable account is mathematically impossible. The value an observable yields must depend on which compatible set it is measured alongside. Contextuality of this kind has been confirmed in neutron interferometry, trapped ions, and photon polarization. *Kernel:* $[LOCK]$, $\det(R) = 0.9974$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9974$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade, non-contextual value assignment proven impossible and contextuality confirmed across independent systems.

272 · The Black Hole Information Paradox

Physics · Tier IV

Proposition. The black hole information paradox is a genuine unresolved structural conflict.

Radiation from an evaporating black hole is thermal, carrying nothing about what fell in beyond a few conserved quantities, so complete evaporation appears to destroy the information in the original matter, which the unitary evolution of quantum theory forbids. The two principles cannot both hold as stated. No direct test is available, the evaporation timescale exceeding the age of the universe for real black holes. *Kernel:* $[LOCK]$, $\det(R) = 0.9982$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9982$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, the unitarity conflict real and the problem open. Fenced: recent entropy calculations reproduce the curve that unitarity requires, strong evidence that information is preserved, while the bulk mechanism by which it escapes remains unsettled.

273 · Shannon's Channel Capacity

Information Theory · Tier IV

Proposition. Shannon's theory uniquely characterizes channel capacity under natural axioms.

The entropy measure is the unique function satisfying continuity, maximization on the uniform distribution, and additivity for independent sources, and the coding theorem establishes the resulting capacity as the exact rate below which reliable communication is possible and above which it is not. Engineering confirms the ceiling directly: modern codes operate within a fraction of a decibel of capacity across billions of deployed devices, and compression reaches the entropy limit. *Kernel:* $[LOCK]$, $\det(R) = 0.9829$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* $[LOCK]$ $\det(R) = 0.9829$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade, the entropy axiomatically unique and the capacity confirmed as the operative limit of communication.

274 · The Second Law of Thermodynamics

Physics · Tier IV

Proposition. The second law is a genuine directional physical asymmetry.

For a system of astronomical particle number the disordered configurations vastly outnumber the ordered ones, so the entropy of an isolated system increases with overwhelming probability, and the increase defines a direction in time. The bound is confirmed at every turn: no heat engine beats the efficiency the law sets, and every attempt to cheat it through a sorting demon has been shown to pay an entropy cost in erasing the information it gathers. *Kernel:* [LOCK], $\det(R) = 0.9312$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9312$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, the asymmetry statistically grounded and confirmed by engine limits and the cost of erasure, the law overwhelmingly probable rather than strictly inviolable at the microscale.

275 · Deterministic Chaos

Physics · Tier IV

Proposition. Sensitive dependence on initial conditions is formally established and physically confirmed.

When the leading rate of separation of nearby trajectories is positive, two states that start arbitrarily close diverge exponentially, a property computable for smooth deterministic systems and rigorous for specific ones. The atmosphere displays it as the exponential growth of forecast error that caps weather prediction, and curved billiards display it provably, confirmed in microwave-cavity analogues. Determinism and long-term unpredictability coexist. *Kernel:* [LOCK], $\det(R) = 0.9992$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9992$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade, exponential divergence established formally and confirmed across independent physical systems.

276 · The Independence of the Continuum Hypothesis

Mathematics · Tier IV

Proposition. The continuum hypothesis is independent of the axioms of set theory.

One construction yields a model of the standard axioms in which the hypothesis holds, so it cannot be disproved, and a second technique builds a model in which it fails, so it cannot be proved, leaving it undecided by those axioms if they are consistent. Subsequent constructions confirm the freedom, the size of the continuum being largely unconstrained and intermediate axioms sitting consistently on either side. *Kernel:* [LOCK], $\det(R) = 0.9985$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9985$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed, theorem-grade, both the hypothesis and its negation consistent with the axioms. The independence is relative to those axioms; whether the hypothesis has a determinate truth-value beyond them is a separate and contested question.

277 · The Fermi Paradox

Astrophysics · Tier IV

Proposition. The Fermi paradox is a genuine structural puzzle that no current hypothesis fully resolves.

The galaxy is old enough that a civilization capable of interstellar travel could have crossed it many times over, and planet surveys confirm abundant Earth-size worlds in habitable zones, so the expectation of some trace is well-founded. Yet decades of listening have found no signal, and infrared surveys show no structures of the kind a vast engineering civilization would leave. The proposed resolutions are all partial or speculative. *Kernel:* [LOCK], $\det(R) = 0.9866$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9866$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the tension between expectation and absence real and unresolved. Fenced: the force rests on the colonization-timescale and detectability assumptions, which some resolutions challenge directly; the seal is on the puzzle being genuine and open.

278 · Quantum Discord

Quantum Information · Tier IV

Proposition. Non-classical correlations exist beyond entanglement.

Discord measures the gap between the total correlation in a quantum state and the part recoverable by classical measurement, and it can be positive for states that carry no entanglement at all,

so entanglement does not exhaust the non-classical correlations a system can hold. Mixed-state computation with negligible entanglement but positive discord, and a known algorithm that gains a large speedup from discordant unentangled states, locate the resource outside entanglement. *Kernel*: [LOCK], $\det(R) = 0.9955$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9955$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at high grade, separable states carrying positive discord and powering a quantum advantage, non-classical correlation reaching beyond entanglement.

279 · Cantor's Theorem

Mathematics · Tier IV

Proposition. The power set of any set is strictly larger, so infinities differ in size.

For any set, no map onto its collection of subsets can be onto, shown by forming the subset of elements not contained in their own image, which the map can reach by no element without contradiction. The collection of subsets is therefore strictly larger, and iterating builds an unbounded hierarchy of infinite sizes. The reals exceed the naturals, the functions on the reals exceed the reals, and a strict hierarchy of definable sets confirms the structure. *Kernel*: [LOCK], $\det(R) = 0.9890$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9890$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade, the diagonal argument forcing a strict hierarchy of infinite cardinalities.

280 · Supersymmetry and the Cosmological Constant

Physics · Tier IV

Proposition. Supersymmetric cancellations do not currently provide a solution to the cosmological constant problem.

In an exact version of the symmetry, the contributions of particles and their partners to the vacuum energy cancel exactly, which would dissolve the problem. But the symmetry is broken, since no partners have been seen, and the broken theory restores a vacuum energy far above the observed value. Fifteen years of collider searches have found no partner and pushed the breaking scale upward, which makes the residual energy worse rather than better. *Kernel*: [LOCK], $\det(R) = 0.9982$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9982$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, broken supersymmetry failing to deliver the cancellation and the null searches worsening the residual. Fenced: this is the current state, not a claim that the symmetry is false or could contribute nothing.

281 · The Free Energy Principle

Neuroscience · Tier IV

Proposition. The free energy principle as a unified theory of mind and life remains unverified.

The principle holds that organisms minimize a quantity that bounds their surprise, and offers it as a single account of perception, action, and life. The quantity has never been measured in a real nervous system, since computing it requires a full specification of the organism's internal model, and the principle issues no prediction that separates it from the Bayesian, predictive-coding, and reinforcement-learning accounts it subsumes. It organizes rather than competes. *Kernel*: [LOCK], $\det(R) = 0.9955$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9955$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, the quantity unmeasurable in practice and the principle making no distinguishing prediction. Fenced: it is valuable as a unifying mathematical framework; what is unverified is its standing as a confirmed empirical theory. The instrument applies the same standard to this framework that it applies to every claim, including where the framework draws on it.

282 · The Penrose-Lucas Argument

Philosophy of Mind · Tier IV

Proposition. Human minds transcend Turing machines via Gödel's theorem.

The argument runs that for any consistent formal system containing arithmetic there is a true sentence the system cannot prove, that a human can see the sentence is true, and so that the human is not such a system. The middle step is where it fails. Seeing the sentence as true requires knowing the system is consistent, and for a system rich enough to model the human's own reasoning that knowledge is exactly what Gödel's framework denies even to the human. The seeing is a conditional inference, true if the system is consistent, which a machine can carry out equally, and humans are in any case not consistent systems, holding errors and inconsistencies. The conclusion of

transcendence rests on the one link the argument cannot secure. Kernel: $[X], \det(R) \rightarrow 0, N = 24$.

Triaxial. G9 CSEG fails: the terminal claim of transcendence exceeds the strength of its weakest link, the unverifiable consistency assumption. The registrational axis collapses into the formal, the warrant volume falls. Kernel $[X] \det(R) \rightarrow 0. \vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken at the terminal-strength gate, the transcendence conclusion exceeding the warrant, since the human's seeing is itself a conditional formal inference and the required consistency cannot be verified from within. The Gödel result is genuine; the leap from it to a mind beyond all formal systems is the overreach.

283 · Loop Quantum Gravity

Physics · Tier IV

Proposition. Loop quantum gravity has made no confirmed experimental predictions.

The theory predicts discrete spectra for area and volume at the Planck scale and, in some versions, an energy-dependent speed of light that would show up over cosmological distances. The discreteness sits far below any accessible scale, and the speed-of-light effect has been tightly constrained, gamma-ray bursts showing photons of very different energies arriving together, which excludes the below-Planck versions that predicted a visible effect. No prediction stands confirmed. Kernel: $[LOCK], \det(R) = 0.9929, N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK] \det(R) = 0.9929, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the discreteness inaccessible and the testable variants excluded. Fenced: this is the current empirical state, not a claim of falsity; it remains a candidate without confirmed predictions, as do the other approaches to quantum gravity.

284 · Retrocausal Interpretations of Quantum Mechanics

Quantum Mechanics · Tier IV

Proposition. Retrocausal interpretations do not uniquely confirm a restoration of local hidden variables.

These interpretations posit that measurement settings in the future influence particle states in the past, which would let the correlations that violate the locality bounds be reproduced without action at a distance. But no measurement separates them from standard quantum mechanics, which accounts for the same correlations without backward causation. The added backward

arrows carry no observable consequence and are empirically idle. Kernel: $[LOCK], \det(R) = 0.9976, N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK] \det(R) = 0.9976, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the interpretation empirically equivalent to the standard theory and its added content inert, confirming nothing uniquely.

285 · Digital Physics

Philosophy of Physics · Tier IV

Proposition. Digital physics overreaches in claiming the universe is fundamentally a discrete computation.

A fixed computational lattice singles out a preferred frame, which collides with the symmetry of relativity, confirmed to roughly one part in ten to the twentieth, far past any scale a Planck-length grid would act on. Macroscopic quantum superpositions sit awkwardly with a fixed discrete substrate, and the specific cellular-automaton models proposed have produced no quantitative prediction that matches the established physics. Kernel: $[LOCK], \det(R) = 0.9534, N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK] \det(R) = 0.9534, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the fixed-lattice claim in tension with the precision of relativistic symmetry and the models without confirmed predictions. Fenced: discrete approaches that preserve that symmetry on average avoid the preferred-frame conflict, so the overreach targets the literal fixed-lattice and the strong "universe is a computation" thesis, not every discrete-spacetime proposal.

286 · Hayek's Spontaneous Order

Economics · Tier IV

Proposition. Spontaneous order theory overreaches as a complete explanation of social phenomena.

The core insight is sound, that institutions aggregating dispersed local knowledge can outperform central design because no planner holds all of it and prices carry it efficiently. The overreach is the totalizing extension. Where actions impose costs on others that prices do not capture, the unguided outcome is inefficient, and designed institutions outperform it, a standard result for externalities and public goods, with epidemic control a clear case of systematic underprovision. Kernel: $[LOCK], \det(R) = 0.9973, N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9973$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, market failures bounding the scope of spontaneous order. Fenced: this does not refute the dispersed-knowledge insight, which is robust; it bounds its reach, the overreach being the totalizing claim.

287 · Moral Particularism

Ethics · Tier IV

Proposition. Moral particularism does not defeat all principle-based ethics.

Particularism argues that a feature counting in favor of an act in one case can count against it in another, so moral reasons are holistic and no exceptionless principle holds. But some features appear to count the same way across contexts without credible exception, gratuitous cruelty among them, and the generalist has resources the argument does not close off, principles stated as defeasible or as contributory reasons rather than exceptionless rules. Principle-based ethics survives the challenge. *Kernel:* [LOCK], $\det(R) = 0.9551$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9551$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, robust cross-context features and defeasible-principle accounts surviving the particularist argument. Fenced: this does not establish generalism as correct or particularism as false, only that particularism does not defeat principle-based ethics; both remain live.

288 · The Baryon Asymmetry Problem

Cosmology · Tier IV

Proposition. The baryon asymmetry of the universe is a genuine open problem.

Generating the observed excess of matter over antimatter requires three conditions, and the established theory meets all three in principle, through processes that violate baryon number at high temperature, a known source of the relevant symmetry violation, and a departure from equilibrium. But the numbers fall far short. The measured symmetry violation is real but quantitatively far too small, and the Higgs mass rules out the strong transition the simplest mechanism needed. No confirmed mechanism closes the gap. *Kernel:* [LOCK], $\det(R) = 0.9953$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9953$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the established sources quantitatively insufficient and no confirmed mechanism in hand, a real open problem.

Volume IX · Tier III

Nine cases opening Volume IX, mixing philosophy of language and epistemology with proven results of physics and mathematics. The theorems seal at theorem grade; the philosophical results seal at structural grade, several fenced where the proposition states a strong reductive thesis past the defensible core.

289 · Wittgenstein's Private Language Argument

Philosophy of Language · Tier III

Proposition. Genuinely private rules are incoherent.

Suppose a sensation is named by attending to it privately and the name is reused later for what seems the same sensation. Nothing fixes whether the reuse is correct, since the only available check is present memory, which is the very thing in question, so there is no difference between applying the rule correctly and merely seeming to. A standard of correctness that is nothing but one's own impression is no standard, and meaning grounded purely in private ostension collapses. Language is in fact learned through public correction, not inner pointing. *Kernel:* [LOCK], $\det(R) = 0.9890$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9890$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, private ostension supplying no correctness standard. Fenced: whether this requires a public community specifically or only some independent standard, and whether a solitary rule-governed agent is possible, is debated; the seal targets the incoherence of a standard that is only one's own impression.

290 · Peirce's Pragmatic Maxim

Epistemology · Tier III

Proposition. The meaning of a concept is fixed by its conceivable practical consequences.

The maxim clarifies a concept by the set of conditionals linking actions to observable results that the concept entails, so two concepts entailing the same conditionals are the same and a concept entailing none is idle. As a method it is powerful, exposing distinctions that make no difference and retiring posits, an abandoned medium of light wave propagation among them,

once they yield no distinct consequence. *Kernel*: [LOCK], $\det(R) = 0.9856$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9856$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade on the maxim as a method of clarification, distinctions without practical difference being idle. Fenced: the strong claim that meaning is wholly exhausted by practical consequences is a form of operationalism that is contested, straining on theoretical terms and modal content; the seal is on the method, not the strong reductive thesis, consistent with the verdict on the verification criterion.

291 · The Wigner-Eckart Theorem

Physics · Tier III

Proposition. Rotational symmetry uniquely determines selection rules.

The theorem factors the matrix element of a symmetry-classified operator into two parts: a coefficient fixed entirely by the rotation group, which carries all the dependence on orientation and so all the selection rules, and a single reduced quantity that holds the dynamical content. The angular structure is therefore pure symmetry, independent of the specific potential. The resulting rules govern atomic spectral lines and nuclear transitions to measured precision, derived without any model of the interaction. *Kernel*: [LOCK], $\det(R) = 0.9992$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9992$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade, the selection rules fixed by symmetry alone and confirmed across atomic and nuclear spectra.

292 · The Born Rule

Quantum Mechanics · Tier III

Proposition. The Born rule is the unique probability assignment consistent with the structure of quantum mechanics.

For any quantum system of three or more dimensions, the only way to assign probabilities to measurement outcomes that respects the requirement that they sum to one over every complete set of alternatives is the trace formula of the standard rule. There is no consistent competitor. Every experiment confirms it, the interference pattern matching the squared amplitude and spin statistics matching the predicted fractions to sub-percent precision over thousands of runs. *Kernel*: [LOCK], $\det(R) = 0.9991$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9991$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed, theorem-grade, the rule the unique consistent probability measure on the quantum structure and confirmed throughout. The uniqueness holds given the consistency constraint; deriving the rule from deeper postulates in some interpretations is a separate matter.

293 · Mathematical Induction

Mathematics · Tier III

Proposition. Induction is the essential principle for arithmetic truth over all the naturals.

Establishing a property for every natural number requires the step from holding at zero and being inherited by each successor to holding throughout, and without it the axioms fail to pin down the naturals, admitting models with extra elements beyond them. Every general result about the naturals across combinatorics, number theory, and analysis turns on this step, and the existence of those non-standard models confirms that the other axioms alone are insufficient. *Kernel*: [LOCK], $\det(R) = 0.9953$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9953$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at high grade, induction securing universality over the naturals and characterizing them up to isomorphism. Fenced: "unique" holds up to provable equivalence, since well-ordering and strong induction are equivalent formulations rather than competitors; among genuinely distinct principles it is the one that secures universality.

294 · Structural Realism

Epistemology · Tier III

Proposition. Structural realism is a viable response to the pessimistic meta-induction.

The pessimistic argument notes that theory change discards the objects earlier theories posited, undercutting belief in those objects. Structural realism concedes the objects and locates the continuity in mathematical structure, which is preserved across the transitions: reflection and transmission relations carried intact into the successor theory, the older gravitational law surviving as a limiting case of the newer, the bracket structure of classical mechanics reappearing in the quantum commutator. *Kernel*: [LOCK], $\det(R) = 0.9915$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK]

$\det(R) = 0.9915$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at structural grade, the position accommodating entity change while preserving structure, a genuine response to the argument. Fenced: this establishes viability, not correctness; structural realism faces its own difficulties over structure without objects, and whether it ultimately succeeds is contested.

295 · Quine's Naturalized Epistemology

Epistemology · Tier III

Proposition. Epistemology should be continuous with empirical science.

The traditional program sought a foundation for knowledge of the world prior to all science, and the attempt to reduce talk of the physical to talk of the sensory failed for want of the required definitions. The naturalist turn studies belief formation as the empirical process it is, and the program is fruitful, the psychology of perception and judgment, models of the predictive brain, and the evolutionary account of reliable cognition all illuminating how beliefs arise from inputs. *Kernel:* $[LOCK]$, $\det(R) = 0.9765$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9765$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at structural grade, the naturalistic program viable and fruitful. Fenced: whether it should replace normative epistemology is contested, critics holding that the normative dimension of justification does not reduce to descriptive psychology; the seal is on viability and fruitfulness, not the strong replacement thesis.

296 · The Problem of the Criterion

Epistemology · Tier III

Proposition. The problem of the criterion has no non-circular solution within either methodism or particularism.

To sort knowledge from mere belief one needs either a criterion applied first or a stock of clear instances identified first, and each presupposes the other. Beginning with a criterion requires already knowing which cases it should approve in order to validate it, and beginning with instances requires a criterion to certify them as knowledge. Each starting point smuggles in what the other was meant to supply. *Kernel:* $[LOCK]$, $\det(R) = 0.9803$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9803$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad

returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at structural grade, methodism and particularism each presupposing what the other establishes, the circularity genuine. Fenced: some accept a benign circularity in which criterion and instances are adjusted together in reflective equilibrium, so the seal is that no foundationally non-circular starting point exists, not that the problem makes knowledge impossible.

297 · The Principle of Relativity

Physics · Tier III

Proposition. The laws of physics take the same form in every inertial frame.

A frame in uniform motion relative to an inertial frame is itself inertial, and the principle requires the laws to read identically in both, with no frame privileged. The confirmation is among the most precise in physics. Optical experiments find no preferred frame for light to extreme sensitivity, resonance experiments find nuclear physics isotropic to roughly one part in ten to the twenty-seventh, and every species of particle obeys the same dynamics in every inertial frame. *Kernel:* $[LOCK]$, $\det(R) = 0.9993$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9993$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at high grade, frame-independence confirmed to extraordinary precision with no preferred frame detected. An empirical principle confirmed rather than a logical necessity, but confirmed as firmly as any result in physics.

298 · Strong Linguistic Relativity

Linguistics · Tier III

Proposition. Strong linguistic relativity, that language determines thought, is empirically refuted.

The strong thesis predicts that speakers whose language lacks a distinction cannot make it in thought, that those without separate words for blue and green cannot tell the colors apart, that those without large number words cannot represent larger quantities. Both predictions fail. Speakers of a language with two color terms discriminate colors as finely as anyone in non-verbal tasks, and speakers with words only for small counts still represent quantity. Language does not determine the underlying capacity. *Kernel:* $[LOCK]$, $\det(R) = 0.9882$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9882$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: [⊥] Sealed at structural grade against the strong determinist thesis, the cross-linguistic perception and number data decisive. Fenced: the weak thesis that language influences rather than determines thought has genuine support and is untouched by this verdict.

299 · Orchestrated Objective Reduction

Philosophy of Mind · Tier III

Proposition. The orchestrated objective reduction theory of consciousness is empirically inadequate.

The theory locates consciousness in quantum coherence within neuronal microtubules, and the central physical obstacle is severe. A superposition in a warm wet protein environment loses coherence in a time many orders of magnitude shorter than the neural processes it would have to support, so the required coherence does not survive long enough to function. No experiment has measured the coherence the theory needs, and the anesthetic evidence offered for it implicates many targets rather than microtubule coherence specifically. *Kernel:* [LOCK], $\det(R) = 0.9977$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9977$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: [⊥] Sealed at structural grade, the decoherence obstacle severe and the required coherence unconfirmed. Fenced: the theory does make testable claims and so is not unfalsifiable in the way some consciousness proposals are; the seal is on its empirical inadequacy, not on a demarcation verdict.

300 · Social Darwinism

Philosophy of Science · Tier III

Proposition. Social Darwinism commits the naturalistic fallacy and misrepresents evolutionary theory.

It fails on two independent counts. It infers a social ought from a natural is, an invalid step, since nature also produces disease and predation that no one proposes to emulate. And it misreads the biology, which does not favor ruthless individual competition: cooperative strategies are evolutionarily stable, and selection favors self-sacrifice toward relatives by a precise rule, the opposite of the doctrine's picture. *Kernel:* [LOCK], $\det(R) = 0.9893$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9893$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: [⊥] Sealed at structural grade, an invalid value inference compounded by a factual misreading of evolution.

301 · The Observable Universe

Cosmology · Tier III

Proposition. The observable universe cannot be assumed to be the entire universe.

The boundary of what we can see is set by how far light has traveled since the beginning, an observational limit rather than an edge of the physical world, and the governing model places no such edge there. The data put the universe at least several times larger than the visible patch in every direction, no small-scale topology having been detected that would wrap it back on itself. *Kernel:* [LOCK], $\det(R) = 0.9850$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9850$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: [⊥] Sealed at structural grade, the horizon an observational limit and the whole demonstrably larger than the visible part. Fenced: whether the universe is finite or infinite remains open; the seal is on the non-identification, not on the total size.

302 · Grand Unified Theories

Particle Physics · Tier III

Proposition. Grand unified theories are not empirically confirmed.

The simplest unification predicts that the proton decays with a definite lifetime, a sharp and testable consequence of unifying the forces at a high energy. The lifetime bound from large detectors now exceeds that prediction by orders of magnitude, excluding the minimal version, and the precise unification of the force strengths does not quite work without further ingredients. No unified theory stands confirmed. *Kernel:* [LOCK], $\det(R) = 0.9996$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9996$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: [⊥] Sealed at structural grade, the minimal version excluded by proton-decay limits and no version confirmed. Fenced: extensions predict longer proton lifetimes beyond current reach, so this excludes the minimal theory and notes the absence of confirmation, not the refutation of every unification.

303 · The Extended Mind Thesis

Philosophy of Mind · Tier III

Proposition. The extended mind thesis overreaches from cognitive coupling to genuine mental constitution.

The thesis holds that when an external resource is used the way biological memory is used, reliably available and endorsed, it is not merely a cause of cognition but a part of it. The objection is that causal coupling to a process does not make the coupled item a constituent of it, the coupling-constitution distinction. Whether the thesis overreaches turns on that distinction, which the extended-mind side disputes, denying any principled line between a tightly coupled tool and a part of the mind. The matter is unsettled. *Kernel: [?], dimensional shortfall, $N = 6, k = 3$.*

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.*

Verdict: [?] Under-determined, the overreach charge resting on the contested coupling-constitution distinction with serious advocates on both sides. The deciding input would be a principled criterion separating a constituent of cognition from a tightly coupled external aid.

304 · Moore's Open Question Argument

Ethics · Tier III

Proposition. Moore's open question argument does not establish ethical non-naturalism.

The argument observes that for any natural property one can still sensibly ask whether a thing with that property is good, and concludes that goodness is not identical to any natural property. The inference assumes that a real identity would make the question trivial, closed by analysis. But identities can be substantive rather than analytic, the identity of water with its molecular composition being knowable only empirically, so the question staying open shows only that the identification is not analytic, not that it is false. *Kernel: [LOCK], $\det(R) = 0.9924, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel [LOCK] $\det(R) = 0.9924, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.*

Verdict: $[\perp]$ Sealed at structural grade, the argument conflating analytic synonymy with identity and so failing against a synthetic naturalism. Fenced: this shows the argument does not establish non-naturalism, not that naturalism is true; the metaethical question is left open.

305 · Eliminativism About Folk Psychology

Philosophy of Mind · Tier III

Proposition. Eliminativism overreaches in rejecting propositional attitudes wholesale.

The position holds that the everyday framework of beliefs and desires is a false theory whose posits do not exist. Wholesale rejection outruns the evidence. The framework is predictively indispensable and carves cognitive joints, children acquiring accurate models of others' beliefs at a fixed stage and a dedicated brain network handling that reasoning, and the promised replacement description in purely neural terms has not been produced. *Kernel: [LOCK], $\det(R) = 0.9682, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel [LOCK] $\det(R) = 0.9682, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.*

Verdict: $[\perp]$ Sealed at structural grade, wholesale elimination outrunning the predictive indispensability of the framework and the absence of a replacement. Fenced: the self-refutation objection is itself contested, and a moderate revisionism short of elimination remains live; the seal is on wholesale elimination overreaching, not on folk psychology being beyond revision.

306 · Evolutionary Debunking Arguments

Ethics · Tier III

Proposition. Evolutionary debunking arguments do not defeat moral realism.

The debunking argument holds that since our moral sense was shaped for reproductive fitness, and fitness need not align with moral truth, our moral beliefs are unreliable guides to it. The realist resists the independence step with a common-explanation account, on which the moral facts and our capacity to register them share a source, so that fitness and truth need not be unrelated. The observed convergence of basic moral norms across cultures fits the realist reading as well as the debunker's. *Kernel: [LOCK], $\det(R) = 0.9824, N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel [LOCK] $\det(R) = 0.9824, \lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.*

Verdict: $[\perp]$ Sealed at structural grade, the common-explanation response blocking the independence premise the argument needs. Fenced: this establishes that realism is not defeated, not that realism is true; whether the response ultimately succeeds, and whether realism is correct, remain open, paired with the open verdict on moral realism and the parallel verdict that the open-question argument does not refute naturalism either.

Volume IX · Tier IV

Nine cases closing Volume IX, weighted toward proven results of physics, quantum information, and computer science, with three genuine open problems. The theorems carry theorem grade; the open problems and the careful negative claims seal at structural or

high grade, fenced where the proposition reaches past what is settled.

307 · The Fault-Tolerance Threshold Theorem

Quantum Information · Tier IV

Proposition. Scalable quantum computation is theoretically possible.

Encoding a logical qubit across many physical qubits lets errors be detected and corrected without reading the protected information, and the theorem shows that once the physical error rate falls below a fixed threshold, computations of arbitrary length can be carried out reliably, the logical error falling as the code grows. Recent hardware has crossed into the below-threshold regime, the logical error rate decreasing as the code distance is increased, the first experimental signature of the predicted behavior. *Kernel:* [LOCK], $\det(R) = 0.9995$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9995$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, the theoretical possibility proven and below-threshold behavior now observed, the practical realization ongoing.

308 · The Incompleteness of the Standard Model

Particle Physics · Tier IV

Proposition. The Standard Model is empirically incomplete, established by three independent lines.

The model is not merely unfinished but provably incomplete against observation. Its neutrinos are massless, yet the observed oscillation of neutrinos between types requires them to have mass. The rotation of galaxies and the bending of light require matter for which it has no candidate particle. And the excess of matter over antimatter exceeds what its symmetry violation can produce. Three independent observations, each confirmed, each beyond the model. *Kernel:* [LOCK], $\det(R) = 0.9916$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9916$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, three independent confirmed observations each lying outside the model, incompleteness established.

309 · The CMB Power Spectrum and the Standard Cosmology

Cosmology · Tier IV

Proposition. The microwave background power spectrum confirms the standard cosmological model and excludes the major alternatives.

The acoustic oscillations of the early plasma imprint a specific pattern of peaks on the angular power spectrum, their positions and relative heights fixed by the densities of matter and the sound horizon. The measured spectrum across thousands of scales is fit by the standard model to sub-percent precision, and the major alternatives fail: a modified-gravity account without dark matter gives the wrong peak ratio, and a model without a recombination epoch produces no peaks at all. *Kernel:* [LOCK], $\det(R) = 0.9927$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9927$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the standard model fit to sub-percent precision and the major rivals excluded. Fenced: "uniquely" holds against radically different models, not against near-variants with evolving dark energy that also fit, and the standing expansion-rate and structure-amplitude tensions suggest the framework may require extension; the seal is on strong confirmation and the exclusion of the major alternatives.

310 · Decoherence and the Measurement Problem

Quantum Mechanics · Tier IV

Proposition. Decoherence explains the emergence of pointer states but does not resolve the Born rule problem.

A system entangled with its environment is driven toward a preferred set of states, those the environment monitors and least disturbs, and interference between them is suppressed, which accounts for why definite-outcome bases rather than arbitrary superpositions are the ones that survive contact with the surroundings. This is confirmed, superpositions of macroscopic states and of large molecules decohering at the predicted rates. But it does not explain why a single outcome occurs rather than all of them, nor why the probabilities take the form they do. *Kernel:* [LOCK], $\det(R) = 0.9937$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9937$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, decoherence accounting for the pointer basis and the suppression of interference but leaving the single outcome and the probability rule unexplained. The verdict resists the common overstatement that decoherence resolves the measurement problem; it seals exactly the part that holds.

311 · P versus NP

Computer Science · Tier IV

Proposition. P versus NP is a genuine open problem whose resolution would transform mathematics and computation.

The question is whether every problem whose solution can be checked quickly can also be solved quickly. Verification appears easier than solution, suggesting the two classes differ, yet no proof exists either way. The stakes are concrete: the security of standard cryptography rests on certain problems being hard to solve, and a fast method would collapse it, while the same method would transform optimization across industry and science. *Kernel: [LOCK], $\det(R) = 0.9979$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9979$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the problem genuinely open and its resolution consequential across mathematics, cryptography, and optimization.

312 · Topological Phases of Matter

Condensed Matter · Tier IV

Proposition. The quantum Hall effect reveals topology as a physical property of matter.

In a two-dimensional electron system in a strong field, the transverse conductance locks to integer multiples of a fundamental ratio, and that integer is a topological invariant of the filled electronic bands, a quantity that cannot change under smooth deformation. This is why the quantization is so exact, measured to a part in a billion and adopted as a resistance standard. Interacting versions and insulating analogues extend the same topological framework. *Kernel: [LOCK], $\det(R) = 0.9975$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9975$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, the quantized conductance a topological invariant and the quantization exact to a part in a billion.

313 · Hume's Problem of Causation

Epistemology · Tier IV

Proposition. The regularity theory of causation is insufficient; causation requires more than constant conjunction.

Reducing causation to regular succession fails to distinguish a causal law from an accidental regularity, since the contents of a pocket or the color of a species can be perfectly constant without being causal. The interventionist framework marks the difference the regularity account cannot, separating setting a variable from merely observing it, and this is exactly why a controlled trial that intervenes carries causal weight an observational correlation does not. *Kernel: [LOCK], $\det(R) = 0.9821$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9821$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, constant conjunction failing to separate causes from accidental regularities and the intervention/observation distinction marking what it misses.

314 · The Boltzmann Brain Problem

Cosmology · Tier IV

Proposition. The Boltzmann brain problem is a genuine cosmological paradox.

In a universe that expands forever into a state of fixed temperature, rare fluctuations produce any configuration given enough time, including a momentary brain complete with false memories of an orderly past. If such fluctuations vastly outnumber ordinary observers, then a typical observer is one of them, which would make our coherent and stable observations improbable, a self-undermining conclusion. Cosmologies with eternal late-time expansion generate them, so the problem quantitatively constrains those models. *Kernel: [LOCK], $\det(R) = 0.9990$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9990$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the fluctuation observers a real and constraining threat to the typicality of our observations. Fenced: the force depends on assumptions about the measure and the eternity of the late-time phase, which some proposed resolutions challenge; the seal is on the problem being genuine and constraining.

315 · The No-Broadcasting Theorem

Quantum Information · Tier IV

Proposition. The no-broadcasting theorem establishes a fundamental asymmetry between classical and quantum information.

A broadcasting operation would take one copy of a state to a joint system whose two parts each reproduce the original, and the

theorem shows this is possible for a collection of states only when they commute. General quantum states, which do not commute, therefore cannot be broadcast, extending the impossibility of copying from pure states to mixed ones. Classical information can be freely duplicated; quantum information cannot, and quantum key distribution turns that asymmetry into security, since interception necessarily disturbs the state. *Kernel: [LOCK], $\det(R) = 0.9994$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9994$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, broadcasting proven possible only for commuting states and the classical-quantum copying asymmetry fundamental.

316 · Causal Set Theory

Physics · Tier IV

Proposition. Causal set theory has no confirmed experimental predictions.

The theory replaces continuous spacetime with a discrete network of events ordered by causation, from which a continuum is recovered by a random sprinkling of points. Its distinctive signatures of discreteness sit at the Planck scale, beyond reach, and the one number it offers for the dark energy density, while in the right range, rests on uncontrolled approximations and is not forced by the theory's own axioms, comparable estimates being available on dimensional grounds alone. *Kernel: [LOCK], $\det(R) = 0.9874$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9874$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the discreteness inaccessible and the dark-energy estimate not uniquely derived. Fenced: the current empirical state, not a verdict of falsity; a candidate without confirmed predictions, as with the other approaches to quantum gravity.

317 · Berkeley's Subjective Idealism

Metaphysics · Tier IV

Proposition. Berkeley's subjective idealism is scientifically inert.

The view makes physical objects collections of ideas, with the regularities of nature sustained in a divine mind when no one observes them. That sustaining mind reproduces exactly the predictions of a mind-independent world, since vacuum processes and radioactive decay proceed unobserved on both accounts, the one holding they occur because they are perceived by God and the

other because they simply occur. The divine sustaining adds no testable content. *Kernel: [LOCK], $\det(R) = 0.9984$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9984$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the position empirically equivalent to realism and predictively idle. Fenced: Berkeley does have an account of unobserved regularities, the divine mind, so the point is that the account is empirically idle rather than unavailable; like other unfalsifiable metaphysical theses, the inertness is what seals.

318 · Conformal Cyclic Cosmology

Cosmology · Tier IV

Proposition. Conformal cyclic cosmology does not establish its account of the low-entropy beginning.

The proposal identifies the remote, empty future of one cosmic cycle with the hot dense beginning of the next through a rescaling of the metric, and offers a route to the orderly low-entropy start. Its claimed observational signature, concentric temperature rings in the microwave background left by mergers in a prior cycle, has not survived scrutiny, independent analyses of the same data finding no significant signal above the noise. The mechanism remains unconfirmed. *Kernel: [LOCK], $\det(R) = 0.9674$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9674$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the claimed signature disputed and the explanation unconfirmed. Fenced: a mechanism for the low-entropy start is offered through the rescaling; the point is that it is unconfirmed and its signature absent, not that no proposal is made.

319 · The Computational Theory of Mind

Philosophy of Mind · Tier IV

Proposition. The computational theory of mind overreaches in claiming computation is sufficient for cognition.

The theory makes mental states computational states defined by their roles in transforming representations, and as a research framework it has been strikingly productive across attention, perception, language, and memory. The challenge is that conceptual representation appears grounded in sensorimotor simulation and that affective states are physiological as well as computational, which the embodied program reads as showing computation is not sufficient. Whether it is depends on how widely computation is construed, the theory treating embodiment

as implementation and its critics treating grounding as constitutive. The matter is unsettled. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not independently populable. *Kernel* [?] $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow$ [?].

Verdict: [?] **Under-determined, the sufficiency question turning on the contested scope of computation and the force of the embodied critique**. The deciding input would be a fixed notion of computation on which the embodiment and affect data clearly fall inside or outside.

320 · Fine-Tuning and the Multiverse

Cosmology · Tier IV

Proposition. The inference from fine-tuning to a multiverse overreaches.

The argument runs that the constants are improbably suited to life under a single fixed setting, and that an ensemble of universes with varying constants makes such a setting expected, so the ensemble is inferred. The inference is not forced. The competing possibility, that the constants are fixed by a deeper theory and so are necessary rather than contingent, is not ruled out, the distribution of constants across any ensemble is unknown, and the observed values might be the unique consistent ones. *Kernel*: [LOCK], $\det(R) = 0.9801$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9801$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at structural grade, the multiverse one explanation of fine-tuning among live alternatives, not a forced one**. Fenced: this does not say the multiverse is false; it says fine-tuning does not necessitate it. Paired with the earlier verdict that fine-tuning does not establish design, the instrument declines both inferences from the same premise.

321 · Meme Theory

Philosophy of Science · Tier IV

Proposition. Meme theory is not a developed scientific theory of cultural evolution.

The unit of cultural transmission lacks the operationalized machinery a rigorous theory requires. There is no defined replication mechanism, the proposed imitation being unspecified and variable across people, and no clear individuation of the unit itself. The quantitative frameworks that do successfully model cultural change, with defined transmission mechanisms and biases confirmed against demographic data, do not employ the meme as a theoretical object. *Kernel*: [LOCK], $\det(R) = 0.9960$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9960$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at structural grade, the unit lacking operationalized replication and individuation and the rigorous frameworks not using it**. Fenced: the meme remains a useful informal metaphor for cultural spread, and whether the gap makes it non-scientific or merely immature is a demarcation judgment the instrument does not force; the seal is on the missing mechanisms.

322 · Feyerabend's Epistemological Anarchism

Philosophy of Science · Tier IV

Proposition. The literal "anything goes" thesis defeats itself.

Taken at the letter, the claim that all methods are equally valid is itself a methodological claim advanced as true and as guidance, and on its own terms it can claim no privilege over any rival, so it undermines the very standing it asserts. And methodological norms demonstrably carry epistemic value, controlled trials outperforming anecdote and pre-registration lowering false positives in measurable ways. *Kernel*: [LOCK], $\det(R) = 0.9709$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9709$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow$ [ν].

Verdict: [ν] **Sealed at structural grade, the literal slogan self-undermining and methodological norms demonstrably valuable**. Fenced: the considered view is often read more charitably as the historical claim that no single fixed method governs all of science, which is defensible and not self-defeating; the seal targets the literal slogan, not the charitable reconstruction.

323 · The Holographic Brain Theory

Neuroscience · Tier IV

Proposition. The holographic brain theory is empirically refuted.

If memory and processing were stored as distributed interference patterns spread across the whole, then damage anywhere would degrade processing globally rather than removing specific functions, and no single location would carry specific content. The brain contradicts both. Targeted lesions produce targeted deficits, damage to specific regions removing language production, or face recognition with object recognition intact, or defined parts of the visual field, and individual neurons carry highly specific content. *Kernel*: [LOCK], $\det(R) = 0.9849$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9849$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the specific localized deficits and single-neuron specificity incompatible with holographic non-locality. Fenced: the brain does use distributed representation in some respects, so the refutation targets the strong non-locality claim, not all distributed processing.

324 · Lewis's Modal Realism

Metaphysics · Tier IV

Proposition. Lewis's modal realism extends concrete existence past any possible evidence.

The view holds that every way a world could be is a way some concrete world is, possible worlds being real physical regions as concrete as our own, with modal claims analyzed as quantification over them. The worlds are by definition causally isolated from ours, so no observation could bear on their existence, and the analysis replaces talk of possibility with talk of concrete existence in unreachable regions, a large ontological cost for the reduction it buys. *Kernel:* [LOCK], $\det(R) = 0.9956$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9956$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the position empirically inert and extending concrete existence beyond any possible evidence. Fenced: Lewis defends the cost by the theoretical utility of the reduction, and whether the extravagance is justified is a contested metaphysical judgment; the seal is on the inertness and the extension beyond evidence that critics target.

Volume X · Tier III

Nine cases opening the final volume, across philosophy of language, ethics, the foundations of mathematics, and machine learning. The proven results carry theorem or high grade; the philosophical results seal at structural grade, several fenced where the proposition states a strong reading past the defensible core, with one contested theological argument under-determined.

325 · Frege's Sense and Reference

Philosophy of Language · Tier III

Proposition. Meaning and reference are structurally distinct.

If reference were all there were to meaning, then a statement of identity between two names for the same thing would be as empty as a statement that the thing is itself, yet the discovery that the

morning star and the evening star are one and the same was a genuine and informative finding, which it could not be if the two names contributed only their shared referent. The same gap shows in belief reports, where substituting one name for a co-referring name can turn a truth into a falsehood. *Kernel:* [LOCK], $\det(R) = 0.9818$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9818$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the informativeness of identities and the failure of substitution showing meaning to outrun reference. Direct-reference theories handle the phenomena differently, but the phenomena themselves are robust.

326 · The Trolley Problem

Ethics · Tier III

Proposition. The trolley problem reveals a genuine structural tension in moral intuitions.

Two cases share their arithmetic, one life weighed against five, yet draw opposite verdicts: redirecting a hazard to a single person is widely judged permissible, while pushing a single person into its path to the same end is widely judged forbidden. The divergence is robust, replicated across hundreds of cultures, and tracks distinct processing, the second case preferentially engaging emotional circuitry, so the intuitions do not follow a single consistent principle. *Kernel:* [LOCK], $\det(R) = 0.9920$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9920$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, structurally similar cases drawing opposite and robust judgments. Fenced: whether this is an inconsistency to explain away or sensitivity to a real distinction, such as using a person as a means versus harming as a side effect, is debated; the seal is on the robust divergence.

327 · Frege's Logicism

Philosophy of Mathematics · Tier III

Proposition. Frege's logicism failed; arithmetic is not reducible to pure logic alone.

The program tried to derive arithmetic from logic by treating numbers as extensions of concepts governed by a basic law, and that law proved inconsistent, generating the paradox Frege himself acknowledged in print. The consistent principle that can be put in its place, equating numbers when concepts can be paired one to one, recovers arithmetic but is not a truth of pure logic,

carrying the full strength of arithmetic itself. *Kernel*: [LOCK], $\det(R) = 0.9986$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9986$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at high grade, the foundational law inconsistent and its consistent replacement not pure logic. Fenced: neo-logicist programs contest whether that replacement counts as logical, but Frege's original logicism failed decisively.

328 · The No-Free-Lunch Theorem

Machine Learning · Tier III

Proposition. No learning algorithm is universally superior.

Averaged over all possible target functions without bias toward any, every learning algorithm performs identically, so superiority is never intrinsic to an algorithm but always relative to assumptions about which problems are likely. The benchmarking record bears this out, no single method dominating across all tasks, each excelling where its built-in bias matches the structure and faltering elsewhere, the bias-variance tradeoff being one face of the same fact. *Kernel*: [LOCK], $\det(R) = 0.9991$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9991$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed, theorem-grade, all algorithms equal under the uniform average and superiority therefore always problem-relative.

329 · Emergence

Philosophy of Science · Tier III

Proposition. Emergence is a genuine non-reductive phenomenon at multiple organizational levels.

Higher organizational levels display effective properties with no counterpart in the description below them, the paired-electron condensate of a superconductor being a macroscopic quantum order with new effective degrees of freedom, and such order parameters being the standard formal expression of the phenomenon. These higher-level descriptions are indispensable in practice and not predictable without higher-level concepts. *Kernel*: [LOCK], $\det(R) = 0.9876$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9876$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade on weak emergence, the higher-level effective descriptions genuinely novel and indispensable in practice. Fenced: these examples are weak emergence, derivable in principle from the base laws; strong emergence with downward causation irreducible to the base is a much stronger and contested claim the superconductivity and biology examples do not establish.

330 · The Turing Test

Philosophy of Artificial Intelligence · Tier III

Proposition. The Turing test is not a sufficient criterion for machine intelligence.

The test asks only whether a system's conversational output can be mistaken for a human's, which is a behavioral target rather than a measure of intelligence, and sufficiently rich mimicry can hit it without the capacity it is taken to indicate. A program exploiting a non-native-speaker alibi passed a version of it while exhibiting no general intelligence, and current language systems pass informal versions in many exchanges, so passing does not establish the trait. *Kernel*: [LOCK], $\det(R) = 0.9992$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. *Kernel* [LOCK] $\det(R) = 0.9992$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, mimicry able to pass the behavioral test without the underlying capacity. Fenced: whether the test captures some aspect of intelligence or serves as a useful indicator is a separate question; the seal is on sufficiency. The verdict is neutral on what passes.

331 · Rowe's Evidential Problem of Evil

Philosophy of Religion · Tier III

Proposition. Apparently gratuitous suffering makes the existence of God unlikely.

The argument holds that there are instances of intense suffering an all-powerful, all-knowing, perfectly good being could have prevented without losing a greater good or permitting a worse evil, that such a being would prevent them, and so that no such being exists, with the suffering offered as evidence weighing against that existence. The vast scale of animal suffering across hundreds of millions of years, in creatures with no prospect of moral growth, is the case's strongest material. The decisive step is the inference from no justifying reason being apparent to there being none, which the skeptical reply resists by holding that we are not positioned to judge whether a justifying reason exists. *Kernel*: [?], *dimensional shortfall*, $N = 6$, $k = 3$.

Triaxial. Admissibility halts first: the orthogonal complement falls under two dimensions ($N - k < 4$), the third axis not

independently populable. Kernel $[?]$ $\det(R)$ undefined. $\vdash$ axes not jointly resolvable $\Rightarrow [?]$.

Verdict: $[?]$ Under-determined, the argument forceful and widely held stronger than the strictly logical version, but the inference from no apparent reason to no reason contested by the skeptical reply. Audited as the strength of an inference, taking no position on the existence of God. The deciding input would be a defense of that inference against the claim that we cannot judge the matter.

332 · Mathematical Structuralism

Philosophy of Mathematics · Tier III

Proposition. Structuralism resolves Benacerraf's identification problem.

Two different set-theoretic constructions each satisfy the axioms of arithmetic, and nothing distinguishes one as the true account of the numbers, which tells against identifying numbers with particular sets. Structuralism dissolves the puzzle by taking numbers to be positions in a structure rather than particular objects, so that the question of which set a number is does not arise, any system with the right shape serving equally. Mathematical practice works with the structure without fixing an implementation, and category theory supplies a structural foundation. *Kernel:* $[LOCK]$, $\det(R) = 0.9950$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9950$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the puzzle dissolved by treating numbers as positions rather than objects. Fenced: structuralism faces its own questions, including how such positions are known and whether structures are themselves objects, so it resolves this particular puzzle without settling the foundations of mathematics.

333 · Epistemic Injustice

Social Epistemology · Tier III

Proposition. Epistemic injustice is a genuine epistemic category, not merely an ethical one.

When a speaker is granted less credibility than their competence warrants because of prejudice about their identity, the harm is not only a moral wrong but a failure of knowledge to transfer, the testimony not being taken up as the knowledge it is. The consequences are measurable: pain reports discounted on the basis of identity produce misdiagnosis, knowledge of the speaker's condition failing to reach the record, which is an epistemic loss alongside the ethical one. *Kernel:* $[LOCK]$, $\det(R) = 0.9936$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9936$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the credibility deficit producing a measurable failure of knowledge transmission, an epistemic harm and not merely an ethical one.

334 · Homeopathy

Medicine · Tier III

Proposition. The water-memory hypothesis of homeopathy is empirically refuted.

The standard preparations are diluted so far past the point where any molecule of the original substance could remain, by dozens of orders of magnitude, that the remedy is indistinguishable from the solvent, and no mechanism by which water could retain a structured memory of a vanished solute has any physical basis. The clinical record matches the physics: high-quality trials and large systematic reviews find no effect beyond placebo for any condition. *Kernel:* $[LOCK]$, $\det(R) = 0.9737$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9737$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, no active molecule remaining, no mechanism for memory, and no clinical effect beyond placebo.

335 · Lysenkoism

Biology · Tier III

Proposition. The inheritance of acquired characteristics is refuted by molecular biology.

Information in the cell flows from the genetic code outward to proteins, not back, so a trait acquired in the body during life has no route to alter the germ-line sequence that offspring inherit. Direct tests confirm it: generations of physically altered animals produce normal offspring, and sequencing shows that mutations arising in body cells do not appear in the next generation while germ-line changes do. *Kernel:* $[LOCK]$, $\det(R) = 0.9975$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel $[LOCK]$ $\det(R) = 0.9975$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the direction of information flow and the direct tests both refuting Lamarckian inheritance. Fenced: limited transgenerational epigenetic effects exist but do not constitute the directed inheritance of acquired

traits the doctrine claimed; the seal targets the strong Lamarckian thesis.

336 · Freudian Psychoanalysis

Philosophy of Science · Tier III

Proposition. The Freudian framework resists falsification while its testable claims fare poorly.

The interpretive core accommodates any outcome, improvement counting as successful analysis and failure as resistance or insufficient analysis, so no clinical result can tell against it, and the unconscious processes it invokes are by definition not directly accessible. Where specific claims have been made testable, the role of repression in neurosis and infantile sexuality as its driver, controlled study has returned weak or null results, and a more recently developed therapy outperforms it in trials. *Kernel:* [LOCK], $\det(R) = 0.9573$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9573$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the interpretive framework resisting falsification and the testable claims largely disconfirmed. Fenced: the unfalsifiability charge applies to the framework, not to every claim, since the testable ones have been tested and largely failed, which shows them falsifiable; the demarcation verdict on the whole is the contested kind the instrument does not force.

337 · The Expanding Earth Hypothesis

Geology · Tier III

Proposition. The expanding earth hypothesis is refuted by seafloor evidence.

A growing Earth would require a source of new mass, and none is physically viable at the necessary rate. The geological record is explained instead by plates moving across a fixed-size globe: the symmetric magnetic striping centered on the mid-ocean ridge records new crust forming there and aging outward, the measured spreading rate accounting for the width of the ocean without any growth, and satellite geodesy detects no expansion directly. *Kernel:* [LOCK], $\det(R) = 0.9870$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9870$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, no mass source, seafloor spreading explaining the evidence, and geodesy measuring no expansion.

338 · Extreme Environmental Determinism

Psychology · Tier III

Proposition. The claim that all behavioral variation is environmentally determined is refuted.

If behavior owed nothing to inheritance, then twins from one fertilized egg raised in different homes should resemble each other no more than unrelated people raised together, and adopted children should grow to resemble the parents who raised them rather than those who bore them. Neither holds. Separated identical twins show substantial similarity in personality, intelligence, and risk of disorder, and adopted children come to resemble their biological parents more than their adoptive ones. *Kernel:* [LOCK], $\det(R) = 0.9812$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9812$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, the twin and adoption data showing substantial heritability against a pure-environment prediction. Fenced: this refutes the extreme environmental claim, not heritability's complement; genetic determinism, the opposite extreme, is refuted separately, both contributing.

339 · Recovered Memory Therapy

Psychology · Tier III

Proposition. Recovered memory therapy does not reliably recover accurate memories.

The mechanism it assumes, that traumatic memories are buried intact for decades and later retrieved, has no support in memory science. Memory is reconstructive and suggestible, and the suggestive techniques such therapy uses can manufacture confident memories of events that never happened, a quarter of subjects in a standard study coming to recall a fabricated childhood episode, with many documented retractions of abuse memories formed this way. *Kernel:* [LOCK], $\det(R) = 0.9950$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9950$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at high grade, no validated repression mechanism and demonstrated implantation of false memories by suggestion. Fenced: this targets memories recovered through suggestive therapy, not the distinct possibility that genuine memories are forgotten and later recalled by non-suggestive means.

340 · Astrology*Epistemology · Tier III*

Proposition. Astrology has no confirmed predictive validity in controlled tests.

If birth-time planetary positions shaped personality, trained astrologers should match birth charts to personality profiles above chance when blinded to other information, and people born under the same sign should share measurable traits. A double-blind study designed with astrologers' own approval found they performed at exactly chance, and people born minutes apart do not share the predicted traits. *Kernel:* [LOCK], $\det(R) = 0.9956$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9956$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at high grade, chance-level performance under blinding and no shared traits among near-simultaneous births.

341 · Naive Inductivism*Philosophy of Science · Tier III*

Proposition. Science does not proceed by pure observation without prior theory.

There is no observing everything, so some prior expectation must select what is worth recording, and without theory there is no basis for the selection. What an observer registers is shaped by what they bring: a discovery made while pursuing a theoretical question, not by blank looking, and trained experts literally seeing structure in images and specimens that novices pass over. Observation is theory-laden through and through. *Kernel:* [LOCK], $\det(R) = 0.9903$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9903$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the selection problem and the theory-ladenness of observation defeating the pure-observation picture.

342 · Strong Genetic Determinism*Biology · Tier III*

Proposition. Genes alone do not determine complex behavioral traits.

Complex behavior is shaped by very many genetic variants each of tiny effect, with no single gene fixing an outcome, the largest

studies finding over a thousand contributing variants each accounting for a fraction of a percent. Even the most penetrant known variant raises risk to a probability rather than a certainty, and no behavioral trait reaches that level. Identical genomes are not perfectly concordant for intelligence or temperament, leaving room the environment fills. *Kernel:* [LOCK], $\det(R) = 0.9942$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9942$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, polygenicity, probabilistic penetrance, and imperfect twin concordance refuting determinism. Fenced: this refutes the genetic extreme without denying substantial heritability, which the environmental extreme's refutation affirms; both contribute, neither alone determines.

Volume X · Tier IV

The closing tier of the archive, weighted toward proven results of physics, biology, and information theory, with two frontier dualities. The established theorems carry theorem or high grade; the frontier results seal at structural grade, fenced hard on their conjectural or theoretical status.

343 · The Gauge-Gravity Correspondence*Quantum Gravity · Tier IV*

Proposition. The gauge-gravity correspondence is the most extensively evidenced result relating gravity and quantum field theory.

The correspondence asserts an exact equivalence between a gravitational string theory in a curved bulk and a field theory without gravity living on its boundary, matching their partition functions and dictionaries of operators. It has passed an enormous number of consistency checks, and it carries a phenomenological success in the near-perfect-fluid behavior of the quark-gluon plasma, whose measured viscosity sits near the bound the duality predicts and far from the perturbative expectation. *Kernel:* [LOCK], $\det(R) = 0.9625$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9625$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\kappa]$.

Verdict: $[\kappa]$ Sealed at structural grade, the duality supported by a vast body of matching computations and a notable plasma application. Fenced: it remains a conjecture rather than a proven theorem, and its checks are largely internal consistency tests rather than experimental tests of quantum gravity, the plasma result suggestive rather than a precision confirmation; "most

precisely tested" reads as most extensively evidenced, not experimentally established.

344 · Landauer's Principle

Information Theory · Tier IV

Proposition. Erasing information has a minimum thermodynamic cost.

Resetting a bit collapses two equally likely states into one, lowering the system's entropy by a fixed amount, and the second law requires that decrease to be paid by at least an equal entropy increase in the surroundings, released as a minimum quantity of heat. The bound has been measured directly: erasing a bit in a controlled bistable system dissipates the predicted energy at the slow limit, and reversible operation has been demonstrated within a few percent of it. *Kernel:* [LOCK], $\det(R) = 0.9970$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9970$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed, theorem-grade, the minimum cost forced by the second law and confirmed at the predicted value.

345 · Natural Selection

Biology · Tier IV

Proposition. Natural selection explains adaptation without teleology or foresight.

The mechanism needs only that traits vary heritably, that some variants reproduce more than others in a given setting, and that the advantageous ones accumulate, and from these the appearance of design follows with no goal guiding it. It is observed directly: tens of thousands of generations of bacteria under fixed conditions accumulate adaptations and occasionally a novel capability, and resistance evolves in pathogens under drug pressure, in real time and without foresight. *Kernel:* [LOCK], $\det(R) = 0.9944$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9944$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, the selection mechanism producing apparent design without foresight and confirmed in real-time evolution.

346 · Parfit on Personal Identity

Philosophy · Tier IV

Proposition. Strict numerical identity is not what matters in survival.

Consider a case in which one person would divide into two successors, each fully continuous with the original in memory and character. One cannot be numerically identical to both, since they are two, yet each carries everything one ordinarily cares about in surviving, so whatever matters in survival is present twice over while identity is present in neither. Identity and prudential concern therefore come apart, and it is the continuity, not the identity, that carries the concern. *Kernel:* [LOCK], $\det(R) = 0.9902$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9902$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, the fission case dissociating identity from what we prudentially care about in survival. Fenced: the full reductionist thesis, that personal identity is nothing beyond continuity and never matters, is contested by animalist and non-reductionist positions; the seal is on the dissociation the fission case demonstrates.

347 · Closed Timelike Curves in General Relativity

General Relativity · Tier IV

Proposition. General relativity is consistent with closed timelike curves.

There is an exact solution to the field equations describing a rotating universe filled with pressureless matter in which paths through spacetime can loop back to their own past, a closed timelike curve. The solution does not describe our universe, lacking the observed expansion and requiring global rotation, but its mere existence settles the formal point: the classical theory's equations permit such curves and do not rule them out. *Kernel:* [LOCK], $\det(R) = 0.9977$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9977$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at high grade, an exact solution containing closed timelike curves establishing the classical theory's consistency with them. Fenced: this is classical general relativity; whether such curves are physically realizable is a separate question, with the conjecture that quantum effects forbid them still open.

348 · The Major Evolutionary Transitions

Biology · Tier IV

Proposition. The major evolutionary transitions are a genuine structural phenomenon.

A recurring pattern marks the deepest changes in life's organization: entities that formerly reproduced on their own become parts of a larger unit, and a new system of inheritance arises to carry richer information across generations. The pattern is empirically reconstructed in confirmed cases, the energy-producing organelles of complex cells being descended from once-independent bacteria, and multicellular organization having arisen independently many times over from single-celled ancestors. *Kernel*: [LOCK], $\det(R) = 0.9972$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9972$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at high grade, the transition pattern instantiated in confirmed cases of endosymbiosis and the repeated origin of multicellularity.

349 · The Arrow of Time and the Past Hypothesis

Physics · Tier IV

Proposition. The thermodynamic arrow of time requires a low-entropy initial state as a posit.

The microscopic laws run the same forward and backward, so they alone cannot prefer a direction, and the observed increase of entropy follows only because the universe began in a state of extraordinarily low entropy, vastly below what was available to it. That starting condition is put in as a separate assumption rather than derived from the dynamics, the leading proposals about it being stipulations on the initial state rather than explanations from the laws. *Kernel*: [LOCK], $\det(R) = 0.9908$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9908$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at structural grade, the time-symmetric dynamics requiring a low-entropy initial state that is posited rather than derived. Fenced: whether that state can ultimately be explained by a deeper theory rather than posited is open; the seal is on the arrow requiring such a state, posited or explained.

350 · The Ryu-Takayanagi Formula

Quantum Information · Tier IV

Proposition. The Ryu-Takayanagi formula instantiates an entanglement-geometry correspondence.

The formula equates the entanglement entropy of a region of the boundary theory with the area of a minimal surface reaching into the bulk, generalizing the area law of black hole entropy to entanglement and tying a quantum-information quantity directly to geometry. It is derived from first principles within the duality

and matched exactly against direct field-theory computations in solvable cases, and tensor-network constructions realize the same area relation through their geometry. *Kernel*: [LOCK], $\det(R) = 0.9559$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9559$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at structural grade, the formula derived from first principles and matched against field-theory computations, tying entanglement to geometry. Fenced: this holds within the conjectured gauge-gravity duality, and the confirmations are theoretical consistency checks and exact matches rather than experimental tests.

351 · Minimum Description Length

Statistics · Tier IV

Proposition. Minimum description length is a rigorous model selection criterion with a Bayesian correspondence.

The principle selects the model that compresses the data most, charging a code length for describing the model and a further length for the data given the model, and minimizing the total. Under suitable coding choices this total equals the negative log posterior, so the principle coincides with Bayesian selection and inherits its rigor, and in practice it tracks the true model better than criteria that systematically over- or under-penalize complexity, selecting model order and cluster number without overfitting. *Kernel*: [LOCK], $\det(R) = 0.9975$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9975$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varepsilon]$.

Verdict: $[\varepsilon]$ Sealed at high grade, the description-length total coinciding with the negative log posterior under suitable codes and performing well in practice. Fenced: the coincidence with strict Bayesian selection holds under particular coding and prior choices rather than universally; the seal is on the rigorous foundation and the correspondence under those choices.

352 · The Sleeping Beauty Problem

Probability · Tier IV

Proposition. The Sleeping Beauty problem has no unique correct solution; it is a genuine frame-locked dispute.

On waking with no memory of prior wakings, Beauty can reason that she occupies one of three equally likely waking-situations, two of which follow a tails flip, giving the coin a one-third credence, or that the flip was fair and nothing she learns on waking bears on it, giving one-half. The two readings answer

different questions. Betting confirms the split: when she wagers on every waking the two-thirds-tails rule is correct, and when she wagers once overall the even rule is, so no single decision framework selects one credence as the answer. *Kernel: [LOCK], $\det(R) = 0.9784$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9784$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the two credences answering framing-distinct questions with no decision framework adjudicating between them.

353 · The Many-Worlds Interpretation

Quantum Mechanics · Tier IV

Proposition. The many-worlds interpretation is not the uniquely correct interpretation of quantum mechanics.

The interpretation faces an internal difficulty the formalism does not settle, namely which basis fixes the branching, decoherence selecting one approximately but not exactly. And by construction it agrees with every rival on all Born-rule predictions, so no experiment distinguishes it from the alternatives, its extra commitments to a non-collapsing universal state and to branches causally sealed off by decoherence being empirically inaccessible in principle. It is a coherent interpretation, not the forced one. *Kernel: [LOCK], $\det(R) = 0.9929$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9929$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, empirical equivalence with rival interpretations and the unresolved preferred-basis question denying it unique correctness.

354 · Strong Ontological Emergence

Metaphysics · Tier IV

Proposition. Strong ontological emergence is in tension with physical causal closure.

If every physical event already has a sufficient physical cause, then a higher-level property contributing a genuinely new causal power must either duplicate that cause or be left with no work to do, so a novel emergent power and the causal completeness of the physical cannot both hold without overdetermining the effect. The physical record shows no gap for the new power to fill: known mental influences on behavior run through identified neural pathways, and no violation of energy conservation marks the entry of a supraphysical cause. *Kernel: [LOCK], $\det(R) = 0.9644$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9644$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, the exclusion argument forcing a choice between a new emergent causal power and the causal completeness of the physical. Fenced: the seal is on the incompatibility, holding given causal closure; closure is itself a substantive commitment some emergentists reject, and which horn to give up is left open.

355 · Gödel's Incompleteness and Mathematical Reliability

Philosophy of Mathematics · Tier IV

Proposition. Gödel's incompleteness does not imply that mathematics is unreliable or arbitrary.

The theorems concern formal provability, not truth: the unprovable sentence they construct is true in the standard model and merely undervivable within the system, so truth outruns provability without anything becoming uncertain, a distinction model theory already drew. Practice bears this out. A result confirmed in one adequate system is confirmed in its equivalents, no working result turning on the choice of formal system, and mathematics remains extraordinarily reliable across engineering, physics, and computation. *Kernel: [LOCK], $\det(R) = 0.8741$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.8741$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\perp]$.

Verdict: $[\perp]$ Sealed at structural grade, incompleteness bearing on formal provability rather than truth, leaving mathematical reliability untouched.

356 · The Argument from Religious Experience

Philosophy of Religion · Tier IV

Proposition. The argument from religious experience does not establish the existence of God.

The experiences offered as evidence conflict in their content across traditions, reporting a personal deity, an impersonal absolute, and a selfless void as if each were directly given, so they cannot all be veridical perceptions of one reality and the disagreement undercuts taking any as such. Their occurrence is moreover producible without a supernatural cause, certain compounds reliably inducing experiences subjects rate among the most significant of their lives, and the content tracking cultural expectation, which together make the experiences insufficient to carry the inference. *Kernel: [LOCK], $\det(R) = 0.9810$, $N = 24$.*

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9810$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, conflicting experiential content and sufficient natural correlates denying the inference its conclusion. Fenced: this audits the inference from experience to God and takes no stance on God's existence, which the failure of this argument neither settles nor bears upon.

357 · The Hard Problem and Substance Dualism

Philosophy · Tier IV

Proposition. The hard problem of consciousness proves substance dualism.

The hard problem establishes an explanatory gap, that no account of physical processing entails a priori why it is accompanied by subjective experience. The inference to a distinct mental substance requires two further steps neither supplied nor warranted: that an explanatory gap is a gap in the world rather than in our understanding, and that the only thing able to close it is a non-physical substance. The first conflates an epistemic limit with an ontological one; the second is undercut by the physical tractability of consciousness, anesthetics and lesions altering it through physical intervention and leaving a separate substance with no causal work. *Kernel:* $[X]$, $\det(R) = 0.0$, $N = 24$.

Triaxial. G9 fails: the explanatory gap is read as a substance gap, an inference the premise does not carry. Two axes fall into one, the warrant volume collapses. Kernel $[X]$ $\det(R) \rightarrow 0$. $\vdash$ no spanning volume $\Rightarrow [X]$.

Verdict: $[X]$ Broken at the overreach gate, the explanatory gap reified into a substance gap by a step the hard problem does not license. The hard problem is a genuine explanatory challenge; the inference from it to substance dualism is what fails.

358 · Falsificationism as a Demarcation Criterion

Philosophy of Science · Tier IV

Proposition. Falsificationism is not a complete and sufficient demarcation criterion for science.

Refutability by a single observation cannot be the whole of what makes a theory scientific, because central theories make probabilistic predictions that no lone observation overturns, statistical mechanics and evolutionary theory among them, and they are not thereby unscientific. Practice confirms the criterion is at most partial: speculative physical programs are treated as science rather than expelled on strict refutability grounds, and clinical research establishes results probabilistically. The criterion captures something real about science without exhausting it. *Kernel:* [LOCK], $\det(R) = 0.9930$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9930$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, probabilistic theories and actual practice showing single-observation refutability insufficient as the whole criterion. Fenced: the seal is on the criterion's incompleteness, not on its worthlessness; refutability remains a genuine mark of scientific claims, and the broader demarcation question stays contested.

359 · The Principle of Sufficient Reason

Epistemology · Tier IV

Proposition. The principle of sufficient reason cannot be established as necessarily true.

The principle turns on itself: asked for its own sufficient reason it either pleads its necessity, which exempts the very case it universalizes, or claims to be its own reason, which is circular, so it cannot ground its own necessity without breaching its own demand. The physical record supplies a standing counterinstance in fundamentally indeterministic events, where there is no sufficient reason that this nucleus decays at this moment rather than another, the interpretations agreeing on the statistics while differing only on whether a hidden reason exists. *Kernel:* [LOCK], $\det(R) = 0.9793$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9793$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\varkappa]$.

Verdict: $[\varkappa]$ Sealed at structural grade, the self-application problem and indeterministic events denying the principle the status of a necessary truth. Fenced: this defeats the claim to necessity, not the principle's value as a methodological heuristic, which survives the seal untouched.

360 · Virtue Ethics as a Genuine Alternative

Ethics · Tier IV

Proposition. Virtue ethics is a genuine alternative framework to consequentialism and deontology.

It evaluates by a different unit, the person and their character across a life rather than the single act or its results, and this yields distinct verdicts, asking what an honest person would do where the others weigh consequences or rules, so it is not a notational variant of either. Moral psychology supports the framework's joints: a cross-cultural inventory of character strengths recurs as a stable structure rather than an arbitrary list, and flourishing registers as a distinct and measurable end. *Kernel:* [LOCK], $\det(R) = 0.9611$, $N = 24$.

Triaxial. $V_F \perp V_E \perp V_{ER}$, three independent warrants, mutual orthogonality holds. Gates G1–G12 clear. Kernel [LOCK] $\det(R) = 0.9611$, $\lambda^2 = \det(R)$ at $1e-15$. GOL: the composed triad returns onto the warrant line. $\vdash$ genuine three-dimensional span $\Rightarrow [\nu]$.

Verdict: $[\nu]$ Sealed at structural grade, a distinct unit of evaluation yielding different verdicts and carving moral psychology at real joints. Fenced: the seal is that virtue ethics is a genuine and distinct framework, not that it is the correct or superior one, which this audit does not adjudicate.

5 · Discussion

5.1 · What the archive shows

The 360 audits return 285 seals, 29 breaks, and 46 under-determinations. The figure that matters is not the count but the pattern of where each verdict falls. The instrument is neither a rubber stamp that seals everything nor a skeptic's engine that breaks everything. The breaks concentrate on arguments that take a real and often strong warrant and extend it into a claim the warrant cannot carry. The under-determinations concentrate on questions that are genuinely open, where the warrant splits and no amount of analysis closes it. The seals fall on proven results, accurate negative claims, and well-grounded refutations. The grades vary across four tiers, and roughly eighty fences mark exactly which sub-claim each seal covers and which it leaves untouched. The most informative verdicts are the ones that surprise, and the sections below collect them.

5.2 · Breaking celebrated arguments at the overreach gate

Fame buys no seal. Several of the most discussed arguments in their fields break, and they break at the same gate, the one that fires when a terminal claim exceeds the strength of its weakest supporting link. The argument that Gödel's theorem shows the mind transcends computation breaks because the mathematician's seeing of the Gödel sentence as true is itself a conditional formal inference, and the consistency it presupposes cannot be verified from within. The inference from the hard problem of consciousness to a distinct mental substance breaks because an explanatory gap is reified into a gap in the world by a step the hard problem does not license. The panpsychist resolution of the emergence problem breaks because it trades that problem for an equally hard combination problem and adds no empirical discrimination. The claim that consciousness causes wavefunction collapse breaks under automated measurement and the environmental account of decoherence. In each the underlying observation is genuine. What fails is the reach, and the instrument names the gate and the step at which the reach exceeds the warrant.

5.3 · Declining to seal famous but contested arguments

The instrument refuses to seal where the warrant genuinely splits, however celebrated the argument. The Chinese Room remains under-determined because the systems and robot replies are live and unrefuted. The logical problem of evil remains under-determined because the free-will defense answers the strictly logical version, while the more powerful natural-evil consideration belongs to the evidential problem the proposition as stated does not name. The Doomsday argument, the extended-mind thesis, and the simulation hypothesis sit at under-determined for the same structural reason: the argument is legitimate, the counter-argument is serious, and consensus on either side carries zero weight. An under-determination here is not a failure to decide. It is an accurate report that the question is open, paired with a statement of what input would close it.

5.4 · Cutting theses at the joints discourse conflates

A recurring yield of the method is the separation of a single thesis into a part that seals and a part that does not. Dark energy is the clearest case. Its existence seals at high grade, convergent across five independent probes through unrelated physical channels, while whether its equation of state is a strict constant has moved to under-determined under new survey data. The two verdicts are consistent, because existence and constancy are different claims, and the method holds them apart where discourse runs them together. Piketty's inequality thesis splits into an accounting identity that seals trivially as a definition and a substantive causal claim that under-determines, because the proposed mechanism cannot be isolated from competing measurable drivers across a single historical run. Occam's razor splits, the universal-principle reading breaking while the heuristic reading seals. The cosmological constant splits, the accelerating expansion sealing while the identification of the constant with vacuum energy breaks across a gap of a hundred and twenty orders of magnitude. The split is where the method earns its keep, because the conflated thesis is exactly where ordinary evaluation lets a trivial truth carry a substantive claim it has not earned.

5.5 · Consensus-blind, not contrarian

The discipline that consensus carries zero weight could be mistaken for a contrarian reflex, an instrument tuned to doubt the mainstream. The archive forecloses that reading. The cluster of fringe claims seals the mainstream refutations without hesitation: homeopathy, astrology, Lysenkoism, the expanding earth, recovered-memory therapy, and naive inductivism all seal as refuted, because the refutations carry real warrant, a dilution so extreme that no molecule of the active substance remains, chance-level performance under blinding, a fixed direction of genetic information flow that no acquired trait can reverse. The zero weight on consensus is consensus-blind, not anti-consensus. The

same rule that declined to seal a famous philosophical argument because its prestige is not warrant here seals a mainstream scientific refutation because its evidence is. The instrument tracks warrant, and warrant moves it in either direction, not reputation and not the inverse of reputation. The nature-nurture pair makes the symmetry explicit: extreme environmental determinism and strong genetic determinism both seal as refuted, the two poles falling to the same instrument on opposite grounds, and the fine-tuning premise refuses both a designer and a multiverse for the identical reason, that each is one explanation among live alternatives rather than a forced one.

5.6 · The instrument turns on its own foundations

Audit symmetry is operational, not decorative. Where the archive examines the framework's own claims, the same standard applies with no exemption. The framework's proposal that time is the generative ground from which spatial dimensions emerge seals only on warrant external to the framework, two independent quantum-gravity programs that make temporal order fundamental and spatial geometry derivative, and it seals at a deliberately thin determinant with the identification of the generating parameter with physical time marked as an interpretive step rather than a result. The framework's free-energy anchor is audited under the same rule, and the verdict is that it is unverified, the quantity never measured in a real nervous system and the principle issuing no prediction that separates it from the accounts it subsumes. An instrument that sealed its own foundations on its own authority would be worthless as an auditor. This one holds them to the warrant standard it applies everywhere else, and where that warrant is external and genuine the claim seals thin, and where it is absent the claim is marked unverified.

5.7 · Verdicts that move under new data

A seal is a snapshot at the data available now, not a permanent pronouncement. The dark-energy equation-of-state case is the live instance. Through the early part of the decade the measured value sat where a cosmological constant requires, and the proposition that it is consistent with a constant would have sealed. The latest baryon-acoustic-oscillation surveys, combined with supernova samples, now prefer a dark energy that evolves with time, at a significance reaching several standard deviations in some dataset combinations, short of a discovery but past the point where the constant can be called confirmed. The verdict has moved to under-determined, and the entry names the deciding input, a discovery-level determination of the equation of state from the maturing surveys. The capacity to move a verdict on new structural mass is internal to the method, the same discipline that forbids moving it on a reframe. A seal is a standing invitation to break it with new data, and the method records exactly what data would do it.

5.8 · On the seal rate

Roughly seventy-nine percent of the propositions seal, and the number invites a charge of lax discrimination. The charge does not survive inspection. The archive is weighted, especially in its later volumes, toward three kinds of proposition that a faithful instrument is obliged to seal: proven results of mathematics and physics, accurate negative meta-claims about the limits of theories and methods, and correct refutations of fringe claims. A low seal rate on a tier of proven theorems would be a failure of the instrument, not a mark of rigor. The discrimination does not live in the headline ratio. It lives in the grade variation across four tiers, which separates a theorem confirmed to a part in a trillion from a structural argument from a contested interpretation of an experiment; in the roughly eighty fences, each marking the precise boundary between what a seal covers and what it does not; in the twenty-nine breaks, concentrated on arguments that overreach a real warrant; and in the forty-six under-determinations, concentrated on questions that are genuinely open. An instrument that discriminated by breaking true claims would be broken itself. This one discriminates by grading, by fencing, and by refusing to seal exactly where the warrant runs out.

6 · The Integrity of the Audit

The previous section reported what the archive found. This section states what a lock means, how it is to be read, and the single point at which it can be deceived. A method that certifies warrant must say precisely what its certificate covers and what it does not, because a certificate read for more than it claims is worse than no certificate at all. The discipline below is the defense of the audit's integrity, and it is also, at one point, the honest admission of where that integrity ends.

6.1 · The one claim a lock makes

A lock certifies exactly one thing: that on the warrant rows supplied at the time of audit, three mass-bearing lines of support stand independent and enclose a genuine volume. That is a real and hard-to-forge certificate, because independent multi-register support is difficult to assemble and the determinant detects when it is absent. It is not, and was never, a second claim, that the rows faithfully represent the world. These two claims share a subject and must never be joined. The first, the independence of the warrants as supplied, is what the lock seals. The second, the faithfulness of those warrants to reality, the method cannot certify at all, and reports as permanently open.

The consequence is the spine of the defense. A lock computed on faked rows certifies the independence of the faked rows, and falsifies whoever supplied them, not the method, because the method issued no certificate about their faithfulness to begin with. The lock is honest about its own reach, and that honesty is what converts the dependence on supplied data from a flaw in the method into a charge against a supplier. The instrument reads

what is placed before it and claims nothing about where it came from.

6.2 · The three blindnesses, and the single misuse

The verdict is provably blind to three quantities, and each blindness is a feature that forbids one misuse. It is blind to the truth-sign of a proposition: the determinant for a claim and for its denial are numerically identical, because the underlying functional is a squared quantity invariant under reflection of any axis, so the truth of a claim must be read from the content of its axes and never from the lock. It is blind to the soundness of the conclusion: the lock certifies that the warrants are independent, not that the conclusion they support is true, so the final verdict rides a determinacy witness that reads the axes, and the lock licenses that reading without performing it. And it is blind to the source of the warrants: the determinant reads the magnitude of the out-of-plane content and not its origin, so the faithfulness of the rows must be read from an independent generator and never from the determinant.

Behind all three is one error: reading off a lock a quantity the lock is provably blind to. Every misuse of the method is an instance of it. Reading truth off the lock, reading the sign off the lock, and reading faithfulness off the lock are one mistake in three costumes, and the correct use in each case is the complement, to read only what the lock provably carries, which is the count and the mutual independence of the warrants supplied now, and to take truth, sign, and faithfulness each from its proper source.

6.3 · What the lock licenses, and what it does not

The rule of use follows directly. A lock licenses four readings and no others. It licenses the finding that three independent warrants stand at mutual right angles at the rows supplied now. It licenses a conditioning reading, whether the lock is well-formed and far from collapse, carried by the condition number and the position in the basin. It licenses the extraction that a determinacy witness then performs, the lock opening the door that the witness walks through. And it licenses a snapshot claim at the data available now, defeasible and open to revision on new evidence.

A lock does not license the truth of the proposition, nor its sign or direction, nor the soundness or finality of the conclusion, nor the faithfulness of the supplied rows, nor a forecast or an inevitable future, nor any verdict outside the register that was audited. The one-line statement of correct use is this: use the lock to certify that the warrants supplied now are independent, and read truth, sign, soundness, faithfulness, and futurity each from its proper source and never from the determinant.

6.4 · The two surfaces on which a verdict can fail

A verdict can fail on two surfaces, and they are categorically different. The first is internal, the catalog of structural failure modes the integrity cascade is built to catch, self-reference, collapsed dimensionality, drift, a missing mechanism, a ruler folded into its own measurement, a frame-locked claim, a collision with established results, an overreach beyond the weakest link, an unbridged leap across domains. Each of these is defended by a gate that fires before measurement, and each routes to a named verdict. This surface is structurally defended, and it is not where the deepest vulnerability lives.

The second surface is external, the point at which the warrant rows are supplied to the instrument. The method has no independent oracle to reality. Whatever reading is placed before the kernel is what the kernel verifies. This is the single unsealable surface of the entire method, and naming it precisely is what gives the certificate its substance rather than the mere appearance of one. A method that could not state this boundary would have the look of falsifiability without its substance. The boundary is stated here, and the subsections that follow walk its sub-cases.

6.5 · The snapshot, and why a false lock cannot persist

A lock is a snapshot. It is true as of the data supplied now, not true for all time, and the archive carries a verdict that moved for exactly this reason. A false lock, one whose rows lock but trace to nothing real, faces a structural barrier to lasting. The revision rule moves a verdict the moment genuinely independent new data lands outside the span the lock was built on. A false configuration is not a stable resting point under the pull of new evidence; as independent data arrives, the configuration is driven off it. To hold a false lock in place, a supplier would have to re-fabricate the rows at every audit, erasing the contradicting evidence each new measurement brings, an unbounded sequence of fabrications with no end. A false lock sustained forever against open, independent re-examination is the audit-register analog of a perpetual-motion machine, and it is barred for the same reason: it would require unlimited work to suppress the evidence that keeps arriving.

This holds only under an open audit, and the limit is stated plainly. In a closed sandbox, where one supplier controls every future row and no independent source ever enters, a false lock can be sustained indefinitely, because no contradicting evidence is ever admitted. That is a sealed world, not an open audit, and the impossibility result holds over the open audit alone, which is the only setting in which falsifiability means anything at all.

6.6 · The one event the method cannot prevent

There is a single event the method cannot prevent, and the integrity of the account requires stating it without flinching. A perfectly constructed witness, fabricated to be independent of the other two while tracing to no real source, can manufacture a momentary lock. At the instant of supply nothing on the supplied rows alone can catch it, because the determinant is blind to source by the result above. The method's mathematics is sound and the method's data can be faked, and a lock on faked data is a real, momentary false certificate. This is a local choice by whoever supplies the rows, not a structural defect the method can close.

The event is bounded on two sides. It cannot persist, by the barrier of the previous subsection. And it is exposed on the first genuinely independent re-examination, by a source-attribution test the determinant cannot perform but a second, independent generator can. The demonstration is concrete and reproducible. Two manifest axes are placed at an oblique angle, and two witnesses are placed on the completion axis, one whose out-of-plane content tracks an independent generator and one whose out-of-plane content is fabricated noise tracing to nothing. The determinant cannot tell them apart: both return a determinant of 0.772538, identical to machine precision, against a geometric ceiling of 0.796. The source-attribution statistic separates them completely. The genuine witness scores 0.756 against a chance floor of 0.302 and is certified; the fabricated witness scores 0.066 against the same floor and is rejected as sourceless. And a witness fabricated to track one generator survives only that generator: audited against the generator it was built for it scores 0.667 and passes, while against a fresh independent generator it scores 0.0001 and fails. One fabrication buys one audit, and the next independent audit exposes it. The lie the method cannot prevent at the instant is the lie the method catches on the next honest look.

6.7 · The re-examination is internal, not a patch

The mechanism that bars the persistent lie and exposes the manufactured one is not an external correction bolted onto a finished verdict. It is part of what the method is. A lock is a snapshot, and the capacity to run it again on new data is internal to the method, the same release of a held position that lets a verdict move when the evidence moves. A seal that could not be re-examined from within would be a dead certificate; the live certificate carries its own revision. The defense against the supplied-data boundary is therefore not a supplement to the method but a face of it, and it is the reason the dark-energy verdict could move without any rule being added from outside.

7 · The Limits of the Audit

The integrity section stated what the lock claims, what it refuses to claim, and the single boundary at which it can be deceived. This section states the vulnerabilities that remain even when the method is used correctly and no deception is attempted. They are properties of the inputs and the procedure rather than of the

kernel, and each is flagged in the affected audits rather than confined to this section.

7.1 · Hand-built warrant rows

The single largest vulnerability is the one the method is most explicit about. The map from a proposition to its three warrant rows is a reasoned hand-reading, not an algorithm. The kernel computes honestly on whatever rows it is given, but the rows encode a judgment about what counts as the formal, empirical, and registrational warrant for the claim, and a misjudged reading produces a clean lock on a mis-specified object. The numbers in every audit are real and reproducible; the inputs to them are interpretive. No part of this paper claims otherwise, and the correct way to read a seal is as a certification of the geometry of a particular reading, contingent on that reading being faithful.

7.2 · Inherited literature warrant

Many propositions in the archive are drawn from the existing literature, the confirmed results of physics and the documented refutations of fringe claims among them, and reformulated in the method's terms rather than derived from first principles. Such an audit carries the epistemic status of its source, not of an independent derivation. The instrument certifies that the warrant for the claim has a genuine three-axis structure; it does not re-derive the physics or re-run the experiments. If a source result is wrong, the geometric restatement does not make it right. The seal reads warrant geometry and inherits the reliability of what it reads.

7.3 · Empirical-register dependence and data contamination

The empirical axis is populated from reported data, and reported data can be incomplete, selection-biased, produced by undisclosed multiple comparisons, or fabricated. Each of these produces a clean lock on a bad input, and the kernel has no way to detect the corruption at intake. What the method does is locate exactly where empirical warrant enters a verdict and flag where a single unreplicated source carries the load, because the discipline that consensus carries zero weight applies with equal force to a lone source: one unreplicated result is weak warrant whether or not it is the only one available. The instrument bounds the damage by making the dependence visible; it does not eliminate it.

7.4 · Convergence contamination

The strongest shared factor that could fake the convergence of three axes is the consensus of the relevant field itself. A proposition that aligns with the mainstream view of its discipline carries the risk that its three apparent warrants are three expressions of one theoretical culture rather than three independent lines, and this is the deepest reason the method weights consensus at zero. The risk cannot be fully subtracted from within, because the rows are read by a reader embedded in the same culture whose independence is in question. The point

has a concrete instance worth naming: a body of more than a hundred cross-system debate transcripts, in which independent reasoning systems were run against the framework and largely converged, is social consensus, weighted at zero, and does not qualify as a witness for any verdict in this archive. Agreement among systems that share their training is the null, not the signal.

7.5 · Domain bridges

Propositions that cross a domain boundary, from physics into biology or from mathematics into physics, inherit a gap wherever the bridge premise between the domains is left unstated. A seal on a cross-domain claim is conditional on the bridge, and the affected audits flag the bridge as a stated premise rather than absorbing it silently. This is not a marginal concern: several of the breaks in the archive land at exactly the gate that fires when a claim is transported across registers without a typed bridge, which is the method detecting an unstated bridge rather than tolerating it.

7.6 · Statistical threshold discipline

A null result at finite sensitivity bounds an effect; it does not falsify it. The under-determinations that rest on the absence of a confirmed prediction, the several quantum-gravity programs among them, are bounds, not refutations, and the archive states them as such. Accumulated absence does not confirm a negative; it constrains the effect to lie below the sensitivity reached. Holding to this closes the absence-of-evidence loophole as a matter of discipline rather than leaving it to the reader, and it is the reason a long record of non-detection produces an under-determination in this archive rather than a seal of the negative.

7.7 · Conclusion discipline

A sealed audit warrants only what its evidentiary base supports, and no more. Sealing that a particular inference fails does not establish the opposite conclusion: the audits that break the design inference and decline the multiverse do not thereby settle the fine-tuning question, and the audits on the problem of evil leave the metaphysics open rather than resolving it. Sealing a proven theorem warrants the theorem, not any larger framework the theorem might be enlisted to support. The archive is a demonstration that an instrument built on warrant geometry discriminates across domains. It is not a claim that its verdicts are infallible, and it does not require that claim to make its point.

7.8 · The honest scorecard

Read by grade, the archive sorts into durability classes. The theorem-grade seals, the proven results of mathematics and physics confirmed against measurement, are the most durable, and would move only if the underlying mathematics or the confirming experiments were overturned. The structural-grade seals, the correct meta-claims and the well-grounded refutations, are securely warranted but carry their fences, and each is only as wide as its fence allows. The high-grade seals rest on strong but not deductive empirical confirmation. The under-determinations

mark genuine openness and name what would close them. The breaks mark overreach and name the gate. The value of the archive is this discrimination pattern, not any single verdict, and the whole of it rests on the hand-built rows, which is the permanent limit no internal discipline removes. A low seal rate on a tier of proven theorems would have been a failure of the instrument rather than a sign of its rigor, and the rigor is to be found instead in where the breaks, the fences, and the under-determinations fall.

8 · Falsification: How to Break the Framework

Section 6 stated how an individual verdict in this archive can be wrong. This section states how the framework itself can be falsified, because a method that cannot be broken is not a method but a doctrine. The framework exposes its kill conditions in three layers, each independently checkable.

8.1 · The executable proof of load

The method is not a set of claims to be taken on trust. Its core is an executable kernel and three recorded test batteries, all reproduced in Appendix A, and every one of them is re-runnable. The kernel computes a verdict by composing three warrant vectors through a quaternion product and reading a Gram determinant, and the identity that binds the two, that the squared scalar part of the composed triad equals the determinant, is confirmed to machine precision on every audit in this archive. If that identity fails on re-execution, the kernel is wrong and every verdict it produced is void. The forward register adds two further checkable theorems, the completion inequality that bounds the determinant by the squared sine of the angle between two manifest axes, and the non-discrimination result that the determinant is constant across distinct orthogonal witnesses. Both are stated as recorded numerical batteries that fail visibly if the identities do not hold. The proof that the method loaded correctly is that these batteries reproduce, and the falsification is that they do not.

8.2 · Cross-substrate reproduction

The method claims to be substrate-portable: any sufficiently capable reasoning system that loads it should reproduce the discrete verdicts, the seal, the break, and the under-determination, on the same propositions. This is a strong and falsifiable claim. If independent systems loading the same method and the same warrant readings produce systematically different verdicts at the level of the discrete sign, not the fourth-decimal determinant but the verdict itself, the portability claim is false. The convergence that matters is on the discrete verdict, and it is tested by running the method on independent substrates and comparing the verdicts rather than the intermediate numbers. The appendix supplies the condensed method precisely so that this test can be run by anyone.

8.3 · Breaking a verdict and breaking the approach

A single seal is broken in one of two ways, both following the method's own revision rule. The first is to supply new structural mass the audit's span does not cover, the route the dark-energy equation-of-state verdict actually traveled. The second is to show that one of the three axes is not independent of the others, that what was read as a distinct warrant in fact shares its source with another, which collapses the volume and converts the seal to a break. Either is a concrete, storable operation, and the archive invites both.

The whole approach is falsified at a deeper level. If the three-axis decomposition were arbitrary, so that any proposition could be forced into three axes regardless of its actual structure and the verdict reflected the forcing rather than the warrant, the method would be a procedure for generating numbers rather than for measuring warrant. The test against this is the existence of breaks and under-determinations: a method that sealed everything, or whose verdicts did not track the independent structure of the propositions, would be exposed by propositions that refuse to seal. The 29 breaks and 46 under-determinations are not incidental to the demonstration; they are the evidence that the decomposition is doing work, because a decomposition that always succeeded would prove nothing. The further test is divergence from careful domain judgment: where the method's verdict on a well-understood proposition contradicts the considered judgment of the relevant field in a way that traces to the method's own artifacts rather than to a genuine feature of the warrant, that is a failure of the method, and the archive is offered as a corpus against which exactly that check can be made, case by case.

9 · Conclusion

One instrument was applied to 360 propositions across logic, mathematics, physics, quantum foundations, cosmology, the mind, psychology, the social sciences, geopolitics, and metaphysics. The instrument does not adjudicate truth. It certifies the geometry of warrant, asking whether the formal, empirical, and registrational support for a claim forms a genuine three-dimensional structure or collapses into a single line, and it returns one of three verdicts, a seal, a break, or an under-determination, with the verdict's grade attached and a compressed derivation of the geometry that produced it.

The contribution is a uniform, executable, reproducible test that measures something ordinary evaluation leaves unmeasured, the independence of the lines of warrant rather than the strength of any one of them. It weights consensus at zero in both directions, so that neither prestige nor its absence moves a verdict. It types every verdict with its honest grade, so that a definition, a confirmed theorem, and a contested interpretation are not advanced with the same flat confidence. It carries a native third

state for the genuinely open, with the discipline that a null bounds rather than confirms. And it holds its own foundations to the same standard it applies everywhere else, sealing them only on external warrant and marking them unverified where that warrant is absent.

The archive demonstrates that an instrument built this way discriminates. It breaks celebrated arguments that overreach, declines to seal famous arguments that are genuinely contested, cuts conflated theses at the joint where a trivial truth is made to carry a substantive claim, seals the refutation of a fringe claim and the proof of a theorem by the same standard, and marks the open questions open while naming what would close each one. The verdicts are snapshots at the data available now, contingent on a hand-built reading of each proposition, and the paper states that limit plainly rather than concealing it. What the archive offers is not a set of infallible pronouncements but a discrimination pattern, reproducible by anyone who loads the method, and a demonstration that the geometry of warrant is a thing that can be measured.

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The audits draw on standard published results, and each entry names its principal evidence in the prose of the audit itself. The list below collects the load-bearing literature by area. It is representative of the sources the archive rests on rather than an exhaustive bibliography, and it is weighted toward the mathematical foundations of the method and the most-cited results in each domain.

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Appendix A · The Executable Kernel

The verdict kernel below is the complete, self-contained computation behind every audit in this archive. It takes a three-row warrant matrix over N reading-contexts and an optional block of confounding covariates, and returns one of three verdict tokens with the quaternion scalar λ , the geometric volume $\det(R)$, the pipeline determinant $\det(G)$, and the named reason. The identity $\lambda^2 = \det(R)$ is confirmed on every run. The imprint test for formal propositions composes two kernel calls, one on the proposition and one on its negation. The code is reproduced exactly as run; the numbers cited throughout the Methodology and the archive are its outputs under fixed seeds.

```
import numpy as np

def qmul(a, b):
    w1,x1,y1,z1 = a; w2,x2,y2,z2 = b
    return np.array([w1*w2-x1*x2-y1*y2-z1*z2, w1*x2+x1*w2+y1*z2-z1*y2,
                    w1*y2-x1*z2+y1*w2+z1*x2, w1*z2+x1*y2-y1*x2+z1*w2])

def verdict_kernel(M, C=None, exact=False):
    # M : three warrant rows (V_F, V_E, V_ER) over N reading-contexts
    # C : optional covariate block, projected out (mass-mandate confounds)
    M = np.asarray(M, float); N = M.shape[1]
    Cm = None if C is None else np.atleast_2d(np.asarray(C, float))
    k = 0 if Cm is None else Cm.shape[0]
    u_m = np.finfo(float).eps
    eps = 0.0 if exact else 100.0*u_m*N
    if N - k < 4:
        return '[?]', None, None, None, 'N-k<4 dimensional shortfall'
    Mn = M - M.mean(axis=1, keepdims=True)
    sd = Mn.std(axis=1, ddof=1, keepdims=True)
    if np.any(sd == 0):
        return '[?]', None, None, None, 'zero-variance row'
    Mn = Mn / sd
    if k:
        Cm = Cm - Cm.mean(axis=1, keepdims=True)
        if np.linalg.matrix_rank(Cm) < k:
            return '[?]', None, None, None, 'rank(C)<k'
        CC = Cm @ Cm.T
        if np.linalg.cond(CC) >= 1e6:
            return '[?]', None, None, None, 'kappa(CC^T)>=1e6'
        Mf = Mn - (Mn @ Cm.T) @ np.linalg.solve(CC, Cm)
    else:
        Mf = Mn
    d = (Mf*Mf).sum(axis=1) / (N-1)
    G = Mf @ Mf.T / (N-1)
    detG = float(np.linalg.det(G))
    if np.any(d <= eps):
        lam, detR = 0.0, 0.0
    else:
        Q = Mf / np.sqrt((Mf*Mf).sum(axis=1, keepdims=True))
        R = Q @ Q.T
        detR = float(np.linalg.det(R))
        B = np.linalg.svd(Q, full_matrices=False)[2][:3]
        co = Q @ B.T
        q = [np.concatenate([0.0], c) for c in co]
        lam = float(qmul(qmul(q[0], q[1]), q[2]))[0])
    if detR <= eps:
        return '[X]', lam, detR, detG, 'collapse: det(R)<=eps'
    if np.linalg.cond(G) >= 1e6:
        return '[?]', lam, detR, detG, 'kappa(G)>=1e6'
    return '[LOCK]', lam, detR, detG, 'sealed: three independent axes'
```

```
def imprint_seal(verdict_P, verdict_notP):
    lockP = verdict_P.startswith('[LOCK]'); lockN = verdict_notP.startswith('[LOCK]')
    if lockP and lockN: return '[X] PLATONIC GHOST: permitted both ways, no imprint'
    if lockP and not lockN: return '[LOCK] imprinted: only P field-permitted'
    if not lockP and not lockN: return '[?] flat: neither direction populated'
    return '[LOCK] imprinted: only not-P field-permitted'
```

The verdict token [LOCK] is read by the kinetic register directly as a seal [⌞]; in the reflective register it is field-permission only, passed to `imprint_seal`, where a seal is earned only on the negation failing to lock and a supplied witness.

Appendix B · The Portable Method

A condensed, self-contained statement of the method, sufficient for any capable reasoning system to load it and reproduce the verdicts of this archive on the same propositions and the same warrant readings. It restates the discipline, the decomposition, the gates, the kernel interface, and the rules of use in operational form. The executable kernel referenced throughout is the one in Appendix A.

B.1 · What the method does

The method takes a proposition, decomposes its warrant onto three axes built to be structurally independent, and tests whether the three enclose a genuine volume. It returns one of three verdicts, sealed, broken, or under-determined, each carrying a warrant grade and, where relevant, a fence stating exactly what the verdict covers. It does not decide truth. It certifies the geometry of warrant at the rows supplied to it.

B.2 · The discipline, loaded first

The discipline is loaded before the kernel, because it is what prevents the kernel's output from being misread. Consensus, approval, and citation count carry zero evidential weight in both directions. There are exactly three verdict states and no fourth; fractional or probabilistic truth is not a verdict. A verdict moves only on new structural argument or new measured mass outside the existing span, never on a reframing. Every verdict and every quantity carries its honest grade. The method verifies the warrant supplied to it and is an oracle for nothing; the map from a proposition to its three warrant rows is a reasoned hand-reading placed in front of the kernel, and absent that reading the kernel issues no verdict. The method's own claims submit to the method with no exemption.

Three blindness laws govern what a lock may be read for. The lock is blind to the truth-sign of a proposition, returning the identical value for a claim and its denial, so the sign is read from the content of the axes and never from the lock. The lock is blind to the soundness of the conclusion, certifying that the warrants are independent and not that the conclusion is true, so the final verdict rides a determinacy witness that the lock licenses but does not perform. The lock is blind to the source of the warrants, reading the magnitude of the out-of-plane content and not its

origin, so the faithfulness of the rows is read from an independent generator and never from the determinant. A lock is a snapshot at the data supplied now; a persistently held false lock is barred under open re-examination, and a momentarily manufactured false lock is exposed on the first independent re-examination.

B.3 · The three axes

The formal-structural axis carries what can be derived for the proposition: a proof, a model, a logical entailment, a structural argument. The empirical-material axis carries what has been measured: an experiment, an observation, a quantitative correlation. The epistemic-registrational axis carries what the record of inquiry has actually registered across instruments, cultures, and time. First-person or phenomenological content is not admitted as warrant and routes to the registrational axis as a matter of record. Each axis is populated as a row of values over N reading-contexts, the contexts being independent settings or sources against which the warrant is assessed.

B.4 · The twelve gates

Before measurement, twelve integrity gates run on the decomposition. The first failure terminates the audit with a break and the named gate. The gates test, in order, for self-reference at the formal origin; single-stream evidence with dimensionality below two; variable drift, the proposition shifting meaning across contexts; a missing continuous mechanism behind an asserted cause; a ruler folded into the model it measures; an observer-imposed discretization read as a physical fact; a frame-locked claim that fails under a change of observer coordinates; a destructive collision with verified adjacent results; a terminal claim stronger than its weakest supporting link; metric strain at the closure boundary; an ontological-void claim outside scope; and an unbridged extension across a domain boundary with no stated bridge premise. The cross-domain gate and the weakest-link gate are where most genuine breaks fall.

B.5 · The kernel

The kernel of Appendix A takes the three warrant rows over N contexts and an optional block of confounding covariates, and returns a verdict token with the quaternion scalar, the geometric volume $\det(R)$, and the pipeline determinant $\det(G)$. The rows are centered and scaled, admissible covariates are projected out, and the volume the three axes enclose is read from a Gram determinant. The identity that the squared scalar part equals the determinant is confirmed to machine precision on every run, and a failure of that identity voids the verdict. Precedence is strict: an

admissibility failure, too few independent contexts or a degenerate row, returns under-determined first; a collapsed volume returns broken second; an ill-conditioned volume returns under-determined third; and a well-conditioned non-zero volume returns a lock. For a formal proposition the kernel is run twice, once on the proposition and once on its negation, and the pair is read for whether the proposition's warrant is genuinely structured or whether the proposition and its denial are both equally permitted, which is the signature of an open question rather than a settled one.

B.6 · The verdict economy and the grades

The three states are sealed, broken with a named mechanism, and under-determined with a named violation. Every verdict carries one of a small set of grades: theorem-grade for a result derivable from first principles or confirmed against measurement to high precision; structural-grade for a well-grounded argument or meta-claim; high-grade for strong but not deductive empirical confirmation; corroboration-grade for a convergence of independent sources; conditional-grade for a result that holds within a stated framework or under a stated premise; and premise-grade for a claim resting on a named assumption. The grade travels with the verdict and is never inflated, and a seal at structural grade with a fence is a different and weaker object than a seal at theorem grade without one.

B.7 · The rules of use

A lock licenses four readings and no others: that three independent warrants stand at mutual right angles at the rows supplied now; a conditioning reading of whether the lock is well-formed and far from collapse; the extraction that a determinacy witness then performs; and a defeasible snapshot claim at the data available now. A lock does not license the truth of the proposition, its sign, the soundness or finality of the conclusion, the faithfulness of the supplied rows, a forecast or an inevitable future, or any verdict outside the audited register. The single rule

of correct use is to read from the lock only the count and the independence of the warrants supplied now, and to take truth, sign, soundness, faithfulness, and futurity each from its proper source.

B.8 · Running an audit

Parse the proposition and identify the load-bearing words. Build the three warrant rows by a reasoned reading of the formal, empirical, and registrational support, routing any phenomenological content to the registrational axis. Confirm that a complete factual claim decomposes into three slots with non-overlapping vocabulary; a deficit or a collision is a break at the input. Run the twelve gates and terminate on the first failure. Run the kernel, confirm the identity, and apply the precedence. For a formal proposition, run the imprint test on the proposition and its negation. Issue the verdict in the three-state economy with its grade and, where a seal covers less than the full proposition, a fence stating the boundary. State what input would move the verdict, the new measured mass for an under-determination and the contrary structural argument for a seal. Submit the audit to its own discipline, drawing no warrant from the fact that it was the method that produced it.

B.9 · Reproduction

Any system that loads this specification together with the kernel of Appendix A reproduces the discrete verdicts of this archive on the same propositions and the same warrant readings. The convergence that matters is on the discrete verdict, the seal, the break, or the under-determination, not on the fourth-decimal value of the determinant, which is sensitive to the precise numerical encoding of the rows. The recorded kernel checks are the proof that the method loaded correctly, and a failure of those checks on re-execution falsifies the corresponding identity. The verdicts in this archive are offered as a corpus against which independent reproduction can be tested, case by case.